Topology Notes

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1. Topological Spaces and Continuity

Definition (topology): A *topology* on a set X is a collection $\mathcal{T}$ of subsets of X that satisfy the following properties:

a) $\emptyset$ and X are in $\mathcal{T}$.
b) The union of an arbitrary collection of sets in $\mathcal{T}$ is in $\mathcal{T}$.
c) The union of any finite collection of sets in $\mathcal{T}$ is in $\mathcal{T}$.

Then the pair $(X, \mathcal{T})$ is a *topological space*, sometimes just denoted X if $\mathcal{T}$ is clear from context. The sets in $\mathcal{T}$ are called the *open sets* of X.

Example (standard topology): The standard topology $\mathcal{T}$ on $\mathbb{R}^n$ is given by

$$\{U \subseteq \mathbb{R}^n : \text{for all } x \in U, \text{there exists } \varepsilon > 0 \text{ such that } B_\varepsilon(x) \subseteq U\}.$$

Example ((in)discrete topology): Two obvious topologies for any set X are given by the discrete topology $\mathcal{P}(X)$ (every subset of X) and the indiscrete topology $\{\emptyset, X\}$.

Example (particular point topology): Fix $p \in X$. Then

$$\{U \subseteq X : p \in U\}$$

is a topology on X.

Example (co-finite topology): Let X be any nonempty set. Then the co-finite topology is given by

$$\mathcal{T} = \{U \subseteq X : X \backslash U \text{ is fininte}\} \cup \{\emptyset\}.$$

Clearly a) is satisfied, since $|X \backslash X| = 0$. Suppose $\{V_\alpha\}_{\alpha \in J}$ is a collection of sets in the $\mathcal{T}$, indexed by some jet J. Then

$$X \backslash \bigcup_{\alpha \in J} V_\alpha \subseteq X \backslash V_{\alpha'},$$

where $\alpha' \in J$ is arbitrary. Since the right side is finite, so is the left, so $\mathcal{T}$ is closed under arbitrary unions, and so b) is satisfied. Now suppose $V_1, \ldots, V_n$ are a collection of sets in $\mathcal{T}$. Then

$$X \backslash \bigcap_{i=1}^{n} V_i = \bigcup_{i=1}^{n} X \backslash V_i.$$

The right side if the union of finite sides, so it's finite, and thus the left is as well. Thus $\mathcal{T}$ is closed under finite intersection, so c) is satisfied. Thus $\mathcal{T}$ is indeed a topology.

Example (co-countable topology): Let X be a nonemtpy set. Then the co-countable topology is given by

$$\mathcal{T} = \{U \subseteq X : X \backslash U \text{ is countable}\} \cup \{\emptyset\}.$$

Remark

To show a topology is closed under finite intersection, we only have to show that it's closed under pairwise intersection, after which finite intersection follows by induction.

Definition (finer/coarser): Let X be a set, and suppose $\mathcal{T}_1$ and $\mathcal{T}_2$ are topologies on X. Then $\mathcal{T}_1$ is *refined* by $\mathcal{T}_2$ if $\mathcal{T}_1 \subseteq \mathcal{T}_2$. We also say that $\mathcal{T}_2$ is *finer* than $\mathcal{T}_1$, or that $\mathcal{T}_1$ is *coarser* than $\mathcal{T}_2$.

Example: On $\mathbb{R}^n$, the discrete topology refines the usual topology, which refines the indiscrete topology.

Example: Topologies are not necessarily comparable, for example the topology that contains all subsets that contained a fixed point x is neither of a subset of nor contains the usual topology on $\mathbb{R}^n$.

1.1. Topological Bases

Given a collection of sets $\mathcal{A}$, we use the shorthand $\bigcup \mathcal{A}$ to denote $\bigcup_{U \in \mathcal{A}} U$.

Definition (basis): If X is a set, a *basis* for a topology on X is a collection $\mathcal{B}$ of subsets of X that satisfies the following properties:

a) For each $x \in X$, at least one basis element contains B contains x (this is equivalent to $\bigcup \mathcal{B} = X$).
b) If $x \in B_1 \cap B_2$ of two basis elements B_1, B_2, then there exists $B_3 \in \mathcal{B}$ such that $x \in B_3 \subseteq B_1 \cap B_2$.

Basis elements are called *basis open sets*.

Proposition (topology generated by basis): Suppose $\mathcal{B}$ is a basis of X. Let

$$\mathcal{T} := \{U \subseteq X : \text{for all } x \in U, \text{there exists } B \in \mathcal{B} \text{ such that } x \in B \subseteq U\}.$$

Then $\mathcal{T}$ is a topology, called the *topology generated by* $\mathcal{B}$, and is denoted $\mathcal{T}_{\mathcal{B}}$.

Proof: Clearly $\emptyset, X \in \mathcal{T}$, since the condition on the collection holds vacuously for $\emptyset$, and the first condition of a basis implies that X also satisfies the condition on the collection.

Now suppose $\{V_\alpha\} \subseteq \mathcal{T}$. Let $x \in \bigcup_\alpha V_\alpha$. Then $x \in V_{\alpha'}$ for some α. Since $V_{\alpha'} \in \mathcal{T}$, there exists $B \in \mathcal{B}$ such that $x \in B \subseteq V_{\alpha'} \subseteq \bigcup_\alpha V_\alpha$. This holds for all x in the union, so $\bigcup_\alpha V_\alpha \in \mathcal{T}$, so $\mathcal{T}$ is closed under arbitrary union.

Now suppose $V_1, V_2 \in \mathcal{T}$, and let $x \in V_1 \cap V_2$. Then $x_1 \in V_1, V_2$, and so by the collection condition, there exists $B_1, B_2 \in \mathcal{B}$ such that $x \in B_1 \subseteq V_1, x \in B_2 \subseteq V_2$. This implies $x \in B_1 \cap B_2 \subseteq V_1 \cap V_2$. Then by the second condition for a basis, there exists $B_3 \in \mathcal{B}$ such that $x \in B_3 \subseteq B_1 \cap B_2 \subseteq V_1 \cap V_2$. Since this holds for all x in the intersection, $V_1 \cap V_2 \in \mathcal{T}$, and so $\mathcal{T}$ is closed under pairwise intersection, and thus finite intersection. Thus $\mathcal{T}$ is a topology, as desired. ■

Proposition (equivalent way to generate topology from basis): Suppose $\mathcal{B}$ is a basis of X. Then $\mathcal{T}_{\mathcal{B}}$ is given by

$$\mathcal{T}' = \left\{\bigcup \mathcal{C} : \mathcal{C} \subseteq \mathcal{B}\right\}.$$

Proof: Suppose $V \in \mathcal{T}'$. Then $V = \bigcup_\alpha B_\alpha$ for $\{B_\alpha\} \subseteq \mathcal{B}$. Let $x \in V$. Then $x \in B_{\alpha'}$, for some α', and thus $x \in B_{\alpha'} \subseteq V$. This holds for all $x \in V$, so $V \in \mathcal{T}_{\mathcal{B}}$. Thus $\mathcal{T}' \subseteq \mathcal{T}_{\mathcal{B}}$. Now suppose $V \in \mathcal{T}_{\mathcal{B}}$. By definition, for each $x \in V$, there exists $B_x \in \mathcal{B}$ such that $x \in B_x \subseteq V$. Thus $\bigcup_{x \in V} B_x = V$, so $V \in \mathcal{T}'$, and thus $\mathcal{T}_{\mathcal{B}} \subseteq \mathcal{T}'$. Since we have both inclusions, the collections are equal, as desired. ■

Example (discrete topology basis): For X, the basis $\mathcal{B} = \{\{x\} : x \in X\}$ generates the discrete topology. This is because the union of any collection of elements in $\mathcal{B}$ gives a subset of X, which is in the discrete topology, and any subset can be written as the union of its individual elements.

Example (standard topology basis): The set of open intervals in $\mathbb{R}$ forms a basis for the standard topology.

Example: $\mathcal{T}_{\mathcal{T}} = \mathcal{T}$.

Example (lower limit topology): Let $\mathcal{B} = \{[a, b) \subseteq \mathbb{R} : a < b\}$. Then $\mathcal{B}$ is a basis of $\mathbb{R}$, and generates a topology $\mathcal{T}$ known as the lower limit topology. The topological space $(\mathbb{R}, \mathcal{T})$ is often called the *Sorgenfrey line*, and is denoted $\mathbb{R}_\ell$.

Proposition (basis from topology): Suppose X is a topological space, and let $\mathcal{C}$ be a collection of open sets of X such that for each open $U \subseteq X$ and for each $x \in U$, there exists $C \in \mathcal{C}$ such that $x \in C \subseteq U$. Then $\mathcal{C}$ is a basis for the topology of X.

Proof: First we show that $\mathcal{C}$ is a basis for X. Clearly $\mathcal{C}$ covers X, since for $x \in X \subseteq X$, the condition implies there exists $C \in \mathcal{C}$ that contains x. Now suppose $C_1, C_2 \in \mathcal{C}$, and let $x \in C_1 \cap C_2$. Since these two are open, there intersection is open, so by the condition, there exists an open $C_3 \in \mathcal{C}$ such that $x \in C_3 \subseteq C_1 \cap C_2$. Thus the second condition for a basis is satisfied.

Now we show that this basis actually generates the topology on X. Suppose V is open in X. Then for each $x \in V$, there exists $C_x \in \mathcal{C}$ that contains x, and thus we can write $V = \bigcup_{x \in V} C_x$, so $V \in \mathcal{T}_{\mathcal{C}}$. Now suppose V is the union of a subcollection of $\mathcal{C}$. Since they're all open sets, their union is open in X, and so the open sets generated by $\mathcal{T}_{\mathcal{C}}$ are open in X. Thus the topologies are the same. ■

Proposition (comparing bases): Let $\mathcal{B}$ and $\mathcal{B}'$ be bases for topologies $\mathcal{T}$ and $\mathcal{T}'$ on X. Then the following are equivalent:

a) $\mathcal{T}'$ is finer than $\mathcal{T}$.
b) For each $x \in X$ and each basis element $B \in \mathcal{B}$ containing x, there exists $B' \in \mathcal{T}'$ such that $x \in B' \subseteq B$.

Proof: Suppose $\mathcal{T}' \supseteq \mathcal{T}$, $x \in X$, and $B \in \mathcal{B}$ contains x. Then B will be in $\mathcal{T}$, since it's a basis element for it, and thus $B \in \mathcal{T}'$. Thus we can take $B' = B$. Since this holds for arbitrary x and B, a) implies b).

Now suppose b) is true, and let $V \in \mathcal{T}$. Then we can write $V = \bigcup_\alpha B_\alpha$ for $\{B_\alpha\} \subseteq \mathcal{T}$. Note that we can write $B_\alpha = \bigcup_{\gamma_\alpha} B'_{\gamma_\alpha}$ for $\left\{B'_{\gamma_\alpha}\right\} \subseteq \mathcal{B}'$ by condition b), so we have $V = \bigcup_\alpha \left(\bigcup_{\gamma_\alpha} B'_{\gamma_\alpha}\right)$. Since this is the union of a collection of basis elements in $\mathcal{B}'$, V is an open set in $\mathcal{T}'$. Thus $\mathcal{T} \subseteq \mathcal{T}'$, as desired. ■

Definition (subbasis): A *subbasis* $\mathcal{S}$ for a topology on X is a collection of subsets of X whose union equals X.

Proposition: Let $\mathcal{B}$ be the collection of all finite intersections of elements in a subbasis $\mathcal{S}$ for a topology on X, or more explicity

$$\mathcal{B} = \left\{ \bigcap_{i=1}^{n} S_i : S_i \in S \right\}.$$

Then $\mathcal{B}$ is a basis for a topology on X.

Proof: Since $\mathcal{S}$ covers X, clearly so does $\mathcal{B}$. Now suppose $B_1 = S_1 \cap \cdots \cap S_m$ and $B_2 = S_1' \cap \cdots \cap S_n'$ are elements in $\mathcal{B}$, and suppose $x \in B_1 \cap B_2$. Then $x \in S_i, S_j'$ for some $i \leq m, j \leq n$. In particular, $x \in S_i \cap S_j' \subseteq B_1 \cap B_2$, and so $\mathcal{B}$ satisfies the second basis condition. ■

Example: Subasses are basically a way to express bases. For example, the subbasis $\{(-\infty, a) : a \in \mathbb{R}\} \cup \{(b, \infty) : b \in \mathbb{R}\}$ generates the basis of open intervals in $\mathbb{R}$, which is a basis for the standard topology on $\mathbb{R}$.

1.2. Order Topology

Definition (order topology): Let X be an ordered set with more than one element. Let $\mathcal{B}$ be the collection of all sets of the following type:

- All open intervals $(a, b) = \{x \in X : a < x < b\}$ in X.
- If X has a smallest element a_0, then all intervals $[a_0, b)$.
- If X has a largest element b_0, then all intervals $(a, b_0]$.

It's easy to see that every element of X lies in a set of one of these types. Furthermore, the intersection of any sets of these types is again one of them or empty.

Definition (ray): If X is an ordered set, and a is an element fo X, there are four subsets of X called *rays* determined by a:

$$\begin{aligned}(a, +\infty) &= \{x : x > a\},\\ (-\infty, a) &= \{a : x < a\},\\ [a, +\infty) &= \{x : x \geq a\},\\ (-\infty, a] &= \{x : x \leq a\}.\end{aligned}$$

The first two are *open rays*, and the second two are *closed rays*.

Proposition: Let $\mathcal{S}$ be the set of all rays in an ordered set X. Then $\mathcal{S}$ is a subbasis for the order topology on X.

Proof: It's easy to see that all open rays are open in the order topology, so $\mathcal{S}$ generates a topology coarser than the order topology. We can also form each open interval using $\mathcal{S}$: intersect $(-\infty, b)$ and $(a, +\infty)$. Thus $\mathcal{S}$ is a subbasis for the order topology. ■

1.3. Product Topology on $X \times Y$

Definition (product topology): Let X and Y be topological spaces. The *product topology* on $X \times Y$ is the topology generated by the basis

$$\mathcal{B} = \{U \times V : U \text{ and } V \text{ are open in } X \text{ and } Y \text{ respectively}\}.$$

It's easy to check that $\mathcal{B}$ is indeed a basis.

Example: Suppose $\mathbb{R}$ is equipped with the standard topology. Then the $\mathbb{R} \times \mathbb{R}$ product topology is generated by a basis of what are essentially unions of rectangles in the plane.

Proposition (product of bases): If $\mathcal{B}_1$ is a basis on X and $\mathcal{B}_2$ is a basis on Y, then

$$\mathcal{B}' = \{B_1 \times B_2 : B_1 \in \mathcal{B}_1, B_2 \in \mathcal{B}_2\}$$

is a basis for the product topology of $X \times Y$.

Proof: Clearly $\mathcal{B}'$ covers $X \times Y$, since $\mathcal{B}_1$ covers X and $\mathcal{B}_2$ covers Y, so we can take the product of all the covering sets. Similarly the second condition for a basis follows by applying it to each individual basis. We note this basis generates the same topology as the product topology since we can first form any open $U \in X$ by forming it using sets in $\mathcal{B}_1$ times some set $B_2 \subseteq V \subseteq$

Y, then do the same thing for $\mathcal{B}_2$. This implies that $\mathcal{B} \subseteq \mathcal{T}_{\mathcal{B}'}$, and so will generate the product topology. ■

Definition (projections): Let $\pi_1 : X \times Y \to X$ be defined by

$$\pi_1(x, y) := x$$

and $\pi_2 : X \times Y \to Y$ be defined by

$$\pi_2(x, y) := y.$$

Proposition: The collection

$$\mathcal{S} = \{\pi_1^{-1}(U) : U \text{ is open in } X\} \cup \{\pi_2^{-1}(V) : V \text{ is open in } Y\}$$

is a subbasis for the product topology on $X \times Y$.

Proof: Note that $\pi_1^{-1}(U) = U \times Y$ and $\pi_2^{-1}(V) = X \times V$. Suppose $A \in X$ is open and $B \in Y$ is open. Then $\pi_1^{-1}(A) \cap \pi_2^{-1}(B) = A \times B$, so the set of finite intersections of elements in $\mathcal{S}$ contain the basis for the product topology on $X \times Y$. Clearly any set of finite intersections of sets from $\mathcal{S}$ will also product $U \times V$ for open U and V in X and Y respectively, so $\mathcal{S}$ does generate the basis for the product topology, as desired. ■

1.4. Subspace Topology

Definition (subspace topology): Let $(X, \mathcal{T})$ be a topological space. If Y is a subset of X, the collection

$$\mathcal{T}_Y = \{Y \cap U : U \in \mathcal{T}\}$$

is the *subspace topology* on Y.

Again it's easy to check that this is a toplogy.

Example: Let $Y = (0, 1) \cup [2, 3]$ and suppose $\mathbb{R}$ is equipped with the standard topology. Then the subspace topology induced on Y is just the open sets of $\mathbb{R}$ intersected with Y. Thus $Y \cap (0, 1) = (0, 1)$ is open in Y, and $Y \cap (1, 4) = [2, 3]$ is also open in Y.

Proposition (basis of subspace topology): If $\mathcal{B}$ is a basis for the topology of X, then

$$\mathcal{B}_Y = \{Y \cap B : B \in \mathcal{B}\}$$

is a basis for the subspace topology on a subset $Y \subseteq X$

Proof: It's easy to see that this is a basis. Then note that since the union of a collection of sets in $\mathcal{B}$ can make any open $U \in X$, the union of those same sets intersected with Y can make $Y \cap U$. Thus this basis generates the subspace topology on Y, as desired. ■

Proposition (open sets relative to space and subspace): Let Y be a subspace of X. If U is open in Y and Y is open in X, then U is open in X.

Proof: Since U is open in Y, it can be written as $V \cap Y$, where V is open in X. Since Y is open in X, the intersection of these two sets is open in X, and thus U is open in X. ■

Proposition (subspace of product): If A is a subspace of X and B is a subspace of Y, then the product topology on $A \times B$ is the same as the topology $A \times B$ inherits as a subspace of $X \times Y$.

Proof: The set $U \times V$ is a basis element in $X \times Y$, for open $U \in X, V \in Y$. Thus $(A \times B) \cap (U \times V)$ is a basis element of the subspace topology on $A \times B$. Then we have $(A \times B) \cap (U \times V) = (A \cap U) \times (B \cap V)$, and since $A \cap U, B \cap V$ are open sets for the subspace topologies on A and B, their product is a basis element for the product topology on $A \times B$. Thus the two have the same bases, and thus generate the same topology on $A \times B$. ■

Example (subspace doesn't necessarily inherit order topology): Let Y be the subset of $[0, 1\} \cup \{2\}$ of $\mathbb{R}$. In the subspace topology, $\{2\}$ is open. However, in the order topology, any basis element containing 2 is of the form $(a, 2]$ for some $a \in Y$. Any such a will yield more than one element in the interval.

Let $I = [0, 1]$. The dictionary order topology on $I \times I$ does not have the set $\{1/2\} \times (1/2, 1]$ as an open set. However, if we give $I \times I$ the subspace topology as a subset of $\mathbb{R} \times \mathbb{R}$ with the dictionary order topology, then the same set is open, as it's the intersection of $I \times I$ and $\{1/2\} \times (1/2, 2)$, which is open in $\mathbb{R} \times \mathbb{R}$.

The above counterexample don't occur if we require the subspace to be convex, where here convex means that if $a, b \in Y$ and $a < b$, then $(a, b) \in Y$, where Y is a subset of X.

Proposition: Let X be an oredered set with the order topology, and let Y be a convex subset of X. Then the order topology on Y is the same as the topology Y inherits as a subspace of X.

Proof: Consider the ray $(a, +\infty)$ in X. If $a \in Y$, then $(a, +\infty)$ will be a ray in Y by convexity. If $a \notin Y$, then it's either an upper or lower bound for Y. In the first case, the intersection is empty. In the second case, the intersection is all of Y. The same logic applies to $(-\infty, a)$. Since the rays for a subbasis for X, the sets $(-\infty, a) \cap Y$ and $(a, +\infty) \cap Y$ form a subbasis for the subspace topology on Y. Since each of these are rays in Y, they are open in order topology on Y. Thus the order topology contains the subspace topology.

For the other direction, note that any open ray of Y equals the intersection of an open ray of X with Y, so it is open in the subspace topology of Y. Since the open rays form a subbasis for Y, the order topology is contained in the subspace topology. ■

1.5. Closed Sets and Limit Points

1.5.1. Closed Sets

Definition (closed): A subset A of a topological space X is said to be *closed* if $X \backslash A$ is open.

Proposition: In a topological space X,
a) $\emptyset$ and X are closed,
b) Arbitrary intersections of closed sets are closed,
c) Finite unions of closed sets are closed.

Proof: Follows from open set rules and DeMorgan's laws. ■

Proposition: Let Y be subspace of X. Then a set A is closed in Y if and only if it equals the intersection of a closed set of X with Y.

Proof: Suppose A is closed in Y. Then $Y \backslash A$ is open in Y. Thus $Y \backslash A = Y \cap V$, where $V \subseteq X$ is open. Then $X \backslash V$ is closed in X, and $A = Y \cap (X \backslash V)$, as desired.

Now suppose $A = Y \cap C$, where C is closed in X. Then $X \backslash C$ is open in X, so $Y \cap X \backslash C = Y \backslash A$ is open in Y by definition of the subspace toplogy. Thus A is closed in Y. ■

Proposition (closed sets relative to space and subspace): Let Y be a subspace of X. If A is closed in Y and Y is closed in X, then A is closed in X.

Proof: We have that $Y \backslash A$ is open in Y, and thus can be written as $Y \cap V$ for some open $V \subseteq X$. Since $X \backslash Y$ is open in X, the set $X \backslash Y \cup V = X \backslash A$ is open in X. Thus A is closed in X, as desired. ■

1.5.2. Closure and Interior of a Set

Definition (interior): Given a subset A of a topological space X, the *interior* of A is defined as the union of all open sets contained in A, and is denote $\operatorname{int}(A)$.

Definition (closure): Given a subset A of a topological space X, the *closure* of A is defined as the intersection of all closed sets containing A, and is denoted $\overline{A}$.

Proposition: Let Y be a subspace of X, let A be a subset of Y, let $\overline{A}$ denote the closure of A in X. Then the closure of A in Y equals $\overline{A} \cap Y$.

Proof: By definition, we have

$$\overline{A} = \bigcap \{C \subseteq X : C \text{ is closed in } X \text{ and contains } A\}.$$

Letting A' denote the closure of A in Y, we again have by definition

$$A' = \bigcap \{C' \subseteq Y : C' \text{ is closed in } Y \text{ and contains } A\}.$$

Since closed sets in Y can be written as the intersection of Y with a closed set in X, we can write the above as

$$\begin{aligned} A' &= \bigcap \{C \cap Y : C \text{ is closed in } X \text{ and contains } A\} \\ &= Y \cap \left(\bigcap \{C \subseteq X : C \text{ is closed in } X \text{ and contains } A\}\right) = Y \cap \overline{A}, \end{aligned}$$

where the second equality holds since closed sets in X give rise to closed sets in Y by intersecting them with Y. ■

Proposition (closure from open sets and basis): Let A be a subset of the topological space X.

a) Then $x \in \overline{A}$ if and only if every open set U containing x intersects A.
b) If $\mathcal{B}$ is a basis for the topology on X, then $x \in \overline{A}$ if and only if every basis element B containing x intersects A.

Proof:

a) Suppose $x \in \overline{A}$. If $x \in A$, then any open set containing x trivially intersects A at x, so suppose $x \notin A$. If U is an open set containing x that doesn't intersect A, then $X \backslash U$ is a closed set not containing x that has A as a subset. But then by the definition of closure, that would imply that $x \notin \overline{A}$, contradiction. Thus every open set containing x intersects A.

   Now suppose every open set contaning x intersects A, and suppose $x \notin \overline{A}$. Thus there exists some closed set C not containing x that contains A. Then $X \backslash C$ is an open set containing x that doesn't intersect A, contradiction.

b) This follows from part a). Namely, the if direction follows since basis elements are open, and the only if direction follows since the basis elements can be unioned to make any open set.

■

Definition (topological neighborhood): If U is an open set of a topological space X containing x, then U is a *neighborhood* of x.

Corollary (restatement of previous proposition): $x \in \overline{A}$ if and only if every neighborhood of x intersects A.

Example: Conside the subspace $Y = (0, 1]$ of $\mathbb{R}$, and the subset $\left(0, \frac{1}{2}\right)$. Then its closure in $\mathbb{R}$ is $\left[0, \frac{1}{2}\right]$, and its closure in Y is $\left(0, \frac{1}{2}\right]$.

Example: Consider $A = (1, 2)$ in the lower limit topology. Note that the open set $[2, 3)$ contains 2 but doesn't intersect $(1, 2)$, so $2 \notin \overline{A}$. However, every open set in the basis of the topology (the interval $[a, b)$) that contains 1 clearly intersects A, so $1 \in \overline{A}$. Thus $\overline{A} = [1, 2)$.

Definition (dense): A subset D of a topological space X is *dense* in X if $\overline{D} = X$.

Proposition: If $(X, \mathcal{T})$ is a topological space and $D \subseteq X$, show that D is dense in X if and only if every nonemtpy open set of X intersects D.

Proof: Suppose D is dense, and let $x \in X$ be arbitrary. Since $x \in \overline{D} = X$, every neighborhood of x intersects D. Since every nonempty open set is a neighborhood of some point x, every nonempty open set intersects D, as desired.

Now suppose every nonempty open set intersects D, and fix $x \in X$. Then every neighborhood of x intersects D, and so $x \in \overline{D}$. This holds for all x, so $\overline{D} = X$. ■

Definition (boundary): Let $(X, \mathcal{T})$ be a topological space and let $A \subseteq X$. Then the *boundary* of A, denote ∂A, is the set

$$\partial A = \{x \in X : \text{every neighborhood of } x \text{ intersects } A \text{ and } X \backslash A\}.$$

Proposition: Let A be a subset of a topological space X. Then

$$\partial A = \overline{A} \cap \overline{X \backslash A} = \overline{A} \backslash \operatorname{int}(A).$$

Proof: Suppose $x \in \partial A$. Then every neighborhood of x intersects A, so by the previous corollary, $x \in \overline{A}$. Similar logic yields $x \in \overline{X \backslash A}$, and so $\partial A \subseteq \overline{A} \cap \overline{X \backslash A}$. If x is in the intersection of the closure and the complements closure, then again by the corollary every open neighborhood must intersect A and $X \backslash A$, so by definition $x \in \partial A$. Thus $\partial A \supseteq \overline{A} \cap \overline{X \backslash A}$. Thus the sets are equal.

Suppose $x \in \partial A$. Then it every neighborhood intersects A, and so $x \in \overline{A}$. Note also that $x \notin \operatorname{int}(A)$, since otherwise there would exist a neighborhood of x contained in A, and so wouldn't intersect $X \backslash A$, thus not being in the boundary. Thus $\partial A \subseteq \overline{A} \backslash \operatorname{int}(A)$. Reversing the reasoning implies the reverse inclusion, so the sets are equal. ■

Proposition: Let A be a subset of a topological space X. Then $\overline{A} = A \cup \partial A$ and $\operatorname{int}(A) = A \backslash \partial A$.

Proof: Suppose $x \in \overline{A}$. If it's not in A, then every neighborhood must intersect both A (as otherwise it wouldn't be in the closure) and $X \backslash A$ (namely at x), and thus $x \in \partial A$, implying $\overline{A} \subseteq A \cup \partial A$. Now suppose $x \in A \cup \partial A$. If $x \in A$, then $x \in \overline{A}$. If $x \in \partial A$, then by definition every neighborhood of x intersects A, so $x \in \overline{A}$. This $A \cup \partial A \subseteq \overline{A}$.

Suppose $x \in \operatorname{int}(A)$. Then there exists a neighborhood of x contained in A. Since this neighborhood doesn't intersect $X \backslash A$, $x \notin \partial A$. Thus $x \in A \backslash \partial A$, so $\operatorname{int}(A) \subseteq A \backslash \partial A$. Now suppose $x \in A$ by $x \notin \partial A$. Then some neighborhood of x does not intersect $X \backslash A$, and thus will be contained in A, implying $x \in \operatorname{int}(A)$. Thus $\operatorname{int}(A) \supseteq A \backslash \partial A$. ■

Proposition: Let A be a subset of a topological space X. Then $X = \operatorname{int}(A) \sqcup \partial A \sqcup \operatorname{int}(X \backslash A)$, where $\sqcup$ means a disjoint union.

Proof: Clearly subsets of A and $X \backslash A$ will be disjoint, so the interiors of both are disjoint. Any $x \in \partial A$ will also be disjoint from both by definition. If $x \in X$, then it has to be in one of these, as either some neighborhood lies in A, some neighborhood lies in $X \backslash A$, or all neighborhoods intersect both. ■

1.5.3. Limit Points

Definition (limit point): If A is a subset of a topological space X, and if $x \in X$, then x is a *limit point* of A if every neighborhood of x intersects A at a point other than x.

> **Remark**
>
> This is equivalent to saying that $x \in \overline{A \backslash \{x\}}$.

Example: Consider $A = (0, 1] \cup \{2\}$. Then $[0, 1]$ is the set of limit points.

Proposition: Let A be a subset of a topological space X, and let A' be the set of all limit points of A. Then $\overline{A} = A \cup A'$.

Proof: Suppose $x \in \overline{A}$. If $x \in A$, then clearly $x \in A \cup A'$. If $x \notin A$, then every neighborhood of x must intersect A at a point other than x, since otherwise x wouldn't be in the closure. Thus $x \in A'$, and so $\overline{A} \subseteq A \cup A'$.

Now suppose $x \in A \cup A'$. Clearly $x \in A \Rightarrow x \in \overline{A}$, so suppose $x \in A'$. Then every neighborhood of of x intersects A by definition, so $x \in \overline{A}$, meaning $A \cup A' \subseteq \overline{A}$. We have the inclusion in both directions, so the sets are equal. ■

Corollary: A subset of a topological space is closed if and only if it contains all its limit points.

Proof: If A is closed, then $\overline{A} = A$, since it's the smallest closed set that contains A, and so must contain all its limit points. If A contains all its limit points, then by the previous proposition, $\overline{A} = A \cup A' = A$, and thus is closed. ■

1.5.4. Hausdorff Spaces

Hausdorff spaces are nice in the sense that in these spaces, points are topologically distinguishable because open sets can "separate" them. In particular, sequence convergence in Hausdorff spaces makes much more sense, as they can only converge to at most one point.

Definition (Hausdorff space): A topological space X is called a *Hausdorff space* (or said to be *Hausdorff*) if for each pair of distinct $x_1, x_2 \in X$, there exist neighborhoods of x_1 and x_2 that are disjoint.

Proposition: Every finite set in a Hausdorff space X is closed.

Proof: We show that $\{x\}$ is closed. The general claim follows by showing it for singletons and then taking finite unions. Suppose $y \neq x$. Then since X is Hausdorff, there exists a neighborhood of x that is disjoint from a neighborhood of y. Since the neighborhood of y clearly doesn't

intersect $\{x\}$, y is not a limit point of $\{x\}$, and thus won't be contained in $\overline{\{x\}}$. This holds for all $y \neq x$, so $\overline{\{x\}} = \{x\}$. ■

Definition (topological convergence): Suppose $(x_n) \in X$ is a sequence in a topological space. Then x_n converges to $x \in X$ if for every neighborhood U of x, there exists a positive integer N such that $x_n \in U$ for all $n \geq N$.

Proposition: If X is Hausdorff, then a sequence converges to at most one point.

Proof: Suppose (x_n) converges to distinct points $a, b \in X$. Since X is Hausdorff, there exists a neighborhood A of a and B of b such that A and B are disjoint. By the definition of convergence, there exists N_a such that $x_n \in A$ for all $n \geq N_a$ and N_b such that $x_n \in B$ for all $n \geq N_b$. Letting $N = \max\{N_a, N_b\}$, we have that $x_n \in A, B$ for all $n \geq N$, but this is impossible since A and B are disjoint. ■

Proposition: The order topology on an ordered set is Hausdorff, the product of two Hausdorff spaces is Hausdorff, and the subspace of a Hausdorff space is Hausdorff.

Proof: Follows easily from definitions. ■

1.6. Continuous Functions

Definition (continuous): Let X and Y be topological spaces. A function $f : X \to Y$ is said to be *continuous* of for each open subset V of Y, the set $f^{-1}(V)$ is open in X.

Definition (continuous at a point): Let X and Y be topological spaces, and let $x \in X$. If for each neighborhood V of $f(x)$ in Y there is a neighborhood U of x such that $f(U) \subseteq V$, then f is *continuous at* x.

It's clear that if the inverse image of basis elements is open, then f is also continuous, since open sets are the union of basis elements, and the inverse image distributes over unions. It also suffices to the inverse image is open for the elements of a subbasis, since inverse images distribute over intersections as well.

Example: Let $\mathbb{R}_\ell$ denote $\mathbb{R}$ with the lower limit topology. Then the identity function $\mathrm{id} : \mathbb{R} \to \mathbb{R}_\ell$ is not continuous, since the inverse image of $[0, 1)$, which is open in $\mathbb{R}_\ell$, is not open in $\mathbb{R}$. However, $\mathrm{id} : \mathbb{R}_\ell \to \mathbb{R}$ is continuous.

Proposition: Let X and Y be topological spaces, and let $f : X \to Y$. Then the following are equivalent:
a) f is continuous.
b) For every subset A of X, we have $f\left(\overline{A}\right) \subseteq \overline{f(A)}$.
c) For every closed set B of Y, the set $f^{-1}(B)$ is closed in X.
d) f is continuous at x for all $x \in X$.

Proof: First we show that a) $\iff$ d). Suppose a) is true, and fix $x \in X$. Let V be any neighborhood of $f(x)$. Then by continuity, $f^{-1}(V)$ is open in X. Since clearly it will contian x, it's an open neighborhood of x, and by definition, $f(f^{-1}(V)) \subseteq V$. Thus f is continuous at x. Now suppose f is continuous at x for all $x \in X$, and let V be any open set in Y. If $f^{-1}(V) = \emptyset$, then clearly the pre-image is open in X. Otherwise, there exists $x \in X$ such that $f(x) \in V$. Then by continuity at a point, there exists a neighborhood U_x of x in X such that $f(U_x) \subseteq V$. Thus $U_x \subseteq f^{-1}(V)$. Then for all $x \in f^{-1}(V)$, we can union their neighborhoods together, and obtain an open set $U = f^{-1}(V)$. This holds for all open V in Y, so f is continuous.

First suppose a) is true, and suppose $y \in f\left(\overline{A}\right)$. Thus $f(x) = y$ for some $x \in \overline{A}$. Let V be a neighborhood of y. Then $f^{-1}(V)$ is a neighborhood of x. Thus it will intersect A (since x is in the closure of A), which implies that V intersects $f(A)$. Since this holds for any neighborhood of y, we have $y \in \overline{f(A)}$, and thus $f\left(\overline{A}\right) \subseteq \overline{f(A)}$.

Now suppose b) is true. Suppose B is closed in Y, and let $A = f^{-1}(B)$. Then $f(A) = f(f^{-1}(B)) \subseteq B$. Thus, for $x \in \overline{A}$, we have

$$f(x) \in f\left(\overline{A}\right) \subseteq \overline{f(A)} \subseteq \overline{B} = B.$$

Thus $x \in f^{-1}(B)$, so $\overline{A} \subseteq A \Rightarrow \overline{A} = A$. Thus A is closed in X.

Now suppose c) is true. If V is open in Y, then $Y \backslash V$ is closed in Y. Then $f^{-1}(Y \backslash V) = f^{-1}(Y) \backslash f^{-1}(V) = X \backslash f^{-1}(V)$ is closed in X, which implies that $f^{-1}(V)$ is open. Thus f is continuous. ■

Proposition (continuity preserves convergence): Let $f : X \to Y$ be continuous and suppose $(x_n) \in X$ converges to x. Then $(f(x_n)) \in Y$ converges to $f(x)$.

Proof: Let V be an open set in Y containing $f(x)$. Then the inverse image is an open set containing x, and thus by convergence, contains $(x_n)_{n \geq N}$ for some N. Thus V contains $(f(x_n))_{n \geq N}$. Since V is arbitrary, we do indeed have that $f(x_n) \to f(x)$. ■

Proposition (continuous functions): Let X, Y, Z be topological spaces.

a) If $f : X \to Y$ is constant, then f is continuous.
b) If A is a subspace of X, the inclusion function $j : A \to X$ is continuous.
c) If $f : X \to Y$ and $g : Y \to Z$ are continuous, then $g \circ f : X \to Z$ is continuous.
d) If $f : X \to Y$ is continuous, and if A is a subspace of X, then $f|_A : A \to Y$ is continuous.
e) Let $f : X \to Y$ be continuous. If Z is a subspace of Y containing the image set $f(X)$, then the function $g : X \to Z$ obtained by restricting the range is continuous. If Z is a space having Y as a subspace, then the function $h : X \to Z$ obtained by expanding the range of f is continuous.
f) The map $f : X \to Y$ is continuous if X can be written as the union of open sets U_α such that $f|_{U_\alpha}$ is continuous for each α.

Proof:
a) If an open $V \subseteq Y$ contains the constant, then $f^{-1}(V) = X$ is open, otherwise $f^{-1}(V) = \emptyset$ is open, so f is continuous.
b) If $V \subseteq X$ is open, then $f^{-1}(V) = A \cap V$, which is open in A by definition.
c) If $V \subseteq Z$ is open, then $g^{-1}(V)$ is open in Y, so $f^{-1}(g^{-1})(V) = (g \circ f)^{-1}(V)$ is open in X.
d) Follows from using b) and c).
e) Let $V \subseteq Z$ be open. Then it can be written as $U \cap Z$ for open U in Y, so $f^{-1}(U) \cap f^{-1}(Z) = f^{-1}(U) \cap X = f^{-1}(U)$ is open in X. Then since $f^{-1}(U) = g^{-1}(V)$ (which follows easily by checking inclusions), we see that h is continuous. To show that after expanding the range a map is still continuous, apply b) and c) using the inclusion map as the outer function.
f) Let V be open in Y. Then $f^{-1}(V) \cap U_\alpha = \left(f|_{U_\alpha}\right)^{-1}(V)$, which is open in U_α by continuity. Since U_α is open, the set is open in X as well. Then since $f^{-1}(V) = \bigcup_\alpha (f^{-1}(V) \cup U_\alpha)$, $f^{-1}(V)$ is open as well.

■

Lemma (pasting lemma): Let $X = A \cup B$, where A, B are closed in X. Let $f : A \to Y$ and $g : B \to Y$ be continuous. If $f(x) = g(x)$ for every $x \in A \cap B$, then f and g combine to give a continuous function $h : X \to Y$, defined by setting $h(x) = f(x)$ if $x \in A$, and $h(x) = g(x)$ if $x \in B$.

Proof: Suppose C is closed in Y. Then $h^{-1}(C) = f^{-1}(C) \cup g^{-1}(C)$. By the equivalent formulations of continuity, $f^{-1}(C)$ is closed in A, and thus closed in X. Similarly, $g^{-1}(C)$ is closed in X, so $h^{-1}(C)$ is closed in X. Thus h is continuous. ■

Remark

The lemma also works if A and B are open.

Proposition (maps into products): Let $f : A \to X \times Y$ be given by the equation

$$f(a) = (f_1(a), f_2(a)).$$

Then f is continuous if and only if $f_1 : A \to X$ and $f_2 : A \to Y$ are continuous.

Proof: Suppose f is continuous, and let $\pi_1 : X \times Y \to X$ and $\pi_2 : X \times Y \to Y$ be the projection maps. Note that $\pi_1^{-1}(U) = U \times Y$ is open if U is open, and similarly for π_2. Thus they're both continuous, so the compositions $f_1 = \pi_1 \circ f$ and $f_2 = \pi_2 \circ f$ are also continuous.

Now suppose each component function is continuous, and suppose W is open in $X \times Y$. Then we can write it as $\bigcup_\alpha U_\alpha \times V_\alpha$ with U_α, V_α open in X, Y. Then note that $f^{-1}(W) = \bigcup_\alpha f^{-1}(U_\alpha \times V_\alpha)$. If we can show that each of these is open in A, we're done.

By continuity, $f_1^{-1}(U_\alpha) \cap f_2^{-1}(V_\alpha)$ is open. Note that if $x \in f^{-1}(U_\alpha \times V_\alpha)$, then $f_1(x) \in U_\alpha$, $f_2(x) \in V_\alpha$, so x is also in the intersection. If f is in the union, then by the reverse logic $x \in f^{-1}(U_\alpha \times V_\alpha)$. Thus $f_1^{-1}(U_\alpha) \cap f_2^{-1}(V_\alpha) = f^{-1}(U_\alpha \times V_\alpha)$, so we're done. ■

Definition (homemorphism): Let X and Y be topological spaces, and suppose $f : X \to Y$ is a bijection. If both f and the inverse f^{-1} of f are continuous, then f is called a *homeomorphism*, and the spaces X and Y are said to be *homeomorphic*.

Homeomorphisms essentially give a bijection between the open sets of topological spaces, so if something is true for open sets in one space, then it will be true for the open sets in the other space. Such properties are called *topological properties*, as they're preserved under homeomorphisms.

Definition (embedding): Let X and Y be topological spaces, and suppose $f : X \to Y$ is continuous. Let Z be the image of f. If $f : X \to Z$ is a homeomorphism, then f is an *embedding* of X into Y, and we say that X *embeds* in Y.

1.7. More Important Topologies

Here we discuss a few more important topologies.

1.7.1. Product Topology

Definition (J-tuple): Let J be an index set. Given a set X, we define a J-tuple of elements of X to be a function $\boldsymbol{x} : J \to X$. If α is an element of J, we often denote $\boldsymbol{x}(\alpha)$ by x_α (i.e. the coordinate of $\boldsymbol{x}$). We denote the the entire with

$$(x_\alpha)_{\alpha \in J},$$

or just (x_α) if J is understood.

This is just a generalization of coordinate notation where the index set can be uncountable.

Definition (projection mapping): Define the function $\pi_\beta : \prod_{\alpha \in J} X_\alpha \to X_\beta$ by

$$\pi_\beta\left((x_\alpha)_{\alpha \in J}\right) = x_\beta.$$

This is the *projection mapping* associated with index β.

This is also just a generalization of the regular projection maps.

Definition (box topology): Let $\{X_\alpha\}_{\alpha \in J}$ be an indexed family of topological spaces. For the cartesian product $\prod_{\alpha \in J} X_\alpha$, let

$$\mathcal{B} = \left\{ \prod_{\alpha \in J} U_\alpha : U_\alpha \text{ is open in } X_\alpha \right\}$$

be a basis. Then the topology generated by this basis is the *box topology*.

Definition (product topology): Let β denote the collection

$$\mathcal{S}_\beta = \left\{ \pi_\beta^{-1}(U_\beta) : U_\beta \text{ is open in } X_\beta \right\},$$

and let

$$\mathcal{S} = \bigcup_{\beta \in J} \mathcal{S}_\beta.$$

The topology generated by the subbasis $\mathcal{S}$ is called the *product topology*.

Note that a basis element generated by the subbasis above is given by

$$B = \bigcap_{i=1}^{n} \pi_{\beta_i}^{-1}\left(U_{\beta_i}\right),$$

where U_{β_i} are open in X_{β_i}. If the product of the sets is a finite product, then the two bases are equal, so the box topology is equal to the product topology. However, if the product is infinite, then the basis

elements of the product topology are a subset of the basis elements of the box topology. This is because a basis element in the product topology looks like a product of finitely many open sets, while the rest of the coordinates are the full spaces X_α. Basis elements in the box topology can have arbitrarily many coordinates be proper subsets of the spaces.

Proposition: Suppose the topology on each space X_α is given by a basis $\mathcal{B}_\alpha$. The collection of all sets of the form

$$\prod_{\alpha \in J} B_\alpha,$$

where $B_\alpha \in \mathcal{B}_\alpha$ for each α, is a basis for the box topology on $\prod_{\alpha \in J} X_\alpha$.

The collection of all sets of the same form, where $B_\alpha \in \mathcal{B}_\alpha$ for finitely many α and $B_\alpha = X_\alpha$ for all remaining indices, is a basis for the product topology.

Proof: The first part follows since we can make an open set $U_\alpha \subseteq X_\alpha$ using $\mathcal{B}_\alpha$, take the product over all such open sets and get the basis for the box topology. The second one is just the basis generated by the subbasis in the definition of the product topology. ■

Proposition: Let A_α be a subspace of X_α for all $\alpha \in J$. Then $\prod A_\alpha$ is a subspace of $\prod X_\alpha$ if both products are given either the box topology or the product topology.

Proof: This follows easily, since every open set can be written as $A_\alpha \cap U_\alpha$, where U_α is open in X_α, and then taking the product yields that every every open set in $\prod A_\alpha$ can be written as $\prod A_\alpha \cap \prod U_\alpha$. ■

Proposition: If each space X_α is Hausdorff, then $\prod X_\alpha$ is Hausdorff in both the box and product topologies.

Proof: First we look at the box topology. Suppose $(x_\alpha), (y_\alpha) \in \prod X_\alpha$ are distinct. Since each X_α is Hausdorff, there are open sets $U_\alpha, V_\alpha \subseteq X_\alpha$ such that $x_\alpha \in U_\alpha, y_\alpha \in V_\alpha$, and U_α and V_α are disjoint for all α. Then $(x_\alpha) \in \prod U_\alpha, (y_\alpha) \in \prod V_\alpha$, and the two products are disjoint, so the product space is Hausdorff.

For the product topology, we only need to pick disjoint open sets for some coordinate, and we can take X_α for the rest. The product of the sets will still be disjoint, and indeed the product will be open, since only one coordinate in each is not the entire space.

Suppose $(x_\alpha), (y_\alpha) \in \prod X_\alpha$ are distinct. Thus there exists some β such that $x_\beta \neq y_\beta$. Since X_β is Hausdorff, there exist open U_β, V_β in X_β such that $x_\beta \in U_\beta, y_\beta \in V_\beta$, and the sets are disjoint. Then the sets $U_\beta \times \prod X_\alpha$ and $V_\beta \times \prod X_\alpha$ are open in both topologies, disjoint, and contain (x_α) and (y_α) respectively. Thus $\prod X_\alpha$ is Hausdorff. ■

Proposition: Let $\{X_\alpha\}$ be an indexed family of spaces, and let $A_\alpha \subseteq X_\alpha$ for each α. If $\prod X_\alpha$ has either the product or the box topology, then

$$\prod \overline{A_\alpha} = \overline{\prod A_\alpha}.$$

Proof: Suppose (x_α) is a point in $\prod \overline{A_\alpha}$, and let $U = \prod U_\alpha$ be a basis element in either topology that contains (x_α). Since $x_\alpha \in \overline{A_\alpha}$, the set $U_\alpha \cap A_\alpha$ is nonemtpy, so some point y_α lies in it. Then $(y_\alpha) \in U \cap \prod A_\alpha$. Since U is arbitrary, we have that $(x_\alpha) \in \overline{\prod A_\alpha}$.

Now suppose $(x_\alpha) \in \overline{\prod A_\alpha}$. Pick some index β, and let V_β be an arbitrary open set of X_β containing x_β. Since $\pi_\beta^{-1}(V_\beta)$ is open in $\prod X_\alpha$ in either topology, and since it contains (x_α), the intersection of it and $\prod A_a$ will contain some point (y_α). Then $y_\beta \in V_\beta$, so $V_\beta \cap \prod A_\beta \neq \emptyset$, so $x_\beta \in \overline{A_\beta}$. ■

So far, the topologies have acted the same (and as we already noted, are the same for finite products), but the next result shows that we often prefer the product topology, as it does not hold for the box topology.

Proposition: Let $f : A \to \prod_{\alpha \in J} X_\alpha$ be given by the equation

$$f(a) = (f_\alpha(a))_{\alpha \in J}$$

where $f_\alpha : A \to X_\alpha$ for each α. Let $\prod X_\alpha$ have the product topology. Then the function f is continuous if and only if each function f_α is continuous.

Proof: Note that the projection map from the product to X_β is continuous. Then if f is continuous, $\pi_\beta \circ f = f_\beta$ is continuous.

Now suppose each coordinate function is continuous. We show that each subbasis element has an open inverse image, from which the result follows by intersecting and union to get the full topology. A subbasis element is $\pi_\beta^{-1}(V_\beta)$, where V_β is open in X_β. Then $f^{-1}\left(\pi_\beta^{-1}(V_\beta)\right) = \left(\pi_\beta \circ f\right)^{-1}(V) = f_\beta^{-1}(V_\beta)$, which is open by the continuity of the coordinate functions, so we're done. ■

The reason this doesn't work in the box topology is that open sets where infinitely many coordinates are not the full space can't be made via finite intersections.

Example (failure of the box topology): Consider the function $f : \mathbb{R} \to \mathbb{R}^\omega$ given by $f(t) = (t, t, t, \ldots)$. Then each coordinate function $f_n(t) = t$ is continuous. However, consider in $\mathbb{R}^\omega$ the open set $(-1, 1) \times \left(-\frac{1}{2}, \frac{1}{2}\right) \times \left(-\frac{1}{3}, \frac{1}{3}\right) \times \cdots$. The inverse image of this set is $\{0\}$, which is not open in $\mathbb{R}$.

From here on out, when we use product spaces, they will be given the product topology unless otherwise stated.

1.7.2. Metric Topology

Definition (metric): A *metric* on a set X is a function

$$d : X \times X \to \mathbb{R}$$

having the following properties:
a) $d(x, y) \geq 0$ for all $x, y \in X$, and equality holds if and only if $x = y$.
b) $d(x, y) = d(y, x)$ for all $x, y \in X$.
c) $d(x, y) + d(y, z) \geq d(x, z)$ for all $x, y, z \in X$.

Definition (metric topology): If d is a metric on the set X, then the collection of all ε-balls $B_d(x, \varepsilon)$ (or denoted $B_\varepsilon(x)$ when the metric is understood) for all $x \in X$ and $\varepsilon > 0$, is a basis for the *metric topology* on X induced by d.

The first basis condition is obvious. For the second, let $B_1 = B_{\varepsilon_1}(x_1)$, $B_2 = B_{\varepsilon_2}(x_2)$, and suppose $x \in B_1 \cap B_2$. Then $B_{\min(d(x,x_1),d(x,x_2))}(x)$ is contained in the intersection, which can be checked with the triangle inequality.

Proposition: A set U is open in the metric topology induced by d if and only if for each $y \in U$, there is a $\delta > 0$ such that $B_\delta(y, \delta) \subseteq U$.

Proof: One direction follows from taking union for each ball around each point in U. The other direction follows from the triangle inequality. ■

Definition (metrizable): If X is a topological space, X is said to be *metrizable* if there exists a metric d on the set X that induces the topology of X. A *metric space* is a metrizable space X together with a specific metric d that gives the topology of X.

Example: On a set X, define $d(x, y) = 1$ if $x \neq y$, and $d(x, y) = 0$ if $x = y$. It's easy to check this is a metric. Then this metric induces the discrete metric, as $B_{\frac{1}{2}}(x) = \{x\}$ for all $x \in X$, and then taking unions gives you every subset of X.

Example: Let $d(x, y) = |x - y|$ on $\mathbb{R}$. This is the standard metric, and induces the standard topology on $\mathbb{R}$.

Example: Consider the indiscrete topology $\{\emptyset, X\}$ on some set with at least two points. If we had some metric d on X, then $d(a,b) = \varepsilon > 0$ for distinct $a, b \in X$. Then the set $B_{\varepsilon/2}(a)$ contains a but not b, so there exist more open sets than just X and $\emptyset$. Thus the indiscrete topology is not metrizable.

Proposition: Let d and d' be two metric on the set X, and let $\mathcal{T}$ and $\mathcal{T}'$ be the topologies they induce. Then $\mathcal{T}'$ is finer than $\mathcal{T}$ if and only if for each $x \in X$ and each $\varepsilon > 0$, there exists a $\delta > 0$ such that

$$B_{d'}(x, \delta) \subseteq B_d(x, \varepsilon).$$

Proof: Suppose $\mathcal{T}'$ is finer than $\mathcal{T}$. Since $B_d(x, \varepsilon) \in \mathcal{T}$, it will also be in $\mathcal{T}'$. Then the previous proposition gives us the desired conclusion. Now suppose the condition holds. For a given basis element B of $\mathcal{T}$, we can find a ball $B_d(x, \varepsilon)$ within it. Then from the condition, there exists δ such that $B_{d'}(x, \delta) \subseteq B_d(x, \varepsilon)$. Then the lemma about comparing bases applies. ■

Definition (standard bounded metric): Let X be a metric space with metric d. Define $\overline{d} : X \times X \to \mathbb{R}$ by the equation

$$\overline{d}(x, y) = \min\{d(x, y), 1\}.$$

Then $\overline{d}$ is the *standard bounded metric* corresponding to d.

We have to check that this actually is a metric. The first two conditions are obvious. To check the triangle inequality, note that if one $d(x, y), d(y, x)$ is greater than 1, then $\overline{d}(x, z) \leq 1 \leq \overline{d}(x, y) + \overline{d}(y, z)$. If both are less than 1, then $\overline{d}(x, z) \leq d(x, z) \leq d(x, y) + d(y, z) = \overline{d}(x, y) + \overline{d}(y, z)$.

Proposition: The standard bounded metric $\overline{d}$ induces the same topology as d.

Proof: Consider $B_d(x, \varepsilon)$ for $x \in X$ and $\varepsilon > 0$. Then note that $B_{\overline{d}}(x, \min\{\varepsilon, 1\}) \subseteq B_d(x, \varepsilon)$. Now consider $B_{\overline{d}}(x, \varepsilon)$ for $x \in X$ and $\varepsilon > 0$. Note that if $\varepsilon > 1$, then the ball is just the ball centered at x with size 1. Either way, this is just some ball induced by d. Thus the bases of both topologies contain each other, so they induce the same topology. ■

Definition (uniform metric/topology): Given an index set J and given $x, y \in \mathbb{R}^J$, define the *uniform metric* $\overline{\rho}$ on $\mathbb{R}^J$ by

$$\overline{\rho}(x, y) = \sup\left\{\overline{d}(x_\alpha, y_\alpha) : \alpha \in J\right\},$$

where d is the standadrd bounded metric on $\mathbb{R}$. This metric induces the *uniform topology* on $\mathbb{R}^J$.

It's easy to check that this is actually a metric. Note that ρ is well-defined, since the supremum is at most 1 since we're using the standadrd bounded metric.

Proposition: The uniform topology on $\mathbb{R}^J$ is finer than the product topology and coarser than the box topology. If J is infinite, then all three topologies are different.

Proof: Consider some subbasis element $\pi_\beta^{-1}(V_\beta)$ of the product topology, and pick arbitrary $x \in \pi_\beta^{-1}(V_\beta)$. There exists some ball of size ε such that $B_{\overline{d}}(x_\beta, \varepsilon) \subseteq V_\beta$. Then the ball $B_{\overline{\rho}}(x, \varepsilon) \subseteq \pi_\beta^{-1}(V_\beta)$, as for every point $y \in B_{\overline{\rho}}(x, \varepsilon)$, the definition of $\overline{\rho}$ implies that $\overline{d}(x_\beta, y_\beta) < \varepsilon$. Thus $\pi_\beta^{-1}(V_\beta)$ is open in the uniform topology, and thus the product topology is coarser than it. If J is infinite, then the ball $B_{\overline{\rho}}(0, \frac{1}{2})$ is not open in the uniform topology, as any neighborhood of 0 in the product topology will contain at least one component that is $\mathbb{R}$, but the ball forces the α component of it to be contained in $(-\frac{1}{2}, \frac{1}{2})$ for all $\alpha \in J$. Thus no neighborhood of of 0 is contained in the ball, and thus the ball is not open in the product topology.

Now consider some basis element $B_{\overline{\rho}}(x, \varepsilon)$ of the uniform topology, consider arbitrary $y \in B_{\overline{\rho}}(x, \varepsilon)$. Thus $p := \overline{\rho}(x, y) < \varepsilon$, which implies $\overline{d}(x_\alpha, y_\alpha) < p$ for all $\alpha \in J$. Thus the set

$$U = \prod_{\alpha \in J} \left(y_\alpha - \frac{\varepsilon - p}{2}, y_\alpha + \frac{\varepsilon - p}{2} \right)$$

is contained in $B_{\overline{\rho}}(x, \varepsilon)$, as for all $\alpha \in J$ and $z_\alpha \in U_\alpha$, we have $\overline{d}(x_\alpha, z_\alpha) \leq \overline{d}(x_\alpha, y_\alpha) + \overline{d}(y_\alpha, z_\alpha) < p + \frac{\varepsilon - p}{2} = \frac{\varepsilon + p}{2}$. Thus $\overline{\rho}(x, z) \leq \frac{\varepsilon + p}{2} < \varepsilon$, and so $z \in B_{\overline{\rho}}(x, \varepsilon)$. This works for all y in the ball, so $B_{\overline{\rho}}(x, \varepsilon)$ is open in the box topology.

If J is infinite, pick some countable subset of J and label is $\mathbb{N}$. Consider $V = \prod_{\alpha \in J \setminus \mathbb{N}} \mathbb{R} \times \prod_{n \in \mathbb{N}} (-\frac{1}{n}, \frac{1}{n})$, which is open in the box topology since each component is open in $\mathbb{R}$. Pick arbitrary $\varepsilon > 0$ and consider $B_{\overline{\rho}}(0, \varepsilon)$. Then there exists some point in it with nth component $\varepsilon/2$, where n is chosen so that $1/n < \varepsilon/2$. Thus there exists no $\varepsilon > 0$ for which V contains the ball. Thus V is not open in the uniform topology. ■

Proposition: Let $\overline{d}(a, b) = \min\{|a - b|, 1\}$ be the standard bounded metric on $\mathbb{R}$. For two points $x, y \in \mathbb{R}^\omega$, define

$$D(x, y) = \sup \left\{ \frac{\overline{d}(x_i, y_i)}{i} \right\}.$$

Then d is a metric that induces the product topology on $\mathbb{R}^\omega$.

Proof: It's easy to see that D is a metric. First let U be open in the metric toplogy. Fix arbitrary $x \in U$, and choose $\varepsilon < 1$ such that $B_D(x, \varepsilon) \subseteq U$. Pick N such that $\frac{1}{N} < \varepsilon$. We claim that

$$V := (x_1 - \varepsilon, x_1 + \varepsilon) \times \cdots \times (x_N - \varepsilon, x_N + \varepsilon) \times \mathbb{R} \times \mathbb{R} \times \cdot$$

is contained in $B_D(x, \varepsilon)$. Let $y \in V$. Then for all $i \geq N$, we have $\frac{\overline{d}(x_i, y_i)}{i} \leq \frac{1}{N}$, so

$$D(x, y) = \sup \left\{ \frac{\overline{d}(x_i, y_i)}{i} \right\} \leq \max \left\{ \frac{\overline{d}(x_1, y_1)}{1}, \ldots, \frac{\overline{d}(x_N, y_N)}{N}, \frac{1}{N} \right\} < \max \left\{ \varepsilon, \ldots, \frac{\varepsilon}{N}, \frac{1}{N} \right\} = \varepsilon.$$

Thus $y \in B_D(x, \varepsilon)$, which means $V \subseteq B_D(x, \varepsilon)$. Since x was arbitary, the metric topology is contained in the product topology.

Now consider a subbasis element of the product topology. Without loss of generality, suppose it's $U = \mathbb{R} \times \cdots \times U_n \times \mathbb{R} \times \cdots$. Fix arbitrary $x \in U$, and pick $\varepsilon < 1$ such that $\left(x_n - \frac{\varepsilon}{n}, x_n + \frac{\varepsilon}{n}\right) \subseteq U_1$. We claim $B_D\left(x, \frac{\varepsilon}{n}\right) \subseteq U$. Indeed, suppose $y \in B_D\left(x, \frac{\varepsilon}{n}\right)$. Then $\frac{\overline{d}(x_n, y_n)}{n} \leq D(x, y) < \frac{\varepsilon}{n}$. Thus $\overline{d}(x_n, y_n) = |x_n - y_n| < \varepsilon$. Thus $y \in U$, which means $B_D\left(x, \frac{\varepsilon}{n}\right) \subseteq U$. Since y was arbitrary, the product topology is contained in the metric topology. ■

1.7.3. Quotient Topology (FINISH LATER)

Definition (quotient map): Let X and Y be topological spaces, and let $p : X \to Y$ be a surjective map. Then p is said to be a *quotient map* provided a subset U of Y is open in Y if and only if $p^{-1}(U)$ is open in X. p is called an *open map* if it sends open sets to open sets, and a *closed map* if it sends closed sets to closed sets.

Equivalently, we can say p is a quotient map provided a subset C of Y is closed in Y if and only if $p^{-1}(C)$ is closed in X. Note that if p is a continuous surjective map that is either open or closed, then it's automatically a quotient map.

Definition (saturated): A subset $C \subseteq X$ is *saturated* with respect to a surjective map $p : X \to Y$ if C contains every set $p^{-1}(\{y\})$ that it intersects.

In other words, C is saturated if it's the inverse image of some subset of Y. Thus p being a quotient map is equivalent to saying that p is continuous and that it sends saturated open sets to open sets (or saturated closed sets to closed sets).

Example: Let $X = [0, 1] \cup [2, 3]$ be a subspace of $\mathbb{R}$, and let $Y = [0, 2]$ be a subspace of $\mathbb{R}$. The map $p : X \to Y$ given by

$$p(x) = \begin{cases} x & \text{for } x \in [0, 1], \\ x - 1 & \text{for } x \in [2, 3], \end{cases}$$

is clearly continuous and surjective. It's also easy to see that p is closed, so p is surjective. However, p is not open, since $[0, 1]$, which is open in X, it not open in Y.

Proposition: If X is a space and A is a set with $p : X \to A$ being surjective, there exists a unique topology $\mathcal{T}$ on which p is a quotient map.

Proof: Let

$$\mathcal{T} = \{U \subseteq A : p^{-1}(U) \text{ is open in } A\}.$$

This is easily seen to be a topology on A. To show it's unique, suppose A had topology $\mathcal{U} \neq \mathcal{T}$. If $\mathcal{U}$ has a set V that's not in $\mathcal{T}$, that implies that $p^{-1}(V)$ is not open in X, which contradicts p being a quotient map. If $\mathcal{U}$ doesn't have a set V that's in $\mathcal{T}$, this again contradicts p being a quotient map, since $p^{-1}(V)$ is open in X but V is not open in A. ■

Definition (quotient topology): If X is a space and A is a set with $p : X \to A$ being surjective, then there exists a unique topology $\mathcal{T}$ on A for which p is a quotient map. This is the *quotient topology* induced by p.

There is a particular case of these that is important. In essence, this is how we glue points of a topological space together.

Definition (quotient space): Let X be a space, and let X^* be a partition of X into disjoint subsets whose union is X. Let $p : X \to X^*$ be the surjective map that carries each point of X to the element of X^* containing it. In the quotient topology induced by p, X^* is called the *quotient space* of X.

Example: Consider $D = \{(x, y) : x^2 + y^2 \leq 1\}$. Consider the equivalence relation $\sim$ on D for which $a \sim b$ is they both lie on the boundary of D. Thus $\sim$ induces a partition $D/\sim$, which is precisely X^* in the above definition. It's easy to see that $D/\sim$ is homeomorphic to S^2, since we're essentially gluing the boundary of D together.

Proposition (quotient maps with respect to subspaces): Let $p : X \to Y$ be a quotient map and let $A \subseteq X$ that is saturated with respect to p. Let $q : A \to p(A)$ be the map obtained by restricting p to A.

a) If A is either open or closed in X, then q is a quotient map.
b) If p is either an open map or a closed map, then q is a quotient map.

Proof:

a) We do the case where A is open, as the case where it's closed is basically the same, just using the closed set formulation of quotient maps. Since A is saturated, there exists $B \subseteq Y$ such that $p^{-1}(B) = A$. Since A is open saturated, we also have that $p(A) = p(p^{-1}(B)) = B$ is open in Y (this follows since p is surjective). Since p is continuous, it's restriction q will also be continuous. Thus we just need to show that q sends saturated open sets to open sets. Suppose $U \subseteq A$ is open saturated with respect to q. Thus there exists $V \subseteq p(A)$ with $q^{-1}(V) = U$. Since q is the restriction of p, we also see that $p^{-1}(V) = U$. Thus U is saturated with respect to p. Since it's a quotient map, we see that $p(U) = V$ is open in Y. Since $V \subseteq p(A)$, this means that $p(U)$ is open in $p(A)$, as desired.

b) We do the case where p is closed, as the case where it's open is basically the same, just using the open set formulation for quotient maps. q is the restriction of a continuous map, so

it's continuous. Thus we just need to show that q sends closed saturated sets to closed sets. Suppose $C \subseteq A$ is closed saturated with respect to q. Thus there exists closed $D \subseteq Y$ with $q^{-1}(D) = C$. Thus $p^{-1}(D) = C$. Since p is a closed map, $p(C) = p(p^{-1}(D)) = D$ is closed in Y. Thus $D = q(C) \subseteq p(A)$ is closed, as desired.

■

1.8. Problems

Problem: Let $(X, \mathcal{T})$ be a topological space, and let $A \subseteq X$ be a set with the property that for every $x \in A$, there is an open set $U_x \in \mathcal{T}$ such that $x \in U_x \subseteq A$. Show that A is open.

Solution: We claim that $\bigcup_{x \in A} U_x = A$. Clearly if y is in the left, then it's in the right, and if y is in the right, then it will be contained in U_y, and so will be in the left. Thus A is the union of open sets, and thus is also open.

Problem: Let $\{\mathcal{T}_\alpha : \alpha \in I\}$ be a collection of topologies on X. Prove that there is a unique finest topology that is refined by all the $\mathcal{T}_\alpha$'s.

Solution: Let $\mathcal{T} = \bigcap_{\alpha \in I} \mathcal{T}_\alpha$. If $\mathcal{T}$ is a topology, that it's clear that $\mathcal{T} \subseteq \mathcal{T}_\alpha$ for all $\alpha \in I$, and so is refined by all of those topologies. Further, if $\mathcal{T}'$ is refined by those same topologies, then $V \in \mathcal{T}' \Rightarrow V \in \mathcal{T}_\alpha \forall \alpha \in I \Rightarrow V \in \mathcal{T} \Rightarrow \mathcal{T}' \subseteq \mathcal{T}$, and so $\mathcal{T}$ is the finest possible topology.

Clearly $\emptyset, X \in \mathcal{T}$. Suppose $\{U_\beta\}_{\beta \in J}$ is a collection of sets in $\mathcal{T}$. Then by definition, $\{U_\beta\} \subseteq T_\alpha$ for all $\alpha \in I$, and thus $\bigcup_{\beta \in J} U_\beta \in \mathcal{T}_\alpha$ for all $\alpha \in I$. Thus $\bigcup_{\beta \in J} U_\beta \in \mathcal{T}$, so $\mathcal{T}$ is closed under arbitrary unions. We do the same thing to get that $\mathcal{T}$ is clsoed under finite intersections, so it is indeed a topology.

Problem: Let $X = [0,1]^{[0,1]}$ be the set of all functions $f : [0,1] \to [0,1]$. Given $A \subseteq [0,1]$, let

$$U_A := \{f \in X : f(x) = 0 \text{ for all } x \in A\}.$$

Show that $\mathcal{B} := \{U_A : A \subseteq [0,1]\}$ is a basis for a topology on X.

Solution: Clearly $\mathcal{B}$ covers X, since a function with zero set Z will be contained in U_Z. Now suppose $U_A, U_B \in \mathcal{B}$ and $f \in U_A \cap U_B$. Thus $f(x) = 0$ for all $x \in A \cap B$, which means $f \in U_{A \cap B} \subseteq U_A \cap U_B$, so $\mathcal{B}$ satisfies the second basis condition.

Problem: Let $\mathcal{B}$ be a basis on X, and let $\mathcal{T}_\mathcal{B}$ be the topology it generates. Show that $\mathcal{T}_\mathcal{B}$ is the intersection of all topologies that contain $\mathcal{B}$.

Solution: Let

$$\mathcal{T}' = \bigcap_{\substack{\mathcal{T} \text{ is a topology on } X \\ \mathcal{B} \subseteq \mathcal{T}}} \mathcal{T}.$$

Clearly $\mathcal{T}_{\mathcal{B}} \subseteq \mathcal{T}'$, since any topology that contains $\mathcal{B}$ must contain all possible unions of elements of $\mathcal{B}$, which generates $\mathcal{T}_{\mathcal{B}}$. Now suppose $V \in \mathcal{T}'$. Since clearly $\mathcal{T}_{\mathcal{B}}$ satisfies the conditions of the intersection, this implies that $V \in \mathcal{T}_{\mathcal{B}}$, and since we have the inclusion both ways, the topologies are equal, as desired.

Problem: Let $\{\mathcal{T}_\alpha : \alpha \in I\}$ be a collection of topologies on a set X. Prove that there is a unique coarsest topology that refines all the $\mathcal{T}_\alpha$'s.

Solution: We know we can create a basis for each topology, so let $\mathcal{B}_\alpha$ denote the basis that generates $\mathcal{T}_\alpha$. Let $\mathcal{B} = \bigcup_{\alpha \in I} \mathcal{B}_\alpha$. We claim that $\mathcal{T}_{\mathcal{B}}$ is the desired topology. Fix $\mathcal{T}_\alpha$. For any $x \in X$ and any $B_\alpha \in \mathcal{B}_\alpha$, we can pick $\mathcal{B}_\alpha \in \mathcal{B}$ and obtain $x \in B_\alpha \subseteq B_\alpha$. Then the proposition about comparing bases implies that $\mathcal{T}_{\mathcal{B}} \supseteq \mathcal{T}_\alpha$.

Now suppose $\mathcal{T}'$ also refines the collection of topologies. If $V \in \mathcal{T}_{\mathcal{B}}$, then it can be written as

$$\bigcup_{\alpha \in I} \left(\bigcup_{\beta \in J_\alpha} B_{\alpha,\beta} \right),$$

where the inner union is the union of a collection of sets from $\mathcal{B}_\alpha$. Since $\mathcal{T}'$ refines all the topologies, it will contain each of these basis elements, and thus their union, implying that $V \in \mathcal{T}'$. Thus $\mathcal{T}_{\mathcal{B}} \subseteq \mathcal{T}'$, implying it's the coarsest possible topology.

Problem: Fix an infinite subset A of $\mathbb{Z}$ such that $\mathbb{Z} \backslash A$ is also infinite. Construct a topology on $\mathbb{Z}$ in which:

a) A is open.
b) Singletons are never open.
c) For any pair of distinct integer m and n, there are disjoint open sets U and V such that $m \in U$ and $n \in V$.

Solution: We create a basis $\mathcal{B}$ as follows: First add A and $\mathbb{Z} \backslash A$ to the basis. Since A is countable, we can rewrite its elements as $\{a_1, a_2, \ldots\}$. Now add the sets

$$A_{2^k}^r := \{a_n : n \equiv r \pmod{2^k}\}$$

to $\mathcal{B}$. Add similar sets $B_{2^k}^r$ using $\mathbb{Z} \backslash A$ instead. Note that this basis covers $\mathbb{Z}$, since any element in $\mathbb{Z}$ is either in A or $\mathbb{Z} \backslash A$. Then it will be one of a_n or b_m, at which point it's clear that $a_n \in A_{2^n}^n$ and $b_m \in B_{2^m}^m$. Also sets are either disjoint or subsets of each other (since we're creating open sets by partitioning existing open sets), so the second condition for a basis is satisfied. Then the topology generated by $\mathcal{B}$ satisfies the problem conditions, which is easy to check.

> **Remark**
>
> The idea for this construction came from the fact that any topology that satisfies the conditions cannot have finite open sets, since otherwise we could continue to intersect sets containing m and n with the disjoint U and V and obtain smaller open sets until we create a singleton. Thus every open set must be infinite. Then we can declare $\mathbb{Z}\backslash A$ to be open, and we now have basically have to find a topology on A and $\mathbb{Z}\backslash A$ that satisfies the condition, but with free reign to choose the initial infinite subset.

Problem: Show that if Y is a subspace of X and A is a subset of Y, then the topology A inherits as a subspace of Y is the same as the topology it inherits as a subspace of X.

Solution: Let $\mathcal{T}_Y$ be the topology A inherits as a subspace of Y and $\mathcal{T}_X$ denote the topology it inherits as a subspace of X. Then U is open in $\mathcal{T}_Y$ is and only if $U = V \cap A$, where V is open in Y, which means $V = W \cap Y$ for some open W in X. Thus U is open in Y if and only if $U = A \cap W \cap Y = A \cap W$ for some open W in X. Thus the topologies are equal.

Problem: If $\mathcal{T}$ and $\mathcal{T}'$ are topologies on X and $\mathcal{T}'$ is strictly finer than $\mathcal{T}$, how do the two subspace topologies on $Y \subseteq X$ compare?

Solution: Let $\mathcal{Y}$ be the subspace topology from $\mathcal{T}$, and $\mathcal{Y}'$ be the subspace topology from $\mathcal{T}'$. If U is open in $\mathcal{Y}$, then $U = V \cap Y$, where V is open in $\mathcal{T}$, which implies V is open in $\mathcal{T}'$, which implies U is open in $\mathcal{Y}'$. Thus $\mathcal{Y} \subseteq \mathcal{Y}'$. However, the relation isn't necessarily strict. For example, take the usual and discrete topologies on $\mathbb{R}$, and subspace $\{1\}$.

Problem: A map $f : X \to Y$ is said to be an *open map* if for every open set U of X, the set $f(U)$ is open in Y. Show that $\pi_1 : X \times Y \to X$ and $\pi_2 : X \times Y \to Y$ are open maps.

Solution: We show π_1 is open, as π_2 follows similarly. For any open $U \subseteq X \times Y$, we can write it as $\bigcup_\alpha U_\alpha \times V_\alpha$, where $U_\alpha \in X, V_\alpha \in Y$ are open. Then

$$\pi_1\left(\bigcup_\alpha U_\alpha \times V_\alpha\right) = \bigcup_\alpha \pi_1(U_\alpha \times V_\alpha) = \bigcup_\alpha U_\alpha,$$

which is open in X. Thus π_1 is an open map.

Problem: Let X and X' be the same set and nonemtpy, but the first has topology $\mathcal{T}$ and the second topology $\mathcal{T}'$. Do the same for $Y, Y', \mathcal{U}, \mathcal{U}'$. Show that $\mathcal{T} \subseteq \mathcal{T}', \mathcal{U} \subseteq \mathcal{U}'$ if and only if the product topology on $X' \times Y'$ is finer than the product topology on $X \times Y$.

Solution: Suppose $\mathcal{T} \subseteq \mathcal{T}', \mathcal{U} \subseteq \mathcal{U}'$. Then every basis element of $X \times Y$ is a basis element of $X' \times Y'$, so the conclusion holds.

Now suppose $X' \times Y'$ has a finer topology than $X \times Y$. Pick open $U \subseteq X$. Then $U \times Y$ is open in $X \times Y$, and thus open in $X' \times Y'$. Since π_1 is an open map, $\pi_1(U \times Y) = U$ is open in X'. Thus $\mathcal{T} \subseteq \mathcal{T}'$. The same reasoning works for $\mathcal{U}$ and $\mathcal{U}'$.

Problem: Show that the dictionary order topology on $\mathbb{R} \times \mathbb{R}$ is the same as the product topology $\mathbb{R}_d \times \mathbb{R}$, where $\mathbb{R}_d$ is the discrete topology.

Solution: Suppose $(a \times b, c \times d)$ is an open interval in the dictionary order topology. If $a = c$, then $(a \times b, c \times d) = \{a\} \times (b, d)$, which is open in $\mathbb{R}_d \times \mathbb{R}$. If $a < c$, then $(a \times b, c \times d) = \bigcup_{x \in (a,c)} (\{x\} \times \mathbb{R}) \cup (\{a\} \times (b, \infty)) \cup (\{c\} \times (-\infty, d))$, which is the union of open sets in $\mathbb{R}_d \times \mathbb{R}$, and thus open. Since open intervals in the order topology are basis elements, we have that the order topology is refined by $\mathbb{R}_d \times \mathbb{R}$.

Now consider $\{a\} \times (b, c)$, which is a basis element of $\mathbb{R}_d \times \mathbb{R}$. This is the open interval $(a \times b, a \times c)$ in the order topology, and thus is open. Thus $\mathbb{R}_d \times \mathbb{R}$ is refined by the order topology, which implies the two topologies are equal.

Problem: Show that $\overline{A \cup B} = \overline{A} \cup \overline{B}$.

Solution: Since $A, B \subseteq A \cup B$ we see that $\overline{A}, \overline{B} \subseteq \overline{A \cup B}$. Thus $\overline{A} \cup \overline{B} \subseteq \overline{A \cup B}$. Since $\overline{A} \cup \overline{B}$ is closed and since $A \cup B \subseteq \overline{A} \cup \overline{B}$, we see that $\overline{A \cup B} \subseteq \overline{A} \cup \overline{B}$.

Problem: Show that the union of dense sets is dense.

Solution: Follows from $\overline{A \cup B} = \overline{A} \cup \overline{B}$.

Problem: Show that a subset A of a topological space X is open if and only if $\operatorname{int}(A) = A$.

Solution: Suppose A is open. Since by definition $\operatorname{int}(A)$ is the union of all open subsets of A, we must have $\operatorname{int}(A) = A$. Now suppose $\operatorname{int}(A) = A$. Thus for every $x \in A$, by definition there exists a neighborhood of x contained in A. Then by the first problem, A is open.

Problem: Let $(X, \mathcal{T})$ be a topological space, and let D_1, D_2 be dense open subsets of X. Prove that $D_1 \cap D_2$ is dense and open.

Solution: Let U be open in X. Then, since D_1 is dense and open, $U \cap D_1$ is nonempty and open. Doing the same with D_2 yields that $U \cap D_1 \cap D_2$ is nonempty, so $D_1 \cap D_2$ is dense. Since both are open, their intersection will also be open.

Problem: Show that X is Hausdorff if and only if the diagonal $\Delta = \{(x, x) : x \in X\}$ is closed in $X \times X$.

Solution: Suppose Δ is closed in $X \times X$. Then $X \times X \backslash \Delta = \{(x, y) : x, y \in X, x \neq y\}$ is open in $X \times X$. Now suppose $x, y \in X$ are not equal. Since the above is open, there exists open A, B in X such that $(x, y) \in A \times B \subseteq X \times X \backslash \Delta$. Since $A \times B$ doesn't contain any elements of Δ, $A \cap B = \emptyset$. Thus $x \in A$ and $y \in B$ are two disjoint open sets that contain x and y respectively, so X is Hausdorff.

Now suppose X is Hausdorff. Consider (x, y) with $x \neq y$. Then there exists A, B open in X that contain x and y respectively but are disjoint. Thus $A \times B$ is a neighborhood of (x, y) that doesn't intersect Δ, so it's not a limit point of Δ. Thus Δ is closed in $X \times X$.

Problem: Suppose $f : X \to Y$ is continuous. If x is a limit point of the subset A of X, is it necessarily true that $f(x)$ is a limit point of $f(A)$?

Solution: No. Take $f : \mathbb{R} \to \mathbb{R}$ with the usual topology defined by $f(x) = 0$, which is clearly continuous, since the inverse image is either $\emptyset$ or $\mathbb{R}$. Then for $A = \mathbb{R}$, $f(A) = \{0\}$. For any $x \in \mathbb{R}$, it's a limit point of $\mathbb{R}$, but $f(x) = 0$ is not a limit point of $f(A)$, since any open set containing 0 doesn't intersect $f(A)$ anywhere else.

Problem: Let X and X' denote a single set in the two topolgies $\mathcal{T}$ and $\mathcal{T}'$. Let $i : X' \to X$ be the identity function.
a) Show that i is continuous $\iff \mathcal{T}'$ is finer then $\mathcal{T}$.
b) Show that i is a homemorphism $\iff \mathcal{T}' = \mathcal{T}$.

Solution:

a) Suppose i is continuous. Since $f^{-1}(V) = V$, by continuity V is open in X' if its open in X. Thus every open set in X is open in X', so $\mathcal{T} \subseteq \mathcal{T}'$. The other follows by the reverse logic.

b) If i is a homeomorphism, then that implies V is open in X if and only if it's open in X'.

Problem: Let $\{A_\alpha\}$ be a collection of subsets of X, and suppose $U = \bigcup_{A_\alpha} A_\alpha$. Let $f : X \to Y$, and suppose $f|_{A_\alpha}$ is continuous for all α.

a) Show that if the collection $\{A_\alpha\}$ is finite and each set A_α is closed, then f is continuous.
b) Find an example where the collection $\{A_\alpha\}$ is countable and each A_α is closed, but f is not continuous.
c) An indexed family of sets $\{A_\alpha\}$ is said to be *locally finite* if each point x of X has a neighborhood that intersects A_α for only finitely many values of α. Show that if the family $\{A_\alpha\}$ is locally finite and each A_α is closed, then f is continuous.

Solution:

a) Suppose $V \subseteq Y$ is closed. Then $\left(f|_{A_\alpha}\right)^{-1}(V) = f^{-1}(V) \cap A_\alpha$ is closed in A_α, and since A_α is closed in X, this set is closed in X. Then we can union over all α, and since there are only finitely many of them, we see that $f^{-1}(V) = \bigcup_\alpha f^{-1}(V) \cap A_\alpha = \bigcup_\alpha \left(f|_{A_\alpha}\right)^{-1}(V)$ is closed.
b) Consider $f : \mathbb{R} \to \mathbb{R}$, where both have the usual topology, and f is zero for $x < 0$ and 1 for $x \geq 1$. Let $A_0 = [0, \infty]$, $A_{-1} = [-\infty, -1]$, and $A_n = \left[-\frac{1}{n}, -\frac{1}{n+1}\right]$ for $n \geq 1$. Then $\{A_n\}_{n\geq 1}$ cover $\mathbb{R}$, and it's easy to see that $f|_{A_n}$ is continuous for all n (as they're all constant functions). However, since $f^{-1}(\{0\}) = (-\infty, 0)$ is not closed and $\{0\}$ is, f is not continuous.
c) Let $\{A_\alpha\}$ be locally finite and suppose each A_α is closed. We show that the union of any subcollection is closed. Suppose $\{B_\alpha\}$ is a subcollection, and let $B = \bigcup_\alpha B_\alpha$. Suppose $x \in \overline{B} \backslash B$. By local finiteness, there exists a neighborhood U_x of x that intersects finitely many B_α, say $B_{\alpha_1}, \ldots, B_{\alpha_n}$. Thus $U_x \cap \left(X \backslash \bigcup_{k=1}^n B_{\alpha_k}\right)$ is an open set that contains x. However, this set doesn't intersect B at all, which contradicts $x \in \overline{B}$. Thus $\overline{B} \backslash B = \emptyset \Rightarrow \overline{B} = B$. Thus B is closed.

   Now suppose $\{A_\alpha\}$ is locally finite and A_α is closed for all α, and suppose $V \subseteq Y$ is closed. Again we see that $\left(f|_{A_\alpha}\right)^{-1}(V) = f^{-1}(V) \cap A_\alpha$ is closed in X. Since each is a subset of A_α, it's easy to see that the collection $\left\{f^{-1}(V) \cap A_\alpha\right\}$ is locally finite and each set is closed. Then by the above paragraph, $\bigcup_\alpha f^{-1}(V) \cap A_\alpha = f^{-1}(V)$ is closed, as desired.

Problem: Let $f : A \to B$ and $g : C \to D$ be continuous functions. Define $f \times g : A \times C \to B \times D$ by

$$(f \times g)((a, c)) = (f(a), g(c)).$$

Prove that $f \times g$ is continuous.

Solution: Consider the basis element $U \times V$ of $B \times D$. Then $(f \times g)^{-1}(U \times V) = f^{-1}(U) \times g^{-1}(V)$, which is open in $A \times B$. Thus $f \times g$ is continuous.

Problem: Let $A \subseteq X$, and let $f : A \to Y$ be continuous, where Y is Hausdorff. Show that if f can be extended to a continuous function $g : \overline{A} \to Y$, then the extension g is unique.

Solution: Suppose for the sake of contradiction there exist distinct continuous extensions $g_1, g_2 : \overline{A} \to Y$. Since these agree on A, there exists $x \in \overline{A} \backslash A$ for which $g_1(x) \neq g_2(x)$. Since Y is Hausdorff, there thus exist disjoint open U, V that contain $g_1(x), g_2(x)$ respectively. Since each function is continuous, the sets $g_1^{-1}(U)$ and $g_2^{-1}(V)$ are open in $\overline{A}$. Thus $g_1^{-1}(U) \cap g_2^{-1}(V)$ is open in $\overline{A}$. Thus we can write $g_1^{-1}(U) \cap g_2^{-1}(V) = \overline{A} \cap W$, where W is open in X. Since the intersection on the left and $\overline{A}$ contain x, we see that $x \in W$. Thus W is is a neighborhood of x, which implies W intersects A at some point $y \neq x$. By equality of sets, we get $y \in g^{-1}(U), g_2^{-1}(V)$. Since g_1 and g_2 are equal on A, we see that $g_1(y) = f(y) = g_2(y)$. Thus $y \in U \cap V$, contradiction.

Problem: Let $x_1, x_2, \ldots$ be a sequence in $\prod X_\alpha$ with the product topology. Show that this sequence converges to x if and only if $\pi_\alpha(x_1), \pi_\alpha(x_2), \ldots$ converges to $\pi_\alpha(x)$.

Solution: Since the projection map is continuous, we already know that the convergence of the product space sequence implies the convergence of the coordinate sequences.

Now suppose each coordinate sequence converges. Consider a subbasis element $\pi_\beta^{-1}(V_\beta)$, where V_β is open in X_β and contains $\pi_\beta(x)$. By convergence, V_β contains $\left(\pi_\beta(x_n)\right)_{n \geq N}$ for some N. Thus $\pi_\beta^{-1}(V_\beta)$ contains $(x_n)_{n \geq N}$. This holds for arbitrary V_β and any index β. Since this holds for all the subbasis element, it will hold for any neighborhood of x, so we're done.

Problem: Let $\mathbb{R}^\infty$ be the subset of $\mathbb{R}^\omega$ consisting of all sequence that are eventually zero. Find the closure of $\mathbb{R}^\infty$ in both the product, box, and uniform topology.

Solution: First we show that $\mathbb{R}^\infty$ is dense in the product topology. The basic open sets are

$$U_1 \times \cdots \times U_{N-1} \times \mathbb{R} \times \cdots,$$

where each U_i is open in $\mathbb{R}$. For the first $N-1$ sets, we can pick an element from there, and then for all the $\mathbb{R}$ components, we can choose 0. This creates an element of $\mathbb{R}^\infty$. Since this works for arbitrary basis open sets, the intersection of any open set and $\mathbb{R}^\infty$ is nonempty, and thus $\mathbb{R}^\infty$ is dense in $\mathbb{R}^\omega$, implying $\overline{\mathbb{R}^\infty} = \mathbb{R}^\omega$.

Now we do the box topology. Consider some sequence $(x_1, x_2, \ldots)$ that's nonzero for infinitely many terms. Then for those nonzero terms, we can enclose them in an open interval that doesn't contain 0. This gives a basic open set in the box topology that doesn't intersect $\mathbb{R}^\infty$. Thus $\overline{\mathbb{R}^\infty} = \mathbb{R}^\infty$.

We claim that in the uniform topology, $\overline{\mathbb{R}^\infty} = c_0$. Suppose $x \notin c_0$. Thus there exists some infinite set of coordinates for which each coordinate has absolute value greater than $\varepsilon > 0$. Pick $\varepsilon \leq 1$, and consider $B_{\overline{\rho}}(x, \frac{\varepsilon}{2})$. Then for each of those coordinates, 0 is never achieved for any y in the ball. Thus the ball doesn't intersect $\mathbb{R}^\infty$, so it's not in the closure.

Now suppose $x \in c_0$, and pick any $\varepsilon > 0$. If $\varepsilon > 0$, the ball centered at x with size ε is the whole space, so suppose $\varepsilon \leq 1$, and consider $B_{\overline{\rho}}(x, \varepsilon)$. There exists some N such that $x_n < \varepsilon/2$ for all $n \geq N$. Then $B_{\overline{\rho}}(x, \varepsilon)$ contains $(x_1, \ldots, x_N, 0, 0, \ldots) \in \mathbb{R}^\infty$. Thus the ball intersects $\mathbb{R}^\infty$ for any $\varepsilon > 0$, so we're done.

Problem: Show that $\mathbb{R} \times \mathbb{R}$ in the dictionary order topology is metrizable.

Solution: Consider

$$d((a,b),(c,d)) = \begin{cases} |b-d| & \text{if } a = c, \\ 1 + |a-c| & \text{if } a \neq c. \end{cases}$$

It's easy to check that d is a metric on $\mathbb{R} \times \mathbb{R}$. Consider $((a,b),(c,d))$ and fix arbitrary (x,y) inside the interval. If $x = a$, then $B_d\big((x,y), \min\big(y-a, \frac{1}{2}\big)\big)$ is contained in the interval. If $x = c$, then $B_d\big((x,y), \min\big(\frac{1}{2}, d-y\big)\big)$ is contained in the interval. If $a < x < c$, then $B_d((x,y),1)$ is contained in the interval. Thus the order topology is contained in the topology induced by d.

Now consider some open ball $B_d((x,y),\varepsilon)$. If $\varepsilon \leq 1$, then this is equal to $((x,y-\varepsilon),(x,y+\varepsilon))$. If $\varepsilon > 1$, then this is equal to $((x,y-\varepsilon),(x,y+\varepsilon)) \cup \bigcup_{x<r<x+\varepsilon-1}(\{r\} \times (-\infty,\infty))$, which is open. Thus the topology induced by d is containde in the order topology.

Problem: Consider the product, uniform, and box topologies on $\mathbb{R}^\omega$. In which topologies are the following functions from $\mathbb{R}$ to $\mathbb{R}^\omega$ continuous?

$$\begin{aligned} f(t) &= (t, 2t, 3t, \ldots), \\ g(t) &= (t, t, t, \ldots) \\ h(t) &= \left(t, \frac{1}{2}t, \frac{1}{3}t, \ldots\right). \end{aligned}$$

Solution: Since each component is continuous, we know that in the product topology, every function is continuous.

$f(t)$ is not continuous in the box topology, as $f^{-1}((-1,1)^\omega) = \{0\}$ is not open in $\mathbb{R}$. We already saw that g is not continuous in the box topology. h is not continuous in the box topology, since $h^{-1}\big(\prod(-\frac{1}{n^2}, \frac{1}{n^2})\big) = \{0\}$ is not open in $\mathbb{R}$.

f is not continuous in the uniform topology, since $f^{-1}\big(B_{\overline{\rho}}(0,1)\big) = \{0\}$. We claim g is continuous in the uniform topology. Fix $x \in \mathbb{R}^\omega$ and $\varepsilon > 0$, and consider the inverse image of $B_{\overline{\rho}}(x,\varepsilon)$. If $\varepsilon > 1$, then the ball is the whole space, and so the inverse image is $\mathbb{R}$, so we can assume $\varepsilon \leq 1$. If the inverse image is nonempty, suppose $r \in g^{-1}\big(B_{\overline{\rho}}(x,\varepsilon)\big)$. Thus $\overline{\rho}((r,r,\ldots),x) < \varepsilon$. Thus $\sup\{|r - x_i|\} < \varepsilon$ (we can replace $\overline{d}$ with d since we assumes $\varepsilon \leq 1$). Let p be the supremum, and consider $s \in (r - \frac{\varepsilon-p}{2}, r + \frac{\varepsilon-p}{2})$. Then $|s - x_i| \leq |s-r| + |r - x_i| \leq \frac{\varepsilon-p}{2} + p = \frac{\varepsilon+p}{2}$. Taking the supremum yields $\sup\{|s-x_i|\} \leq \frac{\varepsilon+p}{2} < \varepsilon$. Thus

$$\left(r - \frac{\varepsilon - p}{2}, r + \frac{\varepsilon - p}{2}\right) \subseteq g^{-1}\big(B_{\overline{\rho}}(x,\varepsilon)\big),$$

implying the inverse image is open, and thus g is continuous.

We claim h is continuous. We use the same method as before, but replace $\sup\{|r - x_i|\}$ with $\sup\{|\frac{r}{i} - x_i|\}$.

Problem: Given $x \in \mathbb{R}^\omega$ and $0 < \varepsilon < 1$, let

$$U(x, \varepsilon) = (x_1 - \varepsilon, x_1 + \varepsilon) \times (x_2 - \varepsilon, x_2 + \varepsilon) \times \cdots.$$

Show that $U(x, \varepsilon)$ is not open in the unform topology and

$$B_{\overline{\rho}}(x, \varepsilon) = \bigcup_{\delta < \varepsilon} U(x, \delta).$$

Solution: Consider the element $x' = \left(x_i + \varepsilon\left(1 - \frac{1}{i}\right)\right)$. Now pick any $\delta > 0$. There exists some i such that $\delta > \frac{\varepsilon}{i}$. Then $B_{\overline{\rho}}(x', \delta)$ contains

$$\left(x_1, x_2 + \frac{1}{2}\varepsilon, \ldots, x_{i-1} + \frac{i-2}{i-1}\varepsilon, x_i + \varepsilon, x_{i+1} + \frac{i}{i+1}\varepsilon, \ldots\right) \notin U(x, \varepsilon).$$

Thus there exists no open neighborhood of x' contained in $U(x, \varepsilon)$, so it's not open.

Now we prove the second statement. Suppose $x' \in B_{\overline{\rho}}(x, \varepsilon)$, and let $\delta = \sup\{|x_i - x_i'|\}$, which must be less than ε. Then $x' \in U\left(x, \frac{\varepsilon+\delta}{2}\right)$, so x' is contained in the right. Thus the left is contained in the left.

Now suppose x' is in the right. Thus there exists some $\delta < \varepsilon$ for which $x' \in U(x, \delta)$. Thus $|x_i' - x_i| < \delta$ for all i, so $\overline{\rho}(x_i' - x_i) \leq \delta < \varepsilon$. Thus x' is in the left, so the right is contained in the left.

Problem: Let the ℓ^2 topology be the topology that ℓ^2 has generated as a metric space. As a subspace of $\mathbb{R}^\omega$, it also inherits the box, uniform, and product topologies.

a) Show that in ℓ^2, the box topology is strictly finer than the ℓ^2 topology, which is strictly finer than the uniform topology.
b) The set

$$H = \prod_{n \in \mathbb{N}} \left[0, \frac{1}{n}\right]$$

is contained in ℓ^2, and is known as the *Hilbert cube*. Compare the four topologies it has as a subspace of ℓ^2.

Solution:

a) Fix $x \in \ell^2, \varepsilon > 0$, and consider $B_{\ell^2}(x, \varepsilon)$. Pick y in the ball. Then $\sum (x_i - y_i)^2 < \varepsilon^2$. Let S be the sum, and let M be the max of $|x_i - y_i|$ (which must exist since the sum converges). We claim that $U = \prod_{n \in \mathbb{N}} \left(y_i - \frac{1}{K^n}, y_i + \frac{1}{K^n}\right) \subseteq B_{\ell^2}(x, \varepsilon)$ for some suitable K. Suppose $z \in U$. Then we have

$$\begin{aligned}\sum (z_i - x_i)^2 &\leq \sum (|z_i - y_i| + |y_i - x_i|)^2 \\ &= \sum (z_i - y_i)^2 + 2|z_i - y_i||y_i - x_i| + (y_i - x_i)^2 \\ &\leq S + \sum \frac{1}{K^{2i}} + \frac{2}{K^i} \cdot M \\ &= S + \frac{1}{K^2 - 1} + \frac{2M}{K - 1} \\ &= S + \frac{2MK + 2M + 1}{K^2 - 1}.\end{aligned}$$

The numerator is linear and the denominator is quadratic in K, so the term goes to 0 as $K \to \infty$. Thus we can pick K such that the term is less that $\frac{1}{2}(\varepsilon^2 - S)$, and so $d_{\ell^2}(z, x) < \varepsilon$ for all $z \in U$. Thus U is contained in the ball, as desired. Since this works for any $y \in B_{\ell^2}(x, \varepsilon)$, the ball is open in the box topology. Since this works for any open ball, the box topology contains the ℓ^2 topology.

To show the box topology is strictly finer, consider $U = \prod_{n \in \mathbb{N}} \left(-\frac{1}{n^2}, \frac{1}{n^2}\right)$. For any $\varepsilon > 0$, consider $B_{\ell^2}(0, \varepsilon)$. Since there exists some integer N for which $\frac{1}{N^2} < \frac{\varepsilon}{2}$, the point $(0, ..., 0, \frac{\varepsilon}{2}, 0, ...)$ (where $\frac{\varepsilon}{2}$ is in the Nth slot) lies in $B_{\ell^2}(0, \varepsilon)$ but not U. Since this holds for any ε, there doesn't exist an open neighborhood of 0 contained in U, so U is not open in the ℓ^2 topoology.

Now fix $x \in \ell^2, \varepsilon > 0$ and consider $B_{\overline{\rho}}(x, \varepsilon)$ (assume $\varepsilon \leq 1$, since otherwise the ball is the whole space). Pick y in the ball, and let $\delta = \overline{\rho}(x, y)$. We claim $B_{\ell^2}\left(y, \frac{1}{2}(\varepsilon - \delta)\right) \subseteq B_{\overline{\rho}}(x, \varepsilon)$. Indeed, suppose $z \in B_{\ell^2}\left(y, \frac{1}{2}(\varepsilon - \delta)\right)$. Then

$$|z_i - y_i| \leq \sum (z_i - y_i)^2 < \frac{1}{2}(\varepsilon - \delta),$$

so $\sup|z_i - y_i| \leq \frac{1}{2}(\varepsilon - \delta)$. We then have

$$\sup|x_i - z_i| \leq \sup|x_i - y_i| + \sup|y_i - z_i| \leq \delta + \frac{1}{2}(\varepsilon - \delta) < \varepsilon.$$

Thus $\overline{\rho}(x, z) < \varepsilon$, so $z \in B_{\overline{\rho}}(x, \varepsilon)$, as desired.

To show the ℓ^2-topology is strictly finer, consider $B_{\ell^2}(0, 1)$. For any $1 \geq \varepsilon > 0$, consider $B_{\overline{\rho}}(0, \varepsilon)$. Picking N such that $\sqrt{N}\varepsilon/2 > 1$, we see that the point $(\frac{\varepsilon}{2}, \frac{\varepsilon}{2}, ..., \frac{\varepsilon}{2}, 0, ...)$ (nonzero N times) is contained in the uniform topology ball but not the ℓ^2 one. Since this works for any ε, there exists no neighborhood of 0 in the uniform topology that is contained in $B_{\ell^2}(0, 1)$.

b) The set $U = \prod_{n \in \mathbb{N}} \left[0, \frac{1}{n^2}\right)$ is open in the box topology. For any $\varepsilon > 0$, consider $B_{\ell^2}(0, \varepsilon)$. Then there exists an integer N such that $\frac{1}{N+1} \leq \varepsilon/2 < \frac{1}{N}$. Thus $\frac{1}{N^2} < \varepsilon/2 < \frac{1}{N}$. Therefore the point $(0, ..., 0, \varepsilon/2, 0, ...) \in H$, which is nonzero in the Nth slot, lies in $B_{\ell^2}(0, \varepsilon)$ but not U. This works for all ε, so there's no open neighborhood of 0 in the ℓ^2-topology on H. Thus U is not open in the ℓ^2-topology.

We now show that any open ball in the ℓ^2-topology on H is open in the product topology on H. This then shows that H inherits the same topology from all three of them. Pick $x \in H$ and $\varepsilon > 0$, and consider $B_{\ell^2}(x, \varepsilon)$. Pick any $y \in B_{\ell^2}(x, \varepsilon)$, and let $r = d_{\ell^2}(x, y) < \varepsilon$. Let $M = \max|x_i - y_i|$. Now pick N such that

$$\frac{1}{N}, \frac{2M}{N}, \sum_{i > N} \frac{1}{i^2} < \frac{\varepsilon^2 - r^2}{8}.$$

Let

$$U = H \cap \prod_{n \leq N} \left(y_i - \frac{1}{N^2}, y_i + \frac{1}{N^2} \right) \times \prod_{n > N} \mathbb{R},$$

which is open in the product topology on H. We claim $U \subseteq B_{\ell^2}(x, \varepsilon)$. Pick z in the ball. We then have

$$\begin{aligned}
\sum (z_i - x_i)^2 &\leq \sum (|z_i - y_i| + |y_i - x_i|)^2 \\
&= \sum (z_i - y_i)^2 + 2|z_i - y_i||y_i - x_i| + (y_i - x_i)^2 \\
&\leq \frac{N}{N^2} + \sum_{i > N} \frac{1}{i^2} + \frac{2M}{N} + r^2 \\
&< 3 \cdot \frac{\varepsilon^2 - r^2}{8} + r^2 \\
&< \varepsilon^2.
\end{aligned}$$

Thus $d(x, z) < \varepsilon$, so U is in the ball, as desired. Thus every y in the ball has an open neighborhood in the product topology, so the ball is open in the product topology.

2. Connectedness and Compactness

2.1. Connected Spaces (FINISH LIST OF POSSIBLE CONNECTEDNESSES)

Definition (separation/connected): Let X be a topological space. A *separation* of X is a pair U, V of disjoint nonempty open subsets of X whose union is X. The space X is said to be *connected* if there does not exist a separation of X.

Note that the if X is disconnected, the sets U and V are clopen, since $X \backslash U = V$ is closed and vice versa. Conversely, if X has a non trivial clopen subset, then that set and its complement form a separation.

Example (disconnected spaces):
- If $|X| \geq 2$, then the indiscrete topology on X is clearly connected.
- $\mathbb{Q}$ is disconnected, since $\mathbb{Q} \cap \left(-\infty, \sqrt{2}\right)$ and $\mathbb{Q} \cap \left(\sqrt{2}, \infty\right)$ are open in $\mathbb{Q}$ and form a separation.

Proposition: Let X be a space and Y be a subspace. Then nonempty $A, B \subseteq Y$ are a separation of Y if and only if $\overline{A} \cap B = A \cap \overline{B} = \emptyset$ and $A \cup B = Y$ (where the closure is taken in X).

Proof: Suppose A, B form a separation of Y. By definition, $A \cup B = Y$. The closure of A in Y is $\overline{A} \cap Y$. Since A forms a separation of Y, it's clopen in Y. Thus $A = \overline{A} \cap Y$. Taking the intersection of B of both sides yields $\emptyset = \overline{A} \cap B$. Similar logic shows $A \cap \overline{B} = \emptyset$.

Now suppose $\overline{A} \cap B = A \cap \overline{B} = \emptyset$ and $A \cup B = Y$. Note the first two equalities implies that A, B are disjoint, so we just need to show that A, B are either closed or open in Y. Again we note that the closure of A in Y is $\overline{A} \cap Y$. Then $\overline{A} \cap Y = \overline{A} \cap (A \cup B) = A \cup \left(\overline{A} \cap B\right) = A$. Thus A is closed in Y. Similarly B is closed in Y. ■

Proposition: If the sets C and D form a separation of X, and if Y is a connected subspace of X, then Y lies entirely within either C or D.

Proof: Suppose not. Then $C \cap Y$ and $D \cap Y$ would be nonempty open sets in Y. Since $C \cup D = X$, the union of the above two sets would be Y, which means they form a separation of Y, contradiction. ■

Proposition: The union of a collection of connected subspaces of X that have a point in common is connected.

Proof: Suppose $\mathcal{C}$ is a collection of connected subspaces of X and $\bigcap \mathcal{C} \neq \emptyset$, and let p be a point in this intersection. Suppose for the sake of contradiction there existed a separation A, B of $\bigcup \mathcal{C}$. Without loss of generality suppose $p \in A$. For any $Y \in \mathcal{C}$, by the previous result we know it must lie in either A or B. Since $p \in A$, we have that $Y \subseteq A$. Thus $\bigcup \mathcal{C} \subseteq A$, which means B is nonempty, so A, B don't form a separation. ■

Proposition: Let A be a connected subspace of X. If $A \subseteq B \subseteq \overline{A}$, then B is also connected.

Proof: Suppose there exists C, D a separation of B. Thus $C \cap A$ and $D \cap A$ are open in A, and they clearly union to A. Note that $\overline{C \cap A} = \overline{C} \cap \overline{A} = \overline{C}$ (since $C \subseteq \overline{A}$), and similarly $\overline{D \cap A} = \overline{D}$. Note that since C, D form a separation of B, we have $\overline{C} \cap D = C \cap \overline{D} = \emptyset$ by one of previous propositions. Thus $\left(\overline{C \cap A}\right) \cap (D \cap A) = \overline{C} \cap D \cap A = \emptyset$. Similarly, $(C \cap A) \cap \left(\overline{D \cap A}\right) = \emptyset$. Thus by that same proposition, the sets $C \cap A$ and $D \cap A$ form a separation of A, contradiction. ■

This also shows that the closure of a connected subspace is connected.

Proposition: The image of a connected space under a continuous map is connected.

Proof: Suppose $f : X \to Y$ is continuous, and let $Z = f(X)$. Suppose $A, B \subseteq Z$ form a separation of X. Since restricting the range of f to Z still keeps the function continuous, we see that $f^{-1}(A)$ and $f^{-1}(B)$ are disjoint open sets in X that union to it. Thus they form a separation of X, contradiction. ■

Proposition: The finite cartesian product of connected spaces is connected.

Proof: We prove this for two spaces X, Y, and a finite number follows by induction. Pick some base point $(a, b) \in X \times Y$. Note that $X \times \{b\}$ is connected (as it's homeomorphic to X and connectedness is entirely formulated by open sets). Similarly, $\{x\} \times Y$ is connected for any $x \in X$. Since they have the point (x, b) in common, the union $(X \times \{b\}) \cup (\{x\} \times Y)$ is connected. Now

$$\bigcup_{x \in X} (X \times \{b\}) \cup (\{x\} \times Y) = X \times Y$$

is connected, since each set in the union is connected and they all contain (a, b). ■

Corollary: The product of connected spaces is connected.

Proof: Let $\{X_\alpha\}_{\alpha \in J}$ be an indexed family of connected spaces and let

$$X = \prod_{\alpha \in J} X_\alpha.$$

Fix a point $(a_\alpha) \in X$. Let K be a finite subset of J, and let X_K denote the subspace of X consisting of points (x_α) with $x_\alpha = a_\alpha$. Then $X_K \cong \prod_{\alpha \in J} X_\alpha$, which by the previous proposition is connected.

Let Y denote the union of X_K over all finite subsets K. each contain (a_α), and so the union is connected. We claim that $\overline{Y} = X$, and thus X will be connected. Let $U = \prod_{\alpha \in J} U_\alpha$ be an open set in X, and suppose $U_\alpha = X_\alpha$ for $\alpha \in J \backslash K$ for some finite K. Then it's easy to see that U intersects X_K, and thus Y. This shows Y is dense in X, as desired. ■

2.1.1. Connected Subspaces of $\mathbb{R}$

Definition (linear continuum): A simply ordered set L having more than one element is called a *linear continuum* if the following hold:
a) L has the least upper bound property.
b) If $x < y$, there exists z such that $x < z < y$.

Proposition: If L is a linear continuum in the order topology, then L is connected. Furthermore, all intervals and rays in L are connected.

Proof: We prove that if $Y \subseteq L$ is convex, then Y is connected, which proves the rest of the claim. Suppose for the sake of contradiction that A, B form a separation of Y. Pick $a \in A, b \in B$ and suppose without loss of generality that $a < b$. By convexity, $[a, b] \in Y$. Thus $[a, b]$ is the union of disjoint sets $A_0 = A \cap [a, b]$ and $B_0 = B \cap [a, b]$, each open in $[a, b]$. They are also nonempty, so we see they form a separation of $[a, b]$. Let $c = \sup A_0$. We show that it isn't contained in either A_0, B_0, which contradicts them unioning to $[a, b]$.

First suppose $c \in B_0$. Then $c \neq a$, so either $c = b$ or $a < c < b$. Either way, from the openess of B_0 in $[a, b]$, there msut exist some interval $(d, c]$ in B_0. If $c = b$, we have a contradiction, as then d is a smaller upper bound for A_0. If $c < b$, then $(c, b]$ does not intersects A_0, as c is an upper bound for A_0. Then $(d, b] = (d, c] \cup (c, b]$ doesn't intersect A_0, but this again is a contradiction, since d is a smaller upper bound on A_0 than c.

Now suppose $c \in A_0$. Then either $c = a$ or $a < c < b$. Since A_0 is open in $[a, b]$, there exists some interval $[c, e)$ in A_0. Since L is a linear continuum, we can choose $z \in L$ such that $c < z < e$. Then $z \in A_0$, which contradicts the fact that c is an upper bound for A_0. ■

Corollary: $\mathbb{R}$ is connected.

Theorem (intermediate value theorem): Let $f : X \to Y$ be a continuous map, where X is connected and Y has the order topology. If $a, b \in X$ and r is a point between $f(a), f(b)$, then there exists a point $c \in X$ such that $f(c) = r$.

Proof: Suppose otherwise. The sets $f(X) \cap (-\infty, r)$ and $f(X) \cap (r, +\infty)$ are disjoint, nonempty (they contain $f(a), f(b)$ respectively), and each is open in $f(X)$. If there were no point $c \in X$ such that $f(c) = r$, then $f(X)$ would be the union of A, B, and thus A, B would be a separation of $f(X)$, which contradicts the fact that $f(X)$ is connected (since it's the image of a connected set under a continuous function). ■

Definition (path connected): Given points $x, y \in X$, a *path* in X from x to y is a continuous map $f : [a, b] \to X$ of some closed interval in the real line into X such that $f(a) = x$ and $f(b) = y$. A space X is said to be *path connected* if every pair of points of X can be joined by a path in X.

Proposition: The image of a path connected space under a continuous function is path connected.

Proof: Follows from the composition of continuous functions. ■

Proposition: The product of path connected spaces is path connected.

Proof: Let $\{X_\alpha\}_{\alpha \in J}$ be path connected spaces, $X = \prod_{\alpha \in J} X_\alpha$, and fix $(a_\alpha), (b_\alpha) \in X$. For each α, there exists a path $\gamma_\alpha : [0, 1] \to X_\alpha$ between a_α and b_α. Thus the function $\gamma : [0, 1] \to X$ given by $\gamma(t) = (\gamma_\alpha(t))$ is a path from (a_α) to (b_α), as desired. ■

Proposition: Every path connected space is connected.

Proof: Suppose A, B for a separation of X. Pick $x, y \in X$, and suppose $\gamma : [a, b] \to X$ such that $\gamma(a) = x, \gamma(b) = y$ and is continuous. Then $\gamma^{-1}(A), \gamma^{-1}(B)$ are open disjoint nonempty sets (that contain a, b respectively). However, this is impossible since $[a, b]$ is connected. Thus if a space isn't connected, it isn't path connected. ■

Example (topologist's sine curve): Let S be the subset

$$S = \left\{ \left(x, \sin\left(\frac{1}{x}\right) \right) : 0 < x \leq 1 \right\}$$

of $\mathbb{R}^2$. Since this is the image of $(0, 1]$ under a continuous map, S is connected. Thus its closure $\overline{S} = S \cup (\{0\} \times [-1, 1])$ is connected. We show that $\overline{S}$ is not path connected.

Suppose $\gamma : [a, c] \to \overline{S}$ a path from $(0, 0)$ to some point of S. Since $\{0\} \times [-1, 1]$ is closed, $\gamma^{-1}(\{0\} \times [-1, 1])$ has a largest element b. Thus $\gamma : [b, c] \to \overline{S}$ is a path that maps b into $\{0\} \times [-1, 1]$ and maps the other points to S.

For ease, replace $[b, c]$ with $[0, 1]$, and let $\gamma(t) = (x(t), y(t))$. Then $x(0) = 0$ and $x(t) > 0, y(t) - \sin\left(\frac{1}{x(t)}\right)$ for $t > 0$. We show there exists $t_n \to 0$ such that $y(t_n) = (-1)^n$, which contradicts the continuity of γ.

To construct such a t_n, choose u such that $0 < u < x\left(\frac{1}{n}\right)$ such that $\sin\left(\frac{1}{u}\right) = (-1)^n$. Then by the intermediate value theorem, we can find t_n with $0 < t_n < \frac{1}{n}$ such that $x(t_n) = u$.

2.1.2. Local Connectedness and Components

Definition (connected components): Given X, define an equivalence relation on X by setting $x \sim y$ if there exists a connected subspace of X containing both x and y. The equivalence classes are called the *components* of X.

Proposition: The components of X are connected disjoint subspaces of X whose union is X, such that each nonempty connected subspace of X intersects only one of them.

Proof: Since the components are equivalence classes, it follows that they're disjoint and their union is X. To show that connected subspaces only intersect one component, suppose connected $A \subseteq X$ intersects C_1 and C_2, say at x_1 and x_2. Thus $x_1 \sim x_2$, which is obviously impossible. To show each component is connected, pick some $x_0 \in C$. For each $x \in C$, we can find a connected subspace A_x containing x_0 and x. Thus by what we just showed, $A_x \subseteq C$. Thus we have

$$C = \bigcup_{x \in C} A_x.$$

■

In other words, connected subspaces must lie entirely within a singular component.

Definition (path connected components): Define an equivalence realation on space X by defining $x \sim y$ if there is a path in X from x to y. The equivalence classes are called the *path components* of X.

Using the pasting lemma, it's easy to check that this relation is transitive, and so it really is an equivalence relation.

Proposition: The path components of X are path connected disjoint subspaces of X whose union is X, such that each nonempty path connected subspace of X intersects only one of them.

Proof: Same proof as in the components case, just with paths instead. ■

Definition (locally (path) connected): A space X is said to be *locally connected at x* if for every neighborhood U of x, there is a connected neighborhood V of x contained in U. If X is locally connected at each point, then it's said to be *locally connected.*

A space X is said to be *locally path connected at x* if for every neighborhood U of x, there is path connected neighborhood V of x contained in U. If X is locally path connected at each point, then it's said to be *locally path connected.*

Since path connectedness implies connectedness, we see that local path connectedness implies local connecetedness.

Proposition: A space X is locally connected if and only if for every open set U of X, each component of U is open in X.

Proof: First suppose X is locally connected, and pick $U \subseteq X$ open. Let $\{C_\alpha\}$ be the components of U, and fix some component C_i. Pick any $x \in C_i$. Since X is locally connected and U is a neighborhood of x, there exists connected open V containing x inside U. Thus $x \in V \subseteq C_i$. Doing this for all $x \in C_i$ and unioning the resulting open V's together yields C_i, so C_i is open in X.

Now suppose that for every open set $U \subseteq X$, each component of U is open in X. Pick $x \in X$ and a neighborhood V of x. Then x lies in some component C of V that's open in X, so C is a connected neighborhood of x lying in V. Thus X is locally connected at x, and this works for all points. ■

Proposition: A space X is locally path connected if and only if for every open set U of X, each path component of U is open in X.

Proof: Same proof as above. ■

Proposition: If X is a topological space, each path component of X lies in a component of X. If X is locally path connected, then the components and the path components of X are the same.

Proof: Let $C \subseteq X$ path component. Thus C is a path connected space, so it's also connected. Thus C must lie in some larger connected component C'. Now suppose X is locally path connected. This also implies that X is locally connected. Thus by the previous propositions, the components and path components of every open set are open in X. Let $C' \subseteq X$ be a component of X, which must be open. Thus the path components $\{C_\alpha\}$ of C' must be open in X as well, which then implies they're open in C' upon taking intersections. However, this is a contradiction if there's more than one path component, since then C' has a separation. ■

Corollary: Suppose X is locally path connected. Then X is connected if and only if X is path connected.

Proof: We already know path connectedness implies connecetedness, so suppose X is connected. Thus X has one connected component. Then by the previous proposition, X has one path component, and thus is path connected. ■

Corollary: Connected open sets of $\mathbb{R}^n$ are path connected.

Proof: By the previous corollary, we just need to show local path connecetedness. Given a neighborhood of $x \in U$ open and connected, find a ball centered at x contained in the neighborhood and U. Then taking straight lines between points shows the ball is path connected. ■

To finish, we gather examples of every possible combination of connected, path connected, locally connected, and locally path connected.

TABLE OF ALL FOUR PROPERTIES WITH SPACE FOR EACH POSSIBLE SET OF PROPERTIES

Let $\mathbb{N}_{\text{cofinite}}$ be a countably infinte space with the cofinite topology. Let $\mathbb{N}_{\text{cofinite}} \cup \{\infty\}$ be the cofinite topology unioned with some element ∞, where $\{\infty\}$ is open and $U \cup \{\infty\}$ is open, where $U \subseteq \mathbb{N}_{\text{cofinite}}$ is open. Let S_{ex} be the extended topologists sine curve (also known as the Warsaw sine curve). Let I_o^2 denote the ordered square.

Connected	Path Connected	Locally Connected	Locally Path Connected	Examples
✓	✓	✓	✓	$\mathbb{R}$
✓	✓	✓	✗	
✓	✗	✓	✗	$\mathbb{N}_{\text{cofinite}}, I_o^2$
✓	✓	✗	✗	S_{ex}
✗	✗	✓	✓	$[0,1] \cup [2,3]$
✓	✗	✗	✗	$\overline{S}$
✗	✗	✓	✗	$\mathbb{N}_{\text{cofinite}} \cup \{\infty\}$
✗	✗	✗	✗	$\mathbb{Q}, \mathbb{R}_\ell$

2.2. Compact Spaces (DO DINI'S THEOREM)

Definition (compact): A space X is said to be *compact* if every open covering $\mathcal{A}$ of X contains a finite subcover of X.

Proposition: Let Y be a subspace of X. Then Y is compact if and only if every covering of Y by open sets of X contains a finite subcover of Y.

Proof: Suppose Y is compact. Let $\{A_\alpha\}$ be an open covering of Y by open sets of X. Then $\{A_\alpha \cap Y\}$ is an open cover of Y by open sets of Y. Thus there exists a finite subcover from this collection. Then the corresponding sets in $\{A_\alpha\}$ form a finite cover by open sets in X.

Now suppose the converse holds. Let $\{A_\alpha\}$ be a cover of Y by open sets of Y. Thus we can write $A_\alpha = Y \cap B_\alpha$ for B_α open in X. Then $\{B_\alpha\}$ is a cover of Y by open sets of X, and thus has a finite subcover $B_1, \ldots, B_n$. Then $Y \cap B_1, \ldots, Y \cap B_n$ is a finite subcover of Y by open sets in Y. ■

Proposition: Every closed subspace of a compact space is compact.

Proof: Let C be a compact subset of X, and suppose $\mathcal{A}$ is an open cover of C. Adding in $X \backslash C$, which is open, we obtain an open cover of X, which has a finite subcover. Since $C \subseteq X$, this also covers C, so C has a finite subcover. ■

Proposition: Every compact subspace of a Hausdorff space is closed.

Proof: Let C be a compact subspace of a Hausdorff space X. We show that $X \backslash C$ is open. Pick $x \in X$. For each $c \in C$, since X is Hausdorff, we can find A_c, B_c such that $x \in A_c, c \in B_c$, and A_c, B_c are open and disjoint. Clearly $\{B_c\}_{c \in C}$ forms an open cover of C, so there exists a finite subcover $B_{c_1}, \ldots, B_{c_n}$ of C. Let $A = \bigcap_{k=1}^n A_{c_n}$, which is open. Then by construction, A is disjoint from each B_{c_k}, and thus is disjoint from C. This means A is a neighborhood of x contained in $X \backslash C$, and since x was arbitrary, the set is open, as desired. ■

Proposition (continuity preserves compactness): The image of a compact space under a continuous function is compact.

Proof: Let $f : X \to Y$ be continuous and X be compact. Let $\{A_\alpha\}$ be an open cover of $f(X)$ by sets open in Y. Then $\{f^{-1}(A_\alpha)\}$ forms an open cover of X by continuity. Thus $f^{-1}\left(A_{\alpha_1}\right), \ldots, f^{-1}\left(A_{\alpha_n}\right)$ forms a finite subcover of X, and thus $A_{\alpha_1}, \ldots, A_{\alpha_n}$ is a finite subcover of $f(X)$. ■

Proposition: Let $f : X \to Y$ be a bijective continuous function. If X is compact and Y is Hausdorff, then f is a homeomorphism.

Proof: If $A \subseteq X$ is closed, and we know it's compact, and by above, $f(A)$ is compact in Y. Thus $f(A)$ is closed in Y. Thus the inverse image of a closed subset of X under f^{-1} (which just becomes the image of the set) is closed, so f^{-1} is continuous, as desired. ■

Lemma (tube lemma): Consider the product space $X \times Y$, where Y is compact. If N is an open set of $X \times Y$ containing $\{x_0\} \times Y$, then there exists some tube $W \times Y$, where W is a neighborhood of x_0 in X, such that $W \times Y \subseteq N$.

Proof: At each point $(x_0, y) \in N$, since N is open, we can find $U_y \times V_y$ with U_y, V_y being open in X, Y such that $(x_0, y) \in U_y \times V_y$ and $U_y \times V_y \subseteq N$. Thus $\{U_y \times V_y\}_{y \in Y}$ form an open cover of $\{x_0\} \times Y$. This is clearly homeomorphic to Y, which is compact, so $\{x_0\} \times Y$ is a compact subspace of $X \times Y$. Thus we have a finite subcover $U_{y_1} \times V_{y_1}, \ldots, U_{y_k} \times V_{y_k}$. Let $W = \bigcap_{i=1}^k U_{y_i}$, which is open. Clearly $W \times Y$ contains $\{x_0\} \times Y$. Furthermore, the $U_{y_1} \times V_{y_1}, \ldots, U_{y_k} \times V_{y_k}$ form a cover of $W \times Y$, so it's a subset of N, as desired. ■

Proposition (finite product is compact): The finite product of compact spaces is compact.

Proof: We show this for two compact spaces, and the general result follows by induction. Suppose X and Y are compact spaces, and let $\mathcal{A}$ be an open covering of $X \times Y$. For a fixed $x \in X$, the set $\{x\} \times Y$ is compact (since it's homeomorphic to Y), so it has some finite subcover $A_1, \ldots, A_n$. Let $N = A_1 \cup \cdots \cup A_n$, which is open and contains $\{x\} \times Y$. Then by the tube lemma there exists $W_x \times Y$ inside N containing $\{x\} \times Y$ with W_x open in X. Furthermore, the $A_1, \ldots, A_n$ cover $W_x \times Y$.

Then $\{W_x\}_{x \in X}$ form an open cover for X, so there exists a finite subcover $W_{x_1}, \ldots, W_{x_n}$. Thus $W_{x_1} \times Y \cup \cdots \cup W_{x_n} Y$ covers $X \times Y$, and since each $W_{x_i} \times Y$ is covered by some finite number of sets in $\mathcal{A}$, there exists a finite subcover in $\mathcal{A}$ that cover $X \times Y$, as desired. ■

From this, we can generalize the tube lemma:

Proposition: Suppose $A \subseteq X$ and $B \subseteq Y$ are compact. Suppose $A \times B \subseteq N$, where N is open. Then there exist open $U \subseteq X, V \subseteq Y$ such that $A \times B \subseteq U \times V \subseteq N$.

Proof: First we show for each $a \in A$, there exists $U_a \times V_a$ with $U_a \subseteq X, V_a \subseteq Y$ both open such that $\{a\} \times B \subseteq U_a \times V_a \subseteq N$.

For each $(a, b) \in \{a\} \times B$, we can find $U_b \times V_b$ open in $X \times Y$ such that $(a, b) \in U_b \times Y_b \subseteq N$ (since N is open). Thus $\{U_b \times V_b\}_{b \in B}$ forms an open cover of $\{a\} \times B$. Since this is homeorphic to the compact set B, it's also compact, so there exists a finite subcover $U_{b_1} \times V_{b_1}, \cdots, U_{b_n} \times V_{b_n}$. Let $U_a = \bigcap_{k=1}^n U_{b_k}$ and $V_a = \bigcup_{k=1}^n V_{b_k}$, both of which are open. We claim that $U_a \times V_a$ satisfies the lemma. First suppose $(a, b) \in \{a\} \times B$. Clearly $a \in U_a$, and since $b \in V_{b_k}$ for some k, we see that $b \in V_a$. Thus $\{a\} \times B \subseteq U_a \times V_a$. Now suppose $(u, v) \in U_a \times V_a$. By construction, $u \in U_{b_i}$ for all $1 \leq i \leq n$. By construction, we also have that $v \in V_{b_j}$ for some $1 \leq j \leq n$. Thus $(u, v) \in U_{b_j} \times V_{b_j} \subseteq N$. Thus $U_a \times V_a \subseteq N$, as desired.

Using the lemma, we can find an open cover $\{U_a \times V_a\}_{a \in A}$ of $A \times B$ of sets that are contained in N. Then by the compactness of $A \times B$, there is some finite subcover $U_{a_1} \times V_{a_1}, \cdots, U_{a_n} \times V_{a_n}$. Let $U = \bigcup_{k=1}^n U_{a_k}$ and $V = \bigcap_{k=1}^n V_{a_k}$, both of which are open. We claim that $A \times B \subseteq U \times V \subseteq N$. The first contained holds since each V_{a_k} contains B, and the U_{a_k} cover A. Now suppose

$(u, v) \in U \times V$. Again by construction, $v \in V_{a_k}$ for all $1 \leq k \leq n$, and there's some $1 \leq j \leq n$ such that $u \in U_{a_j}$. Thus $(u, v) \in U_{a_j} \times V_{a_j} \subseteq N$. Thus $U \times V \subseteq N$, as desired. ■

Definition (finite intersection property): A collection $\mathcal{C}$ of subsets of X is said to have the *finite intersection property* if for every finite subcollection of $\mathcal{C}$, the intersection of the sets in the subcollection is nonempty.

Proposition: A topological space X is compact if and only if for every collection $\mathcal{C}$ of closed sets in X which have the finite intersection property, the intersection $\bigcap \mathcal{C}$ is nonempty.

Proof: This follows from just noticing the statement above is the contrapositive of the open covering definition of compactness. ■

2.2.1. Limit Point and Sequential Compactness

Definition (limit point compactness): A space X is said to be *limit point compact* if every infinite subset of X contains a limit point.

Definition (sequentially compact): Let X be a topological space. Then X is said to be *sequentially compact* if if every sequence in X has a convergent subsequence.

Proposition: Both compactness and sequential compactness imply limit point compactness.

Proof: First suppose X is compact. We show that if $A \subseteq X$ has no limit points, then A is finite. If A has no limit points, then it vacuously contains all of them, so A is closed. We also see that for each $a \in A$, we can find a neighborhood U_a such that $U_a \cap A = \{a\}$ (since there must exists some neighborhood of a that doesn't intersect A for a not to be a limit point). Then the U_a sets and $X \backslash A$ (which is open since A is closed) form an open cover. By compactness there is a finite subcover. Since $X \backslash A$ doesn't cover A, some finite set of the U_a must cover A, which implies there are only finitely many U_a.

Now suppose X is sequentially compact. Then for any infinite set A of X, we can enumerate some sequence in it. By sequential compactness, some subsequence converges to some point a. Then this point must be a limit point of A. ■

Example (limit point compact but not compact): Let $Y = \{a, b\}$ be a two point set with the indiscrete topology, and consider $X = \mathbb{N} \times Y$. Note that any nonempty set A in X is has a limit

point, by taking neighborhoods the point with first coordinate n, where either (n, a) or (n, b) are in X. However, the open sets $\{n\} \times Y$ for an open cover but have no finte subcover.

Example (compactness and sequential compactness don't contain each other): $[0,1]^{[0,1]}$ is compact by Tychonoff's theorem but not sequentially compact.

$[0, \omega_1)$ is sequentially compact but not compact, where ω_1 is the first uncountable ordinal.

(PROVE LATER?)

Proposition: Let X be metrizable. Then X is compact if and only if it's limit point compact if and only if it's sequentially compact.

Proof: See analysis notes. ■

2.2.2. Local Compactness and One Point Compactification

Definition (locally compact): A space X is said to be *locally compact at* x if there is some compact subspace C of x that contains a neighborhood of x. If X is locally compact at all points, then X is said to be *locally compact.*

Example: $\mathbb{R}$ is locally compact, just take $[x-1, x+1]$.

Definition ((one point) compactification): If Y is a compact Hausdorff space and X is a proper subspace of Y whose closure equals Y, then Y is said to be a *compactification* of X. If $Y \backslash X$ is a single point, then Y is said to be the *one point compactification* of X.

Theorem (one point compactification construction): Let X be a space. Then X is locally compact Hausdorff if and only if there exists a space Y satisfying the following conditions:
a) X is a topological subspace of Y.
b) The set $Y \backslash X$ consists of a single point.
c) Y is a compact Hausdorff space.

If Y and Y' are two spaces satisfying these conditions, then there is a homeomorphism of Y with Y' that equals the identity map on X.

Proof: First we show that this space is unique up to homeomorphism. Define $h : Y \to Y'$ by taking the single point $p \in Y \backslash X$ to the single point $q \in Y' \backslash X$, and let every other element map to itself. We show that h is an open map. Symmetry then implies the inverse map is open, and thus a homeomorphism.

If $p \notin U$ where U is open, then $h(U) = U$. Since U is open in Y and contained in X, it's open in X. Since Y' is Hausdorff, $\{p'\}$ is closed, and thus $Y \backslash \{p'\} = X$ is open. Thus U is open in Y'.

Now suppose $p \in U$. Since $C := Y \backslash U$ is closed Y and Y is compact, C is compact in Y. Since C is contained in X, it's also compact in X. Since X is a subspace of Y', C is also compact in Y'. Since Y' is Hausdorff, we see that C is closed in Y'. Thus $h(U) = U \backslash \{p\} \cup \{p'\} = Y' \backslash C$ is open.

Next, we show if Y satisfies the conditions, then X is locally compact Hausdorff. Since $X \subseteq Y$, we get that it's Hausdorff. Now suppose $x \in X$. Using Hausdorffness, we can choose open disjoint U, V containing x and the single point in $Y \backslash X$. The set $C := Y \backslash V$ is closed in Y, and thus compact in Y. Since C lies in X, it's also compact in X. Since U, V are disjoin, $U \subseteq C$. Thus C contains a neighborhood of x, and so X is locally compact at x.

Finally, suppose X is locally compact Hausdorff. We construct Y to satisfy the conditions. Let ∞ denote the symbol we add, and define $Y := X \cup \{\infty\}$. Topologize Y by defining the open sets to be either (1) open sets in X or (2) sets of the form $Y \backslash C$, where C is compact in X.

First we check this is a topology. $\emptyset$ is a set of the first type and Y is a set of the second type. For finite intersections, we just need to check if the intersection of two open sets is open. We have three cases:

$$\begin{aligned} U_1 \cap U_2 &\quad \text{is type (1)} \\ (Y \backslash C_1) \cap (Y \backslash C_2) = Y \backslash (C_1 \cup C_2) &\quad \text{is type (2)} \\ U \cap (Y \backslash C) = U \cap (X \backslash C) &\quad \text{is type (1).} \end{aligned}$$

For the union of open sets, we have

$$\begin{aligned} \bigcup U_\alpha = U &\quad \text{is type (1)} \\ \bigcup Y \backslash C_\beta = Y \backslash \bigcap C_\beta &\quad \text{is type (2)} \\ \left(\bigcup U_\alpha\right) \cup \left(\bigcup Y \backslash C_\beta\right) = U \cup Y \backslash C = Y \backslash (C \backslash U) &\quad \text{is type (1),} \end{aligned}$$

where the second case holds since the intersection of compact sets is compact, and the third case holds since $C \backslash U$ is closed in C, and thus compact.

Next we show X is a subspace of Y. If $U \subseteq Y$ is open and of type (1), then $U \cap X = U$. If it's of type (2), then $U = Y \backslash C$, and $U \cap X = X \backslash C$. In either case, both are open in X. In the other direction, if U is open in X, then it's an open set of type (1) in Y.

Next we show compactness. Let $\mathcal{A}$ be an open covering of Y. Since ∞ is covering $\mathcal{A}$ contains a set of the form $Y \backslash C$, where C is compact in X. Now look at the intersection of the remaining open sets with X. These then form an open cover of C, and from compactness we can thus recover a finite subcover. The corresponding elements in the original cover give us the finite subcover of Y, along with $Y \backslash C$.

Finally, we show Y is Hausdorff. Since X is Hausdorff, we only need to show that ∞ and a point $x \in X$ can be separated. By local compactness, there exists compact C containing x and a neighborhood U. Then U and $Y \backslash C$ are disjoint neighborhoods of x and ∞ in Y, as desired. ■

Compared to the local definitions of connectedness we gave earlier, the definition of local compactness truly local, since we don't require compact subsets at all scales. Here's a more local formulation that's equivalent in Hausdorff spaces derived by applying the one point compactification:

Proposition: Let X be a Hausdorff space. Then X is locally compact if and only if given x in X, and given a neighborhood U of x, there is a neighborhood V of x such that $\overline{V}$ is compact and $\overline{V} \subseteq U$.

Proof: The backwards direction is easy, since $x \in V \subseteq \overline{V}$ is a compact neighborhood of x. Now suppose X is locally compact. Then U is an open neighborhood of x in the one point compactification $\hat{X}$. Thus $\hat{X} \backslash U$ is closed in $\hat{X}$, and thus compact. Using basically the same proof method when showing that compact subsets of Hausdorff spaces are closed (or the normality of compact Hausdorff spaces; see the next chapter), we can find disjoint open W, V that contain $\hat{X} \backslash U$ and x. Since V is contained in the closed set $\hat{X} \backslash W \subseteq U$, we have $\overline{V} \subseteq U$. Since $\hat{X}$ is compact, $\overline{V}$ is compact as well. Thus $\overline{V}$ is compact in X, as desired. ■

Corollary: Let X be locally compact Hausdorff, and let A be a subspace of X. If A is closed in X or open in X, then A is locally compact.

Proof: Note that A is automatically Hausdorff. First suppose A is open, and pick $x \in A$. By the previous proposition, there exists a neighborhood V of x in X for which $\overline{V}$ is compact and $\overline{V} \subseteq A$. Thus $\overline{V}$ is a compact subset of A containing the neighborhood V of x, and thus A is locally compact at x.

Now suppose A is closed, and pick $x \in A$. Let C be a compact set of X containing some neighborhood U of x. Then $C \cap A$ is closed in C and thus compact in C. Suppose we have some open cover of $C \cap A$ by open sets in A. These open sets are thus open sets of X intersected with A that cover $C \cap A$. Now intersecting those open sets with C gives us an open cover of $C \cap A$ by open sets of C. By compactness, there exists a finite subcover. Using the corresponding open sets in A, we see the original cover has a finite subcover. Thus $C \cap A$ is compact in A.

Since we have $U \cap A$ is open in A, we see $x \in U \cap A \subseteq C \cap A$. Thus A is locally compact at x. ■

Corollary: A space X is homeomorphic to an open subspace of a compact Hausdorff space if and only if X is locally compact Hausdorff.

Proof: First suppose X is locally compact Hausdorff. Then X is homeomorphic to the corresponding open set in the one point compactification $\hat{X}$.

Now suppose X is homeomorphic to an open subspace of a compact Hausdorff space. By the previous proposition, that subspace, is locally compact. Clearly it's Hausdorff, so X must locally compact Hausdorff. ■

2.2.3. Tychonoff's Theorem

Lemma (maximal finite intersection collection): Let X be a set, and let $\mathcal{A}$ be a collection of subsets of X having the finite intersection property. Then there is a collection $\mathcal{M}$ of subsets of X such that $\mathcal{M}$ contains $\mathcal{A}$, $\mathcal{M}$ has the finite intersection property, and such that any collection of subsets of X that properly contians $\mathcal{M}$ does not have the finite intersection property.

Proof: Let $\mathfrak{C}$ denote the set of all collections of subsets of X with the finite intersection property containing $\mathcal{A}$, and partially order $\mathfrak{C}$ by inclusion. Let $\mathfrak{c}$ be a chain of elements in $\mathfrak{C}$. We show that this chain has an upper bound, and thus Zorn's lemma implies that there is a maximal collection $\mathcal{M}$ containing $\mathcal{A}$.

Let $\mathcal{U} := \bigcup_{\mathcal{C}\in\mathfrak{c}} \mathcal{C}$. Clearly $\mathcal{A} \subseteq \mathcal{C}$. We just need to show that $\mathcal{U}$ has the finite intersection property, as then it's clearly an upper bound for $\mathfrak{c}$. Let $C_1, \cdots, C_n \in \mathcal{U}$. Since this collection of sets is finite, the sets all belong to some $\mathcal{C} \in \mathfrak{C}$. Since $\mathcal{C}$ has the finite intersection property, $\cap_{k=1}^n C_k \neq \emptyset$. Thus $\mathcal{U}$ has the finite intersection property. ■

Lemma: Let X be a set, and let $\mathcal{M}$ be a collection of subsets of X that is maximal with respect to the finite intersection property.
a) Any finite intersection of elments of $\mathcal{M}$ is an element of $\mathcal{M}$.
b) If A a subset of X that intersects every element of $\mathcal{M}$, then A is an element of $\mathcal{M}$.

Proof:
a) Suppose otherwise. Let $C = \bigcap_{k=1}^n C_n$ be the intersections of sets in $\mathcal{M}$ that doesn't lie in $\mathcal{M}$, and consider $\mathcal{M} \cup \{C\}$. Then this collection has the finite intersection property, since for any $C_1', \cdots, C_m' \in \mathcal{M}$, we have

$$C \cap \bigcap_{j=1}^m c_j' = \bigcap_{k=1}^n C_k \cap \bigcap_{j=1}^m C_j' \neq \emptyset,$$

since this is a finite collection of $\mathcal{M}$. This contradicts the maximality of $\mathcal{M}$.
b) If not, then add A into the collection. Then for finitely many $C_i \in \mathcal{M}$, we have $\bigcap C_i \in \mathcal{M}$ by a), and thus $A \cap \bigcap C_i \in \mathcal{M} \cup \{A\}$. This again contradicts maximality.

■

Theorem (Tychonoff's theorem): The product of compact spaces is compact in the product topology.

Proof: Let

$$X = \prod_{\alpha\in J} X_\alpha,$$

where each X_α is compact. We prove X is compact using the finite intersection formulation of compactness. In particular, we show that if $\mathcal{A}$ is a collection of subsets of X with the finite intersection property, then

$$\bigcap_{A \in \mathcal{A}} \overline{A} \neq \emptyset.$$

Apply the first lemma to obtian a maximal collection $\mathcal{M} \supseteq \mathcal{A}$ with respect to the finite intersection property. It thus suffices to show that $\bigcap_{M \in \mathcal{M}} \overline{M} \neq \emptyset$.

Consider the collection

$$\{\pi_\alpha(M) : M \in \mathcal{M}\}.$$

Since $\mathcal{M}$ has the finite intersection property, this collection does as well. From the compactness of X_α, we can thus find $x_\alpha \in X_\alpha$ for which

$$x_\alpha \in \bigcap_{M \in \mathcal{M}} \overline{\pi_\alpha(M)}.$$

Let $x = (x_\alpha)_{\alpha \in J} \in X$. We show that $x \in \overline{M}$ for each $M \in \mathcal{M}$, and this will finish the proof. Thus fix some $M \in \mathcal{M}$.

First consider some subbasis element $\pi_\beta^{-1}(U_\beta)$, where U_β is a neighborhood of x_β. Then $\pi_\beta^{-1}(U_\beta)$ contains x. Now pick any $M \in \mathcal{M}$. Since $x_\beta \in \overline{\pi_\beta(M)}$ by construction, U_β intersects $\pi_\beta(M)$ at some point $\pi_\beta(y)$ with $y \in M$. Thus $y \in \pi_\beta^{-1}(U_\beta) \cap M$.

It follows from b) of the previous lemma that every subbasis element containing x belongs to $\mathcal{M}$. Then using a) of the previous lemma, it follows that every basis element containing x belongs to $\mathcal{M}$. Since $\mathcal{M}$ has the finite intersection property, every basis element containing x intersects every element of $\mathcal{M}$. In particular, every basis element containing x intersects M, so $x \in \overline{M}$, as desired. ■

2.3. Problems

Problem: Let $\{A_n\}$ be a sequence of connected subspaces of X such that $A_n \cap A_{n+1} \neq \emptyset$ for all n. Show that $\bigcup A_n$ is connected.

Proof: Since A_1 and A_2 share a point, their union is connected. Since A_2 and A_3 share a point, $A_1 \cup A_2$ and A_3 share a point, so $A_1 \cup A_2 \cup A_3$ is connected. Continuing in this fashion, we see that $E_n = A_1 \cup \cdots \cup A_n$ is connected for all n. Now note that all E_n share a point (namely some point in A_1), so their union, which is $\bigcup A_n$, is connected, as desired. ■

Problem: Let $\{A_\alpha\}$ be a sequence of connected subspaces of X, and let A be a connected subspace of X such that $A \cap A_\alpha \neq \emptyset$ for all α. Prove that $A \cup (\bigcup A_\alpha)$ is connected.

Proof: Suppose C, D is a separation of $A \cup (\bigcup A_\alpha)$. Since A is a connected subspace, it must lie in C or D. Without loss of generality, suppose it lies in C. Then we know that $A \cup A_\alpha$ is a connected subspace for all α. Since A lies in C, this implies that $A \cup A_\alpha$ lies in C for all α. However, this is a contradiction, since this implies D is empty. ■

Problem: Let $A \subset X$. Show that if C is a connected subspace of X that intersects both A and $X \backslash A$, then C intersects ∂A.

Proof: Suppose otherwise. Thus C intersects $(X \backslash A) \backslash \partial A = \operatorname{int}(X \backslash A)$, which is open. Similarly, C intersects $A \backslash \partial A = \operatorname{int}(A)$, which is open. Thus $C \cap \operatorname{int}(X \backslash A)$ and $C \cap \operatorname{int}(A)$ are disjoint open sets of C, and since C doesn't intersect ∂A, we see that their union equals C. Thus the two sets form a separation of C, contradiction. ■

Problem: A space is called *hyperconnected* of there exist no non-empty disjoint open sets.

a) Prove that a hyperconnected space is connected.
b) Prove that a space X is hyperconnected if and only if the closure of every non-empty open set is X.
c) Prove or disprove that a hyperconnected space is necessarily path connected.

Solution:

a) Suppose X is hyperconnected but not connected. Then there exists non-empty open U, V disjoint that form a separation of X. Thus there exists non-empty disjoint open sets, which contradicts X being hyperconnected.
b) Suppose X is hyperconnected, and let $U \subseteq X$ be an non-empty open set. Suppose $C \subseteq X$ is closed and $C \supseteq U$. If C is not equal to the whole space, then $X \backslash C$ is open and non-empty. Since C contains U, this set is also disjoint from U. Thus U and $X \backslash C$ are non-empty disjoint open sets, which contradicts X being hyperconnected. Thus C must be equal to the whole space. Thus the only closed set that contains U is X, so $\overline{U} = X$.

   Now suppose the closure of every non-empty open set is X. If U, V are non-empty disjoint open sets in X, then $X \backslash V$ is a closed set containing U that is not the whole space. Thus $\overline{U} \subseteq X \backslash V \neq X$, contradiction. Thus there cannot exist non-empty disjoint open sets in X, so X is hyperconnected.
c) We claim that $\mathbb{N}$ with the cofinite topology is a hyperconnected space that isn't path connected. First we show $\mathbb{N}$ is hyperconnected. If $U \subseteq \mathbb{N}$ is open, then any subset of $\mathbb{N} \backslash U$ is finite, and thus not open in $\mathbb{N}$, which means there can't exist non-empty disjoint open sets.

   Now we show that $\mathbb{N}$ isn't path connected. Let $m, n \in \mathbb{N}$ be distinct, and suppose for the sake of contradiction that $\gamma : [0, 1] \to \mathbb{N}$ such that $\gamma(0) = m, \gamma(1) = n$ was continuous.

   Lemma: There exists no closed interval $I \subseteq [0, 1]$ such that $\gamma(I)$ is a finite subset of $\mathbb{N}$.

   Proof: Suppose otherwise, and let $S = \gamma(I)$. Restricting γ to I still gives us a continuous function. Pick some $k \in S$, and let $A = (\gamma|_I)^{-1}(\{k\}), B = (\gamma|_I)^{-1}(S \backslash \{k\})$. Since $\{k\}$ and $S \backslash \{k\}$ are finite, they're closed sets in $\mathbb{N}$ with the cofinite topology. Thus A, B are closed in I. Furthermore, A, B are non-empty and disjoint by construction. Thus $A \cup B = I$ is a disjoint union of non-empty closed subsets of I. Thus $B = I \backslash A$ and $A = I \backslash B$ are open sets, and thus nontrivial clopen sets, which implies I is disconnected, contradiction. ■

Now let $C_k := \gamma^{-1}(\{k\})$ for $k \in \mathbb{N}$. Since γ is continuous and $\{k\}$ is closed in $\mathbb{N}$, these sets are closed in $[0, 1]$. We show that they have empty interior. Suppose C_k had non-empty interior. Thus there exists some open interval $(a, b) \subseteq C_k$. Thus

$$\left[a + \frac{b-a}{4}, b - \frac{b-a}{4}\right] \subseteq C_k.$$

Letting this interval be I, we see that $\gamma(I) = \{k\}$. However, this contradicts the lemma. Thus C_k has empty interior.

Thus $(C_k)_{k\in\mathbb{N}}$ is a countable collection of closed sets with empty interior, and by definition, we see that

$$\bigcup_{k\in\mathbb{N}} C_k = [0, 1].$$

However, this contradicsts the Baire category theorem, since $[0, 1]$ is a complete metric space. Thus γ can't be continuous, so there exists no path between m, n, and so $\mathbb{N}$ with the cofinite topology is not path connected.

Problem: What are the components and path components of $\mathbb{R}_\ell$? What are the continuous maps $f : \mathbb{R} \to \mathbb{R}_\ell$?

Solution: We claim that the components are singletons, which implies the path components are singletons as well by the fact that path components lie in components. Suppose $A \subseteq \mathbb{R}_\ell$ has more than two points, say a, b with $a < b$. Then $\left(-\infty, \frac{a+b}{2}\right)$ and $\left[\frac{a+b}{2}, \infty\right)$ for a separation of A, so a and b don't lie in the same component.

Since $f(\mathbb{R})$ must be connected in $\mathbb{R}_\ell$, we see the only continuous maps are constant maps.

Problem: Show that the ordered square I_o^2 is connected and locally connected, but not path connected nor locally path connected. Then find the path components of I_o^2.

Solution: First we show I_o^2 is a linear continuum. If $(a, b) < (c, d)$, then the point $\left(\frac{a+c}{2}, \frac{b+d}{2}\right)$ lies between them so that condition is satisfied. Let $S \subseteq I_o^2$ be nonemtpy. We need to show that S has a least upper bound. Let

$$x := \sup\{a : (\{a\} \times I) \cap S \neq \emptyset\}.$$

This exists since S is nonempty and a is bounded above

Then let

$$y := \sup(\{b : (x, b) \in S\} \cup \{0\}),$$

which is also well defined for the same reasons as above. It's easy to see that (x, y) is an upper bound for S. Now suppose (x', y') is a smaller upper bound. First suppose $x' < x$. This is impossible, since

that would imply x' is an upper bound for $\{a : (\{a\} \times I) \cap S \neq \emptyset\}$. Now suppose $x' = x$ and $y' < y$. Again for the same reason, this is impossible. Thus (x, y) is the least upper bound.

Since I_o^2 is a linear continuum, the space is connected. Local path connectedness follows by taking a closed interval containing the point inside the neighborhood and using the same proof to show it's a linear continuum.

Now we show I_o^2 isn't path connected. Suppose $\gamma : [0, 1] \to I_o^2$ is a path from $(0, 0)$ to $(1, 1)$. By the intermediate value theorem, γ is surjective. For each $r \in [0, 1]$, the interval $\{r\} \times (0, 1)$ is open in I_o^2, so $\gamma^{-1}(\{r\} \times (0, 1))$ is open in $[0, 1]$, and thus contains some rational. Since the $\{r\} \times (0, 1)$ sets are disconnected, there preimages are disconnected as well. Since each preimage contains a rational, there is an injective map from $[0, 1]$ to $\mathbb{Q} \cap [0, 1]$ given by taking r to some $q \in \mathbb{Q} \cap \gamma^{-1}(\{r\} \times (0, 1))$, which is impossible. Thus such a path can't exist.

We show that I_o^2 isn't locally path connected at $\left(\frac{1}{2}, 1\right)$. Indeed, any neighborhood of the point must contain a closed interval of the form $\left[\left(\frac{1}{2}, 1 - \varepsilon\right), \left(\frac{1}{2} + \varepsilon', c\right)\right]$. Using the same proof as above, we can show that the endpoints aren't path connected. Thus we see that any points with different x values aren't path connected. Then it's easy to see that $\{\{r\} \times I : r \in [0, 1]\}$ are the path components.

Problem: Show that if $f : X \to Y$ is continuous, where X is compact and Y is Hausdorff, then f is a closed map.

Solution: If $C \subseteq X$ is closed, then it's compact, and so $f(X) \subseteq Y$ is compact, and thus closed.

Problem: Show that if Y is compact, then the projection $\pi_1 : X \times Y \to X$ is a closed map.

Solution: Let $C \subseteq X \times Y$ be closed, and consider it's complement $X \times Y \backslash C$, which is open. From the tube lemma, we know that for each $x \in X$ for which $\{x\} \times Y \subseteq X \times Y \backslash C$, there exists open $U_x \subseteq X$ such that $\{x\} \times Y \subseteq U_x \times Y \subseteq X \times Y \backslash C$. Let V be the union of the U_x for all such X. Thus $V \times Y$ is open in $X \times Y$. We claim $\pi_1(C) = X \backslash V$, which is closed.

Suppose $x \in \pi_1(C)$. Thus $\{x\} \times Y$ is not contained in $X \times Y \backslash C$. Thus $x \notin V$, since otherwise it would be contained in some $U_{x'}$, and thus we could take a smaller open set containing it, which would implies $\{x\} \times Y \subseteq X \times Y \backslash C$.

Now suppose $x \in X \backslash V$. By the same logic, x can't be contained in some $U_{x'}$. Thus there must exist $(x, y) \in C$, so $x \in \pi_1(C)$.

Problem: Dini's theorem

Problem: A space X is said to be countably compact if every countable open covering of X contains a finite subcollection that covers X. Show that for a T_1 space X, countable compactness is equivalent to limit point compactness.

Solution: First suppose X is not limit point compact. Then there exists an infinite subset A that contains not limit points. Enumerate some sequence $(x_n) \in A$. Since no points in A are limit points, there exists open U_n that's a neighborhood of x_n for which $U_n \cap A = \{x_n\}$. Since A has no limit points, it vacuously contains all of them, and thus is closed. In particular, the U_i and $X \backslash A$ form countable cover of X. However, there doesn't exist a finite subcover, since no U_i contains an infinite number of points of A.

Now suppose X is not countable compact. Thus there exists $\{U_i\}_{i \in \mathbb{N}}$ which has no finite subcover. Pick $x_i \notin U_1 \cup \cdots \cup U_i$. We claim $A = \{x_1, x_2, \ldots\}$ doesn't contain any of its limit points. Fix x_i, and let n be the smallest integer for which $x_i \in A_n := U_1 \cup \cdots \cup U_n$. Thus $x_j \notin A_n$ for $j \geq n$. The set $A_n \cap \bigcap_{j=1}^{n-1} X \backslash \{x_n\}$ is open, and by construction only intersects A at x_n.

3. Countability and Separation Axioms

3.1. Countability Axioms

Definition (first-countable): A space X is said to have a *countable basis at* x if there is a countable collection $\mathcal{B}$ of neighborhoods of x such that each neighborhood of x contains at least one of the elements of $\mathcal{B}$. A space that has a countable basis at each of its points is said to satisfy the *first countability axiom*, or to be *first-countable*.

Proposition (metric spaces are first-countable): Every metric space is first-countable.

Proof: Let M be a metric space, and fix arbitrary $x \in M$. Consider the collection $\left\{B_{1/n}(x)\right\}_{n\in\mathbb{N}}$, and let U be any neighborhood of x. Since we're in a metric space, this implies that there exists a ball of radius r centered at x contained in U. Picking $n \in \mathbb{N}$ such that $\frac{1}{n} < r$, we see that U contains $B_{1/n}(x)$. Thus U contains an element of the collection, so M is first-countable at x. Since x was arbitrary, x is first-countable. ■

Proposition: Let X be a space.

a) Let A be a subset of X. If there is a sequence of points of A convering to x, then $x \in \overline{A}$, and the converse holds of X is first-countable.
b) Let $f : X \to Y$ be a function. If f is continuous, then for every convergent sequence $x_n \to x$ in X, the sequence $f(x_n)$ converges to $f(x)$. The converse holds of X is first-countable.

Proof:

a) The forward direction is obvious (any neighborhood of x must contain a tail of the sequence, and thus intersects A). Now suppose X is first-countable, and for $x \in \overline{A}$, let $\mathcal{B}$ be a countable basis at x. Order these sets $B_1', B_2', \ldots$. Let $B_i = \bigcap_{k=1}^{i} B_k'$, which are open. Note that this is also a countable basis at x, since if any neighborhood of x contain B_k', it also will contain B_k. Since each is a neighborhood of x, we also see that they intersect A at some point x_n. We claim that the sequence (x_n) converges to x. Indeed, let U be any neighborhood of x. Then U contains some B_n. Then by construction, $B_n \supseteq B_{n+1} \supseteq \cdots$. Each set contains $x_n, x_{n+1}, \ldots$, and so U does as well. This works for any neighborhood of x, so indeed $x_n \to x$.

b) We already proved the forward direction, so suppose X is first countable, and that for every $x_n \to x$, we have $f(x_n) \to f(x)$. To show f is continuous, we show that for any $A \subseteq X$ we have $f\left(\overline{A}\right) \subseteq \overline{f(A)}$. Pick $x \in \overline{A}$. By a), there exists a sequence $(x_n) \in A$ that converges to x. By hypothesis, $f(x_n) \to f(x)$. Since each point in the image of the sequence will lie in A, by a) again, we see that $f(x) \in \overline{f(A)}$, as desired.

■

Definition (second-countable): If a space X has a countable basis for its topology, then X is said to satisfy the *second countability axiom*, or to be *second-countable*.

It's easy to see that second-countability implies first-countability.

Example (first-countable but not second-countable): Since $\mathbb{R}^\omega$ in the uniform topology is a metric space, it's first countable. We show that it's not second-countable. First suppose a space X has a countable basis $\mathcal{B}$. Let A be a discrete subspace of X. For each a, we can choose a basis element B_a that intersects A at only a. If $a \neq b$, then $B_a \neq B_b$ by construction. Thus the map $a \to B_a$ is an injection of A into $\mathcal{B}$, so A must be countable.

Now consider the subspace A of $\mathbb{R}^\omega$ consisting of all only 0's and 1's. It has the discrete topology since $\overline{\rho}(a, b) = 1$ for any distinct points in that subspace. Thus $\mathbb{R}^\omega$ can't have a countable basis.

Proposition: A subspace of a first-countable space is first-countable and the countable product of first-countable spaces is first-countable. A subspace of second-countable space is second-countable, and a countable product of second-countable spaces is second-countable.

Proof: Let $A \subseteq X$ be a subspace of a first-countable space, and suppose $x \in A$. Then x has a countable basis in X. Intersecting the elements of this basis with A, we get a countable basis for x in A. Clearly a subspace of a second-countable space is second-countable.

Now suppose $X_1, X_2, \ldots$ be first-countable, and pick points $x_1, x_2, \ldots$. Then there exist countable bases $\mathcal{B}_1, \mathcal{B}_2, \ldots$ at each of these points. Let

$$\mathcal{B} := \left\{ \prod_{i=1}^{\infty} B_i : B_i \in \mathcal{B}_i, B_i \neq X_i \text{ for finitely many } i \right\}.$$

This the countable union of sets that are the finite, and so $\mathcal{B}$ is finite. Let U be an open set in $\prod X_i$ containing $x = (x_1, x_2, \ldots)$. Then there exists $U_1 \times U_2 \times \cdots$, where this is a basis set in the product topology, containing x. Thus finitely many of these are not X_i, and each contains x_i. Thus we can pick $B_i \subseteq U_i$ for those finitely many, and the rest of the B_i can be X_i. Thus $\prod X_i$ is first-countable.

Using the same method, we see that the countable product of second-countable spaces is second-countable. By construction, it is the basis for the product topology. ■

Definition (separable): If a space X has a countable dense set, then X is said to be *separable*.

Definition (Lindelof): Suppose every open cover of a space X has a countable subcover. Then X is said to be a *Lindelof space*.

These defintions are also countability axioms (even if the names don't make them seem like it). Note also that compactness implies Lindelof, but not the other way around (take $\mathbb{R}$ for example).

Proposition: If X is second-countable, then it's separable and Lindelof.

Proof: Let $\mathcal{B}$ be a countable basis for a space X. For each set in $\mathcal{B}$, pick a point x and let A be a collection of all such points. We claim that A is dense in X. Let U be open in X. Then it contains some basis element B. Thus U contains some point x in A by construction. This holds for every open set U, so $\overline{A} = X$.

Now let $\{B_\alpha\}$ be an open cover of X. Define a map $f : \{B_\alpha\} \to \mathcal{B}$ by mapping B_α to some basis set contained in B_α. Clearly $\mathcal{B}$ is an open cover of X, so taking a single set from the preimage of each basis element, we get a countable subcover. ■

Proposition: Let X be a metrizable space.
a) X is second-countable if and only if X is separable.
b) X is second-countable if and only if X is Lindelof.

Proof: We already showed the forward direction unconditionally, so suppose X is metrizable with metric d.

a) Suppose X is separable. Let A be a countable dense subset of X. Define

$$\mathcal{B} := \left\{B_{1/n}(a) : n \in \mathbb{N}, a \in A\right\},$$

which is a countable set. We claim this is a basis for the metric topology X. First we check that it covers $\mathcal{B}$. Pick $x \in X$, and consider $B_{1/2}(x)$. This intersects A at some point a, so $d(x, a) \leq \frac{1}{2}$. Thus $B_1(a) \in \mathcal{B}$ contains x. Thus $\mathcal{B}$ covers X.

Now we show the intersection of two basis elements contains a basis element at all points. Consider two elements $B_{1/m}(a), B_{1/n}(b)$ of $\mathcal{B}$, and pick $x \in B_{1/m}(a) \cap B_{1/n}(b)$. If the intersection is empty, we're done. Otherwise, we can construct a small enough ball around x using the triangle inequality to be contained in the intersection, then pick integer k such that $\frac{1}{k}$ is less than the radius of that ball.

Now we show this basis generates the metric topology. Let $\mathcal{T}_\mathcal{B}$ be the topology generated by $\mathcal{B}$, and let X denote it with the metric topology. Trivially the metric topology is finer than $\mathcal{T}_\mathcal{B}$, so we just need to show that $\mathcal{T}_\mathcal{B}$ is finer than X. Let $B_r(x)$ be a ball in X, and pick $y \in B_r(x)$. Let $\varepsilon = \frac{1}{8}(r - d(x, y))$, and pick $n \in \mathbb{N}$ such that $\frac{1}{n} < \varepsilon$. Then $B_{1/n}(y)$ must intersect A at some point a. Since $d(a, y) < \frac{1}{n}$, we see that the $B_{1/n}(a) \in \mathcal{B}$ contains y. Furthermore, for $b \in B_{1/n}(a)$, we have $d(b, x) \leq d(b, a) + d(a, y) + d(y, x) < \frac{1}{n} + \frac{1}{n} + d(x, y) < \frac{1}{4}(r - d(x, y)) + d(x, y) < r$. Thus $B_{1/n}(a) \subseteq B_r(x)$, so $B_r(x)$ can be written as the union of elements of $\mathcal{B}$. Since $B_r(x)$ was arbitrary, we're done.

b) Now suppose X is Lindelof. The collection

$$\mathcal{B}'_n = \left\{B_{1/n}(x) : x \in X\right\}$$

is clearly an open cover for all $n \in \mathbb{N}$. Thus each has a countable subcover $\mathcal{B}'_n$. Let

$$\mathcal{B} = \bigcup_{n \in \mathbb{N}} \mathcal{B}'_n.$$

This is clearly countable, and we claim this is a basis for the metric topology on X. By construction is covers X, so next we check that it's a basis. Let $A, B \in \mathcal{B}$, and pick arbitrary $x \in A \cap B$. Let r_1 be the size of A and r_2 the size of B. Let $\varepsilon = \min(r_1 - d(x, A_{\text{center}}), r_2 - d(x, B_{\text{center}}))$. Pick $n \in \mathbb{N}$ such that $\frac{1}{n} < \frac{\varepsilon}{4}$. Then by construction, there exists $C \in \mathcal{B}'_n$ such that $x \in C$. For any $y \in C$, we have $d(y, A_{\text{center}}) \leq d(y, x) + d(x, A_{\text{center}}) < \frac{\varepsilon}{4} + d(x, A_{\text{center}}) < r_1$. Similarly, we can show that $d(y, B_{\text{center}}) < r_2$. Thus $C \subseteq A \cap B$, so $\mathcal{B}$ is a basis.

Now we just need to check $\mathcal{B}$ generates the metric topology (since trivially the metric topology generates $\mathcal{T}_{\mathcal{B}}$). Pick arbitrary $B_r(x)$ and $y \in B_r(x)$. Let $\varepsilon = r - d(x, y)$, and pick $n \in \mathbb{N}$ such that $\frac{1}{n} < \frac{\varepsilon}{4}$. Again by construction, there exists $C \in \mathcal{B}_n$ containing y. Using the same argument as before, we can show that $C \subseteq B_r(x)$, so we're done.

■

Example: We show that $\mathbb{R}_\ell$ satisfies all countability axioms except the second. Clearly $\mathbb{Q}$ is a countable dense set, so it's separable. The collection $\left\{\left[x, x + \frac{1}{n}\right)\right\}_{n \in \mathbb{N}}$ is clearly a basis at x, so it's first countable.

Suppose $\mathcal{B}$ is a basis for $\mathbb{R}_\ell$. Thus for each x, there exist $B_x \in \mathcal{B}$ with $x \in B_x \subseteq [x, x+1)$. From this, we see that $x \neq y$ implies $B_x \neq B_y$, sine the minimums of the two sets are x and y. Thus $x \mapsto B_x$ is an injection into $\mathcal{B}$, implying it's uncountable.

To show that it's Lindelof, it suffices to show that every cover by basis elements has a countable subcover (since we can write every cover as the union of basis elements, then we extract a countable subcover, then for each of these sets we pick a set in the original cover than contains it, giving us a countable subcover). Let $\mathcal{A} = \{[a_\alpha, b_\alpha)\}$ be cover, and let $C = \bigcup_\alpha (a_\alpha, b_\alpha)$. Suppose $x \in \mathbb{R} \backslash C$. Thus it must be equal to some a_β. Now choose rational q_x in (a_β, b_β). Thus the interval $(x, q_x) \subseteq C$. It then follows that if $x, y \in \mathbb{R} \backslash C$ with $x < y$, we must have $q_x < q_y$ (since otherwise $x < y < q_y \leq q_x$ implies $y \in (x, q_x) \subseteq C$). Thus $x \mapsto q_x$ is an injection from $\mathbb{R} \backslash C$ into $\mathbb{Q}$, so $\mathbb{R} \backslash C$ is countable. Thus we can pick a countable subcover of it from $\mathcal{A}$, say $\mathcal{A}'$.

Now consider C as a subset of $\mathbb{R}$ with the standard topology. Since $\mathbb{R}$ is second-countable, so is C, and thus from the set (a_α, b_α), we can find some countable cover. Then let $\mathcal{A}''$ be this cover but with the left endpoints included. Then $\mathcal{A}' \cup \mathcal{A}''$ forms a countable cover of $\mathbb{R}_\ell$, as desired.

Example (product of Lindelof spaces isn't Lindelof): Consider $\mathbb{R}_\ell^2$ and the subset $\Delta^- = \{(x, -x) : x \in \mathbb{R}\}$. Consider a cover of Δ^- with the set $[x, x+1) \times [-x, -x+1)$, which intersects Δ^- only at $(x, -x)$. Since Δ^- is closed, $\mathbb{R}_\ell^2 \backslash \Delta^-$ is open, so we can cover it by open sets contained in itself such that they stay open in $\mathbb{R}_\ell^2$. We thus have a cover of $\mathbb{R}_\ell^2$. However, there's no countable subcover, since the only set that contains $(x, -x)$ for each $x \in \mathbb{R}$ is $[x, x+1) \times [-x, -x+1)$.

Example (subspace of Lindelof spaces isn't Lindelof): Consider the topology on $\mathbb{R}$ that declares open sets to be subsets that don't contain 0 and $\mathbb{R}$ itself. Any open cover of this space thus must contain $\mathbb{R}$, so this space is compact. Now consider the subspace $\mathbb{R}\backslash\{0\}$. This subspace inherits the discrete topology, and thus is not Lindelof.

Example (subspace of separable space isn't separable): $\mathbb{R}^2_\ell$ is separable, with a countable dense set given by $\mathbb{Q}^2_\ell$. However, the subspace Δ^- is uncountable discrete, and thus not separable.

Proposition (products of separable spaces): The countable product of separable spaces is separable.

Proof: Let $X_1, X_2, \ldots$ be separable spaces, and let $A_1, A_2, \ldots$ be countable dense spaces for each. Fix some $b_i \in A_i$. Let $X = \prod_{n=1}^{\infty} X_n$ and

$$A := \{(a_1, a_2, \ldots) : a_i \in A_i \text{ and } a_i \neq b_i \text{ for finitely many } i\}.$$

It's easy to see A is countable, so we just need to show it's dense. Let $U_1 \times U_2 \times \cdots$ be a basis set in X. Thus $U_i \neq X_i$ for some finite set K. For $i \in \mathbb{N}\backslash K$, pick $b_i \in U_i$. For $i \in K$, pick $a_i \in X_i$ such that $a_i \in U_i$ (which exists by separability). Thus the basis set intersects A. ■

It actually turns out that if you have continuum many separable spaces, their product is also separable.

3.2. Separation Axioms

We first lay out all relevant definitions for the separation axioms.

Definition (separation adjectives):

a) A space X is said to be T_0 or *Kolmogorov* if for any pair of distinct points $x, y \in X$, there exists an open set containing one but not the other.
b) A space X is said to be T_1 or *Frechet* if for any pair of distinct points $x, y \in X$, there exist open U, V such that $x \in U$ but y isn't and $y \in V$ but x isn't.
c) A space X is said to be T_2 or *Hausdorff* if for any pair of distinct points $x, y \in X$, there exist disjoint open sets U, V such that $x \in U, y \in V$.
d) A space X is said to be *regular* if for a closed set C and point $x \in X$ not in C, there exist disjoint open sets U, V such that $x \in U, C \subseteq V$.
e) A space X is said to be *normal* if for disjoint closed sets C, D, there exist disjoint open sets U, V such that $C \subseteq U, D \subseteq V$.

It's easy to see that $T_2 \Rightarrow T_1 \Rightarrow T_0$.

Example (containments of $T_{0,1,2}$ are strict): Consider a two point set $X = \{0, 1\}$. Let $\emptyset, \{0\}, X$ be the open sets. There are only two distinct points, and the open set $U = \{0\}$ contains only one and not the other. Thus is T_0. However, it's not T_1, since the only open set that contains 1, which is X, also contains 0.

Now consider $\mathbb{N}_{\text{cofinite}}$. Given $n \neq m$, the sets $\mathbb{N}\backslash\{n\}$ and $\mathbb{N}\backslash\{m\}$ are open and contain one point and not the other. However, it's not T_2 since every nonempty open set can miss at most a finite set of points, and thus any two nonempty open sets must intersect.

Proposition (T_1 equivalent formulation): A space X is T_1 if and only if singleton sets are closed.

Proof: First suppose X is T_1. We show that $X\backslash\{x\}$ is open for any $x \in X$. For $y \in X\backslash\{x\}$, since X is T_1, we can find a neighborhood U_y of y that doesn't contain x. Thus $U_y \subseteq X\backslash\{x\}$. Unioning U_y over all $y \in Y$ yields that $X\backslash\{x\}$ is open.

Now suppose singletons are closed, and suppose $x \neq y$. Then $X\backslash\{x\}$ is an open set containing y by not x, and similarly for $X\backslash\{y\}$. ■

Proposition: Suppose X is regular. Then X is T_0 if and only if it's T_1 if and only if it's T_2.

Proof: We just need to show $T_0 \Rightarrow T_2$. Let $a \neq b$ be points in X. By T_0, without loss of generality there exists an open set U containing a but not b. Thus $X\backslash U$ is a closed set containing b but not a. Then since X is regular, there exist disjoint open sets V, W containing a and $X\backslash U$, and thus b. ■

Unfortunately, regular and normal spaces don't behave as well. Regular spaces aren't necessarily Hausdorff, since one point sets need not be closed. For the same reason normal spaces aren't necessarily Hausdorff. In fact, the situation is even worse, for the indiscrete topology is vacuously regular and normal, but not even T_0. To remedy this, we defined the following:

Definition (separation axioms):
a) A space X is said to be T_3 if it's regular and T_1.
b) A space X is said to be T_4 if it's normal and T_1.

From the above proposition, T_1 is interchangeable with T_0 and T_2. It's now easy to see that $T_4 \Rightarrow T_3 \Rightarrow T_2 \Rightarrow T_1 \Rightarrow T_0$. We also have the following reformulations of regular and normal:

Proposition:
a) X is regular if and only if given a point x of X and neighborhood U of x, there is a neighborhood V of x such that $\overline{V} \subseteq U$.
b) X is normal if and only if given a closed set A and an open set U containing A, there is an open set V containing A such that $\overline{V} \subseteq U$.

Proof:

a) First suppose X is regular. Let $x \in X$ and suppose U is a neighborhood of X. Then $X \backslash U$ is closed and disjoint from x, so there exists disjoint closed sets V, V' such that $x \in V$ and $X \backslash U \subseteq V'$. The second containment implies $X \backslash V' \subseteq U$, and since V, V' are disjoint, we have $V \subseteq X \backslash V' \subseteq U$. Since $X \backslash V'$ is closed, we thus have $\overline{V} \subseteq X \backslash V' \subseteq U$, as desired.

   Now we show the other direction. Pick $x \in X$ and let C be a closed set not containing x. Then $X \backslash C$ is a neighborhood of x, so there exists a neighborhood V of x such that $\overline{V} \subseteq X \backslash C$. This then implies $C \subseteq X \backslash \overline{V}$, so C is contained in an open set. Since V is clearly disjoint from $X \backslash \overline{V}$, and since $x \in V$, we see that X is normal.

b) First suppose X is normal. Let $A \subseteq X$ be closed and suppose $U \supseteq A$ is open. Then $X \backslash U$ is a closed set disjoint from A. Since X is normal, there exist disjoint open sets V, V' such that $A \subseteq V, X \backslash U \subseteq V'$. The second containment implies $X \backslash V' \subseteq U$. Since V and V' are disjoint, this implies $V \subseteq X \backslash V' \subseteq U$. Note that since V' is open, $X \backslash V'$ is closed. Thus by the minimality of the closure, $\overline{V} \subseteq X \backslash V' \subseteq U$, as desired.

   Now we show the other direction. Let $A, B \subseteq X$ be two disjoint closed subsets. Then $X \backslash B$ is open and contains A, so by hypothesis, there exists an open set V containing A such that $\overline{V} \subseteq X \backslash B$. Thus $X \backslash \overline{V}$ is open and it's disjoint from V. Since $\overline{V}$ is disjoint from B (it's contained in $X \backslash B$), we see that $X \backslash \overline{V}$ contains B. Thus $V, X \backslash \overline{V}$ are disjoint open sets that contain A, B respectively. Since A, B were arbitrary, we see that X is normal, as desired.

■

Proposition:
a) A subspace of a regular space is regular.
b) A product of regular spaces is regular.

This tells us that the subspaces and products of T_3 spaces are also T_3.

Proof:

a) Let $Y \subseteq X$ be spaces and suppose X is regular. Suppose $y \in Y$ and $C \subseteq Y$ is closed in Y. Thus there exists $C' \subseteq X$ that's closed in X such that $C = Y \cap C'$. Since X is regular, there exist disjoint open sets A, B in X such that $y \in A$ and $C' \subseteq B$. Thus $A' = Y \cap A$ and $B' = Y \cap B$ are disjoint open sets in Y that contain y and C respectively. Thus Y is regular.

b) We use the reformulation of T_3. Let $\{X_\alpha\}_{\alpha \in J}$ be a collection of regular spaces, and let $X = \prod_{\alpha \in J} X_\alpha$. Pick $x = (x_\alpha)_{\alpha \in J}$ and a neighborhood U of x. Since U is open, we can pick a basis element $\prod_{\alpha \in J} U_\alpha$ of the product topology in U containing x. Since each of X_α is regular, and $x_\alpha \in U_\alpha$, we can find a neighborhood V_α of x such that $\overline{V_\alpha} = U_\alpha$ (for the U_α that are the

whole space, we take $V_\alpha = X_\alpha$). Then $\prod_{\alpha \in J} V_\alpha$ is a neighborhood (since only finitely many of the terms will be not the whole space by construction) of x and $\overline{\prod_{\alpha \in J} V_\alpha} = \prod_{\alpha \in J} \overline{V_\alpha} \subseteq \prod_{\alpha \in J} U_\alpha \subseteq U$. Thus X is regular.

■

Example (Hausdorff space that isn't regular): Let $\mathbb{H}$ be the upper half-plane with the subspace topology. Let $L \subseteq \mathbb{R}^2$ be the x-axis. Define a topology on $\mathbb{H} \cup L$ to be generated by the basis consisting of open sets in $\mathbb{H}$ as well as sets $D_\varepsilon(x)$ of the form $D_\varepsilon(x) := (B_\varepsilon(x) \cap \mathbb{H}) \cup \{x\}$, where $B_\varepsilon(x)$ is the open ball of radius $\varepsilon > 0$ in $\mathbb{R}^2$ and $x \in L$. Denote the resulting space by $\mathcal{X}$. It's easy to check that $\mathcal{X}$ is Hausdorff. We claim that it's not T_3.

Note that $C = \left(\left[0, \frac{1}{2}\right) \cup \left(\frac{1}{2}, 1\right]\right) \times \{0\}$ is closed $\mathcal{X}$. This follows by showing that $(\mathbb{H} \cup L) \backslash C$ is open. It's easy to see the points in $\mathbb{H}$ are interior points. For points on L, we just take a basis set, and those sets doesn't intersect any point of L. Thus the set is open. Now we claim that the point $\left(\frac{1}{2}, 0\right)$ and C can't be separated. Indeed, given any basis element of $\left(\frac{1}{2}, 0\right)$, it has some positive radius ε. Then the point $\left(\frac{1}{2} + \varepsilon, 0\right)$ has basis elements with positive radius. Thus these two open sets must intersect. Since any open set containing C must contain $\left(\frac{1}{2} + \varepsilon, 0\right)$, C and the point can't be separated.

Proposition:
a) Every Lindelof regular space is normal.
b) Every metrizable space is normal.
c) Every compact Hausdorff space is normal.

Proof:

a) Let X be regular and Lindelof, and let A, B be two closed subsets of X. For each point in A, since X is regular, we can find a neighborhood U of x not intersecting B. Again by regularity, we can find a neighborhood U_x of x such that $\overline{U_x} \subseteq U$. Since X is Lindelof, and since the U_x cover A, there exists a countable cover $U_{x_1}, U_{x_2}, \ldots$.

   We can do the same for B, obtaining a collection $V_{y_1}, V_{y_2}, \ldots$ that covers B for which $\overline{V_{y_i}}$ is disjoint from A. Then $U := \bigcup U_{x_i}$ and $V := \bigcup V_{x_i}$ cover A, B. Now define

$$U_n' := U_n \backslash \bigcup_{i=1}^{n} \overline{V_i} \quad \text{and} \quad V_n' := V_n \backslash \bigcup_{i=1}^{n} \overline{U_i}.$$

   These are open sets. Note that $\{U_n'\}$ covers A, since by construction no $x \in A$ is contained in $\overline{V_i}$. Similarly, $\{V_n'\}$ covers B. We claim every pairing of sets from each is disjoint. If $x \in U_i' \cap V_j'$ with $i \leq j$, then $x \in U_i, V_j$, but not in $\overline{U_k}$ for any $k \leq j$, which contradicts it being in U_i. Similar analysis works for $i > j$. Thus, letting U' be the union of the U_n's, and similarly defining V', we have a separation of A and B.

b) Let d be a metric on X, and let C, D be disjoint closed subsets. For each point $c \in C$, we can find $B(c, \varepsilon_c)$ that doesn't intersect D (if every such set did, then c would be in the closure of D, and thus D, which is a contradiction). Similarly, we can find $B(d, \varepsilon_d)$ for each point $d \in D$ such that it doesn't intersect C.

Now define

$$U = \bigcup_{c \in C} B\left(c, \frac{\varepsilon_c}{2}\right) \quad \text{and} \quad V = \bigcup_{d \in D} B\left(d, \frac{\varepsilon_d}{2}\right).$$

Clearly each is open and covers C, D. We claim they are disjoint. Indeed, if $x \in U \cap V$, then $x \in B(c, \frac{\varepsilon_c}{2}) \cap B(d, \frac{\varepsilon_d}{2})$ for some $c \in C, d \in D$. Using the triangle inequality, this implies $d(c, d) \leq \frac{\varepsilon_c + \varepsilon_d}{2}$. WLOG, assume $\varepsilon_c \leq \varepsilon_d$. Thus $d(c, d) \leq \varepsilon_d$. This implies that $c \in B_{d, \varepsilon_d}$, which contradicts our construction. Thus $U \cap V = \emptyset$, so we have a separation of C, D.

c) Let X be compact Hausdorff, and suppose A, B are disjoint closed sets in X. Then A, B are compact as well. We use the following lemma:

Lemma: Let $A \subseteq X$ be compact and $c \in X$ not contained in A. Then there exists U, V open and disjoint in X such that $A \subseteq U$ and $c \in V$.

Proof: Since X is Hausdorff, for each point $x \in A$, there exists U_x, V_x open and disjoint in X such that $x \in U_x, c \in V_x$. Thus $\{U_x\}_{x \in A}$ forms an open cover of A. Thus by compactness, there exists some finite subcover $U_{x_1}, \ldots, U_{x_n}$ for some $x_1, \ldots, x_n \in A$. Let $U = \bigcup_{k=1}^n U_{x_n}$ and $V = \bigcap_{k=1}^n V_{x_n}$. Clearly both U and V are open, and by construction V is disjoint from each U_{x_n} (since V_{x_n} is disjoint from it and $V \subseteq V_{x_n}$), so V is disjoint form U. We also have $A \subseteq U$ since the U_{n_i} form a cover of A, and $c \in V$ since $c \in V_{n_i}$. Thus we've constructed U, V as desired. ■

For each $b \in B$, by the lemma there exists U_b, V_b open and disjoint in X such that $A \subseteq U_b, b \in V_b$. Thus $\{V_b\}_{b \in B}$ form an open cover of B, so by compactness there exist a finite subcover $V_{b_1}, \ldots, V_{b_m}$ for some $b_1, \ldots, b_m \in B$. Let $U = \bigcap_{k=1}^m U_{b_m}$ and $V = \bigcup_{k=1}^m V_{b_m}$. Clearly U, V are open and $A \subseteq U, B \subseteq V$. By construction, U is disjoint from V_{b_i} for each i, since $U_{b_i} \supseteq U$ is disjoint to V_{b_i}. Thus U, V are disjoint, and so they separate A and B. Thus X is normal.

Proposition: If X is normal and $A \subseteq X$ is closed, then A is normal.

Proof: Let C, D be two closed sets in A. Since A is closed, both C and D are closed in X. Thus disjoint open sets U, V separate them, and the open sets $U \cap A, V \cap A$ in A separate C, D. ■

Example (subspaces of T_4 spaces aren't necessarily T_4):

Let J be uncountable. By Tychonoff's theorem, the space $[0, 1]^J$ is compact. Furthermore, as the product of Hausdorff spaces, it's Hausdorff, and thus T_4. We claim the subspace $(0, 1)^J$ is not T_4. We prove this by showing that $\mathbb{R}^J \cong (0, 1)^J$ is not T_4. Since $\mathbb{N}$ is closed in $\mathbb{R}$, $X := \mathbb{N}^J$ is closed in $\mathbb{R}^J$. By the previous proposition, it's thus enough to show that $\mathbb{N}^J$ is not T_4. We treat a typical element of X as a function from J to $\mathbb{N}$.

For $x \in X$ and a finite $B \subseteq J$, let $U(x, B)$ denote the set of points y in X for which $y(\alpha) = x(\alpha)$ for all $\alpha \in B$. Let $x' \in U(x_1, B_1) \cap U(x_2, B_2)$. Then $U(x', B_1 \cup B_2)$ lies inside the intersection. Clearly the topology on X is finer than the topology generated by this basis, so consider some

basis element $\prod_{\alpha \in J} U_\alpha$ of X. Let B be the finite subset of J for which $\alpha \in J \Rightarrow U_\alpha \neq \mathbb{N}$. Picking any x in the set, we see that $U(x, B)$ lies inside the basis element, and thus this basis generates the topology on X.

Now define P_n to be the subset of X consisting of points such that $x|_{J \setminus x^{-1}(\{n\})}$ is injective. First we show P_n is closed. Consider $x \in X \setminus P_n$. Thus there exist distinct $\alpha, \beta \in J$ such that $x(\alpha) = x(\beta) \neq n$. The basis set $U(x, \{\alpha, \beta\})$ then lies inside $X \setminus P_n$, and thus the set is open.

Now consider $x \in P_1 \cap P_2$. $x \in P_1$ implies that $x^{-1}(\{2\})$ is finite. However, since $x \in P_2$, this implies J minus finitely many points can inject into $\mathbb{N}$, impossible. Thus $P_1 \cap P_2 = \emptyset$. To finish, we show that any open sets U and V containing P_1 and P_2 respectively must intersect.

We want to construct the following objects: a sequence of of distinct elements $\alpha_1, \alpha_2, \ldots$ in J, a sequence of integers $0 = n_0 < n_1 < \cdots$, sets $B_i = \left\{\alpha_1, \ldots, \alpha_{n_i}\right\}$ for $i \geq 1$, and points $x_i \in X$ defined by

$$\begin{aligned} x_i(\alpha_j) &= j \quad 1 \leq j \leq n_{j-1}, \\ x_i(\alpha) &= 1 \quad \text{otherwise.} \end{aligned}$$

In particular, we want to construct them such that

$$U(x_i, B_i) \subseteq U.$$

We first start with x_1, which ends up being all ones, and thus lies in P_1, and by extension U. Since U is open, there exists a finite subset of J, which we call B_1, with elements $\left\{\alpha_1, \ldots, \alpha_{n_1}\right\}$ such that $U(x_1, B_1) \subseteq U$.

Thus we have x_2, and it's easy to see that it lies in P_1, and thus U. Thus there exists a finite subset of J, say B_2', with elements $\left\{\alpha_{n_1+1}, \ldots, \alpha_{n_2}\right\}$ for which $U(x_2, B_2') \subseteq U$. Let $B_2 = B_1 \cup B_2'$. Then $U(x_2, B_2) \subseteq U(x_2, B_2') \subseteq U$. We can continue this process and obtain the desired sequences and elements.

Let A be the sequence of elements of J constructed previously. Now define $y \in X$ by

$$\begin{aligned} y(\alpha_j) &= j \quad \alpha_j \in A, \\ y(\alpha) &= 2 \quad \text{otherwise.} \end{aligned}$$

It's easy to see that $y \in P_2$, and thus there exists a finite $B \subseteq J$ such that $U(y, B) \subseteq V$. Now choose i such that $B \cap A \subseteq B_i$. We claim that $U(x_{i+1}, B_{i+1}) \cap U(y, B)$ is nonempty, which will imply $U \cap V$ is nonempty.

Define $z \in X$ by

$$\begin{aligned} z(\alpha_j) &= j \quad 1 \leq j \leq n_i, \\ z(\alpha_j) &= 1 \quad n_i < j \leq n_{i+1}, \\ z(\alpha) &= 2 \quad \alpha \in B \setminus A, \\ z(\alpha) &= 3 \quad \text{otherwise.} \end{aligned}$$

It's easy to see that $z \in U(x_{i+1}, B_{i+1})$. For $\alpha \in B \cap A$, we know from $B \cap A \subseteq B_i$ that α is some α_j for $1 \leq j \leq n_i$, and thus $z(\alpha) = y(\alpha)$. For $\alpha \in B \setminus A$, by construction we have $z(\alpha) = y(\alpha)$. Thus $z \in U(y, B)$, as desired.

Example (products of T_4 spaces aren't necessarily T_4 and $T_4 \neq T_3$):

First we show that $\mathbb{R}_\ell$ is T_4. We already know that it's Lindelof and Hausdorff, so we just need to show it's regular. Let x be a point and U a neighborhood. Thus we can find $[a, b)$ with $x \in [a, b) \subseteq U$. Then $\left[\frac{a+x}{2}, \frac{b+x}{2}\right)$ contains x, and its closure $\left[\frac{a+x}{2}, \frac{b+x}{2}\right) \subseteq [a, b) \subseteq U$. Thus $\mathbb{R}_\ell$ is regular.

We now show that $\mathbb{R}_\ell^2$ isn't T_4. It then follows that the product of T_4 spaces need not be T_4, and furthermore, since $\mathbb{R}_\ell^2$ is T_3, it follows that T_3 spaces are not necessarily T_4.

Let $\Delta^- = \{(x, -x) \in \mathbb{R}_\ell^2 : x \in \mathbb{R}_\ell\}$. We claim this set is closed and inherits the discrete topology. The second is easy, since $[x, x+1)$ and $[-x, -x+1)$ are basis elements in $\mathbb{R}_\ell^2$, and thus $U = [x, x+1) \times [-x, -x+1)$ is a basis element in $\mathbb{R}_\ell^2$, since since $\Delta^- \cap U = \{(x, -x)\}$, the singleton is open in Δ^-. It's also easy to see that $\mathbb{R}_\ell^2 \backslash \Delta^-$ is open and thus Δ^- is closed, since we can easily make a small enough square at each point of $\mathbb{R}_\ell^2 \backslash \Delta^-$ that lies inside of it.

Now assume that $\mathbb{R}_\ell^2$ is normal. Let $A \subseteq \Delta^-$, which is closed since Δ^- has the discrete topology. Thus so is $\Delta^- \backslash A$. Since Δ^- is closed in $\mathbb{R}_\ell^2$, we see that A and $\Delta^- \backslash A$ are closed in $\mathbb{R}_\ell^2$. By normality, they can thus be separated by open sets of $\mathbb{R}_\ell^2$.

Let $\mathbb{Q}_\ell^2$ denote the subspace of $\mathbb{R}_\ell^2$ with rational coordinates. Now fix for every nonempty proper subset $A \subseteq \Delta^-$ a separation U_A, V_A of A and $\Delta^- \backslash A$, which exists by b). Let $\Psi : \mathcal{P}(\Delta^-) \to \mathcal{P}(\mathbb{Q}_\ell^2)$ be the function defined by sending $A \mapsto U_A \cap \mathbb{Q}_\ell^2$ for all non-empty proper subsets of Δ^- and $\Psi(\emptyset), \Psi(\Delta^-) = \mathbb{Q}_\ell^2$. We claim Ψ is injective.

If $A \neq \emptyset$, we must have that $\Psi(A) \neq \emptyset$ (since $\mathbb{Q}_\ell^2$ is dense in $\mathbb{R}_\ell^2$). Similarly, if $A \neq \Delta^-$, the set V_A is nonempty and thus contains rationals that U_A does not, so $\Psi(A) \neq \mathbb{Q}_\ell^2$. Now we just need to show that $\Psi(A) \neq \Psi(B)$ for distinct proper nonempty subsets A, B of Δ^-.

Suppose $A \neq B$ are proper nonempty subsets of Δ^-. Without loss of generality, there thus exists $(a, -a) \in A$ that's not in B. Since both U_A, V_B are open and contain $(a, -a)$, there exists some neighborhood of $(a, -a)$ lying in both. By density, this neighborhood intersects $\mathbb{Q}_\ell^2$. Thus some point of $\mathbb{Q}_\ell^2$ that lies in $\Psi(A)$ doesn't lie in $\Psi(B)$, so $\Psi(A) = \Psi(B)$, as desired.

We're now ready to obtain a contradiction. $\mathbb{Q}_\ell^2$ is countable, so it's power set has cardinality $\mathfrak{c}$. Since Δ^- has an obvious bijection to $\mathbb{R}$, it also has cardinality $\mathfrak{c}$. Thus there exists a bijection between the two sets. Let such a bijection f. Then $\Psi \circ f : \mathcal{P}(\Delta^-) \to \Delta^-$ is an injection, which contradicts Cantor's diagonalization argument.

3.3. Urysohn's Lemma and Consequences

Lemma (Urysohn's lemma): Let X be normal, and let $A, B \subseteq X$ be closed, disjoint, nonempty. There exists a continuous function $f : X \to [a, b] \subseteq \mathbb{R}$ such that $f(x) = a$ for $x \in A$ and $f(x) = b$ for $x \in B$.

Proof: Without loss of generality by scaling we can assume that $[a, b] = [0, 1]$. Let $Q := \mathbb{Q} \cap [0, 1]$. We enumerate $Q \backslash \{0, 1\}$ by dictionary ordering on first the denominator and then the

numerator, so $\left\{\frac{1}{2}, \frac{1}{3}, \frac{2}{3}, \frac{1}{4}, \frac{3}{4}, \frac{1}{5}, \cdots\right\}$. We then enumerate Q by $1, 0$, then the enumeration of $Q \backslash \{0, 1\}$.

We define neighborhoods U_q of A for $q \in Q$ inductively on the enumeration of Q. First,

$$U_1 := X \backslash B.$$

Using normality, we can find an open $V \subseteq U_1$ containing A such that $\overline{V} \subseteq U_1$. We refer to this fact as $(*)$. Then we define

$$U_0 := V.$$

Thus $A \subseteq U_0 \subseteq \overline{U_0} \subseteq U_1$.

To define $U_{1/2}$, we apply $(*)$ to $\overline{U_0}$ as the closed set and U_1 as the open set. Thus we obtain an open set $U_{1/2}$ such that

$$A \subseteq \overline{U_0} \subseteq U_{1/2} \subseteq \overline{U_{1/2}} \subseteq U_1.$$

Now we define the rest inductively. Let Q_n be the set of the first n elements of Q. Assume we've defined U_q for all $q \in Q_n$ such that whenever $p < q$, we have that

$$\overline{U_p} \subseteq U_q.$$

We've already done the base case, so this is okay.

We want to consider the last element of $Q_{n+1} = Q_n \cup \{r\}$ and use this to define U_r. Q as a subset of $\mathbb{Q}$ has an order. By construction Q_{n+1} has finitely many elements, so r has a largest element before it r_- and a smallest element after it r_+. Thus $r_- < r < r_+$, so using the induction hypothesis, we can apply $(*)$ to $\overline{U_{r_-}}$ and U_{r_+}. Thus we obtain U_r such that

$$\overline{U_{r_-}} \subseteq U_r \subseteq U_{r_+}.$$

We set this to be U_r in our sequence of sets. We continue doing this inductively.

Now define $U_q = \emptyset$ for $q < 0$ and $U_q = X$ for $q > 1$. It's now easy to see by construction that if $p < q$, then $\overline{U_p} \subseteq U_q$.

Now, for each $x \in X$, define $\mathbb{Q}(x) = \{r : x \in U_r\}$. Since $U_r = \emptyset$ for $r < 0$, we see that $\mathbb{Q}(x)$ is bounded below. Furthermore, by construction it will contain all $r > 1$. Thus $\inf \mathbb{Q}(x)$ exists for all $x \in X$.

We now define $f : X \to [0, 1]$ by $f(x) = \inf \mathbb{Q}(x)$. By construction, we see that for all $x \in A$, x is contained in all U_r with $r \geq 0$. Thus $f(x) = 0$ for all $x \in A$. Similarly, $f(x) = 1$ for all $x \in B$. Since it's easy to check each $\mathbb{Q}(x)$ is bounded below by 0 and can't be greater than 1 since $f(B) = 1$, we see this function is well-defined.

We now show that

$$x \in \overline{U_q} \Rightarrow f(x) \leq q \quad \text{and} \quad x \notin U_q \Rightarrow f(x) \geq q.$$

Indeed, suppose $x \in \overline{U_q}$. Thus $q \in \mathbb{Q}(x)$, so $\inf \mathbb{Q}(x) \leq q$. Similarly, if $x \notin U_q$, then $q \notin \mathbb{Q}(x)$. Furthermore, for every $p < q$, we have $\overline{U_p} \subseteq U_q$, so $p \notin \mathbb{Q}(x)$. Thus $\inf \mathbb{Q}(x) \geq q$.

Now we just need to show that f is continuous. To do this, we show that f is continuous at each point. Let $x \in X$, and let (y, z) be a neighborhood of $f(x)$. We can pick $q, p \in \mathbb{Q}$ such that $f(x) \in (p, q)$. Set $U := U_q \backslash \overline{U_p}$, which is open and nonempty, since $p < q$.

Applying the contrapositives of the inequalities above, since $f(x) > p$, we have $x \notin \overline{U_p}$, and since $f(x) < q$, we have $x \in U_q$. Thus $x \in U$. Now we just need to show that $f(U) \subseteq (y, z)$.

Pick $w \in U$. Thus $w \in U_q$, $w \notin \overline{U_p}$, which implies $w \in \overline{U_q}$ and $w \notin U_p$. Using the facts, we thus obtain $y < p \leq f(w) \leq q < z$, so we're done. ■

Definition: If A, B are two subsets of a space X, and if there's a continuous $f : X \to [0, 1]$ such that $f(A) = \{0\}$ and $f(B) = \{1\}$, we say A and B can be *separated by a continuous function.*

Definition (completely regular): A space X is *completely regular* if for each point x_0 and each closed set A not contaning x_0, there is a continuous function $f : X \to [0, 1]$ such that $f(x_0) = 1$ and $f(A) = \{0\}$.

Note that completely regular implies regular, since we can take the disjoint open sets $f^{-1}\left(\left[0, \frac{1}{2}\right)\right)$ and $f^{-1}\left(\left(\frac{1}{2}, 1\right]\right)$ to cover the closed set and the point.

Definition ($T_{3.5}$): A space X is $T_{3.5}$ if it's completely regular and T_1.

From Urysohn's lemma, we thus see that $T_4 \Rightarrow T_{3.5}$.

Proposition: Suppose X is completely regular. Then X is T_0 if and only if it's T_1 if and only if it's T_2.

Proof: Completely regular implies regular, and we know in regular spaces that T_0 implies T_2, so we're done. ■

Unlike normal spaces, completely regular spaces are nice with respect to subspaces and products.

Proposition: A subspace of a completely regular space is completely regular. A product of completely regular spaces is completely regular.

Thus subspaces and products of $T_{3.5}$ spaces are $T_{3.5}$.

Proof: Let X be completely regular, and let Y be a subspace of X. Let $C \subseteq Y$ be closed and let $x_0 \in Y$ be disjoint from C. Then we can write $C = Y \cap C'$ for some closed $C' \subseteq X$. Then by hypothesis, there exists a continuous $f : X \to [0, 1]$ such that $f(x_0) = 1$ and $f(C') = \{0\}$. Then restricting f to Y gives the desired continuous function.

Let $X = \prod X_\alpha$, where X_α is completely regular. Let $b = (b_\alpha)$ be a point in X, and let A be a closed set of X disjoint from b. Choose a basis element $\prod U_\alpha$ containing b that doesn't intersect A (we can do this since the product of regular spaces is regular). For all but finitely many $\alpha, U_\alpha = X_\alpha$. Call these $\alpha_1, \ldots, \alpha_n$. By complete regularity, we can find continuous functions $f_i : X_{\alpha_i} \to [0, 1]$ such that $f_i\left(b_{\alpha_i}\right) = 1$ and $f_i\left(X_{\alpha_i} \setminus U_{\alpha_i}\right) = \{0\}$ (since $b \in \prod U_\alpha$, $b_{\alpha_i} \in U_{\alpha_i}$). Let

$\varphi_i(x) = f_i\left(\pi_{\alpha_i}(x)\right)$. Then φ_i maps X continuously into $\mathbb{R}$ and vanishes outside $\pi_{\alpha_i}^{-1}\left(U_{\alpha_i}\right)$ by construction. The function $f(x) := \varphi_1(x)\cdots\varphi_n(x)$ is the desired function, since it equals 1 at b and vanished outside $\prod U_\alpha$. ■

Example (space that's T_3 but not $T_{3.5}$):

First we defined the space (which already requires some effort). Define a subset X of $\mathbb{R}^2$ as follows: for each even integer m, let L_m denote the line segment $\{m\} \times [-1, 0]$. For each odd integer n and each integer $k \geq 2$, let $C_{n,k}$ denote the union of the line segments $\left\{n + 1 - \frac{1}{k}\right\} \times [-1, 0]$ and $\left\{n - 1 + \frac{1}{k}\right\} \times [-1, 0]$ and the semicircle $\left\{(x, y) : (x - n)^2 + y^2 = \left(1 - \frac{1}{k}\right)^2, y \geq 0\right\}$. Let $p_{n,k}$ denote the topmost point $\left(n, 1 - \frac{1}{k}\right)$ of $C_{n,k}$. Then let X be the union of all L_m and $C_{n,k}$, along with two extra points a, b.

We topologize X with the following basis sets:

(i) The intersection of X with an open horizontal line segment that contains none of the points $p_{n,k}$.
(ii) A set formed from one of the $C_{n,k}$ with at most finitely many points deleted.
(iii) For each even integer m, the union of $\{a\}$ and the set of points (x, y) of X for which $x < m$.
(iv) For each even integer m, the union of $\{b\}$ and the set of points (x, y) of X for which $x > m$.

It's easy (although tedious) to check that these sets do actually form a basis on X.

Now let $f : X \to \mathbb{R}$ be continuous. Let $S_{n,k}$ denote the set of points p in $C_{n,k}$ for which $f(p) \neq f\left(p_{n,k}\right)$. We claim that $S_{n,k}$ is countable. Consider

$$U_{n,k,m} = f^{-1}\left(\left(f\left(p_{n,k}\right) - \frac{1}{m}, f\left(p_{n,k}\right) + \frac{1}{m}\right)\right).$$

By continuity, $U_{n,k,m}$ is open. Since it contains $p_{n,k}$, there exists some basis set inside of it containing $p_{n,k}$. Of the four basis sets, only (i) couldn't be a possible basis set. Taking the other three cases into account, we see that $C_{n,k}$ with at most finitely many points deleted (none of which are $p_{n,k}$) lies in $U_{n,k,m}$. We then have

$$\begin{aligned} f^{-1}\left(\left\{f\left(p_{n,k}\right)\right\}\right) &= f^{-1}\left(\bigcap_{m=1}^{\infty}\left(f\left(p_{n,k}\right) - \frac{1}{m}, f\left(p_{n,k}\right) + \frac{1}{m}\right)\right) \\ &= \bigcap_{m=1}^{\infty} f^{-1}\left(\left(f\left(p_{n,k}\right) - \frac{1}{m}, f\left(p_{n,k}\right) + \frac{1}{m}\right)\right) \\ &= \bigcap_{m=1}^{\infty} U_{n,k,m}. \end{aligned}$$

Thus $f^{-1}\left(\left\{f\left(p_{n,k}\right)\right\}\right)$ contains $C_{n,k}$ with at most countably many points removed. In particular, this implies $S_{n,k}$ is countable, as desired.

Now choose $d \in [-1, 0]$ such that $y = d$ intersects X at no points in any $S_{n,k}$ (which is possible since the union of them all is countable). Now fix odd n. For each k, we have $f\left(n - 1 + \frac{1}{k}, d\right) = f\left(p_{n,k}\right) = f\left(n + 1 - \frac{1}{k}, d\right)$ by the fact that neither $\left(n - 1 + \frac{1}{k}, d\right)$ note $\left(n + 1 - \frac{1}{k}, d\right)$ lie in $S_{n,k}$. Taking limits yields

$$\lim_{k\to\infty} f\left(n - 1 + \frac{1}{k}, d\right) = f(n - 1, d) = \lim_{k\to\infty} f\left(p_{n,k}\right) = f(n + 1, d) = \lim_{k\to\infty} f\left(n + 1 - \frac{1}{k}, d\right),$$

where the limits hold by continuity and the fact that any open neighborhood of $(n-1, d)$ must contain a set of form (i), and thus contain the tail of the sequence $\left(n-1+\frac{1}{k}, d\right)$, and similarly for $(n+1, d)$. In particular, this tells us that $f(n, d)$ is constant over all even n. Then using continuity again, and the fact that $\lim_{n\to\infty}(n, d) = b$ and $\lim_{n\to-\infty}(n, d) = a$ (again use the basis sets to see that they always contain a tail of the sequence), we see that $f(a) = f(b)$.

We now show that X is regular but not completely regular. First we note that it's easy to check that X is T_1, so singletons are closed. Furthermore, we note that $C_{n,k}\backslash\left\{\left(p_{n,k}\right)\right\}$ inherits the discrete topology.

We first show X is regular using the alternative formulation. Let $x \in X$ and U a neighborhood of x. If $x \in C_{n,k}\backslash\{p\}$, then $x \in \{x\} = \overline{\{x\}} \subseteq U$. If $x = p_{n,k}$ some set of the form $C_{n,k}$ minus finitely many points lies in U, say V. It's easy to check that it's complement is open, and thus it's closed. Thus $x \in V = \overline{V} \subseteq U$. If $x \in L_m$, then U contains a basis set of type (i), say V. In particular, any such basis set is closed, and so again $x \in V = \overline{V} \subseteq U$. If $x = a, b$, then note that the closure of a set of type (iii) or (iv) just involves including L_m, and thus we can easily find a neighborhood of x inside of U with closure inside U. Thus X is T_3.

Let $f : X \to \mathbb{R}$ be contiuous. From previously, we must have $f(a) = f(b)$. However, a is a point and $\{b\}$ is closed, and so the two cannot be separated. Thus X is not $T_{3.5}$.

Example (space that's $T_{3.5}$ but not T_4): Since $\mathbb{R}_\ell$ is T_4, it's also $T_{3.5}$. Thus $\mathbb{R}_\ell^2$ is $T_{3.5}$. However, we know that $\mathbb{R}_\ell^2$ is not normal.

Corollary:

$$T_0 \subset T_1 \subset T_2 \subset T_3 \subset T_{3.5} \subset T_4$$

3.3.1. Urysohn Metrization Theorem

Theorem (Urysohn metrization theorem): Let X be regular and second countable. Then X is metrizable.

Proof: The idea of the proof is to embed X into a space Y that is metrizable. Thus X will be homeomorphic to a subspace of Y, which will be metrizable. In particular, we will embed X into $\mathbb{R}^\omega$. Note that since X is regular and second countable, it's also normal. Thus we cann apply the Urysohn lemma to X.

We first find a collection of continuous functions $\{f_i\}_{i\in\mathbb{N}}$ from X to $[0, 1]$ such that for all $x \in X$ and $x \in U$ open, there exists i such that f_i separates $\{x\}$ and $X\backslash U$. Let $\mathcal{B} = \{B_n\}_{n\in\mathbb{N}}$ be a countable basis for X. For every $(n, m) \in \mathbb{N}^2$ with $\overline{B_n} \subseteq B_m$, define $f_{n,m}$ using the Urysohn lemma to separate $\overline{B_n}$ and $X\backslash B_m$. This is clearly a countable collections.

Now pick $x \in X$ and a neighborhood U of X. We can find a basis element B_m such that $B_m \subseteq U$. By regularity, we can find a neighborhood U' of x such that $\overline{U'} \subseteq B_m$. Then taking another basis element inside U', we find there exists B_n, B_m such that $x \in B_n \subseteq \overline{B_n} \subseteq B_m \subseteq U$. Then by constructions, $f_{n,m}$ separates $\overline{B_n}, X \backslash B_m$, and thus by the above containments, we see it also separates $\{x\}$ and $X \backslash U$. Thus the collection $\{f_{n,m}\}$ is the desired collection. Reindex to be $\{f_i\}_{i \in \mathbb{N}}$.

Now define $F : X \to \mathbb{R}^\omega$ by $x \mapsto (f_1(x), f_2(x), \ldots)$. We claim this is the desired embedding. Continuity follows since each component is continuous and $\mathbb{R}^\omega$ has the product topology. To show injectivity, suppose $x, y \in X$ are distinct. By normality, we can separate these two points by U, V. By constructions, there exists f_i, f_j that separate $\{x\}, X \backslash U$ and $\{y\}, X \backslash V$. Since $f_i(X \backslash U) = \{1\}$ and $y \in X \backslash U$, we see that $f_i(y) = 0$. Since $f_i(x) = 1$ by construction, we see that $F(x) \neq F(y)$. Thus F is injective.

Set $W := F(X)$. To finish, we need to show that F is a homemorphism from X to W. We already showed it's a continuous bijection. We show that F is an open map. It then follows that F^{-1} is continuous.

Let $U \subseteq X$ be open, and pick $w \in F(U)$. By bijiectivity, there exists $x \in U$ such that $F(x) = w$. By construction, there exists f_i such that $f_i(x) = 1$, $f_i(X \backslash U) = \{0\}$. Let $Z := \pi_i^{-1}((0, \infty)) \cap W$, which is open in W. It's easy to see that $w \in Z$. Pick $z \in Z$. Thus $F(y) = z$ and $f_i(y) > 0$. By construction, this implies $y \in U$, and thus $z \subseteq F(U)$. Thus $Z \subseteq F(U)$. In particular, this implies $F(U)$ is open, as desired. ■

3.3.2. Tietze Extension Theorem

3.3.3. Imbeddings of Manifolds

3.4. Problems

Problem: Show that if X is Lindelof and Y is compact, then $X \times Y$ is Lindelof.

Solution: Let $\mathcal{A}$ be an open cover of $X \times Y$. For each $x \in X$, the subset $\{x\} \times Y$ is compact, and thus has some finite subcover $A_1, A_2, \ldots, A_n$. Let $U_x = A_1 \cup \cdots \cup A_n$. Thus $\{x\} \times Y \subseteq U_x$.

Thus by the tube lemma, there exists open $W_x \subseteq X$ that contains x such that $W_x \times Y \subseteq U_x$. By Lindelof, we get some countable subcover $W_{x_1}, W_{x_2}, \ldots$. Since fintely many A_i cover each of these, we get a countable subcover of $X \times Y$, as desired.

Problem: Show that $\mathbb{R}^{[0,1]}$ is separable.

Solution: Note that we can interpret elements in this space as functions from $[0,1]$ to $\mathbb{R}$. Let K_n consist of funtions of the form $q_1\chi_{[a_1,b_1]} + \cdots + q_n\chi_{[a_n,b_n]}$, where $q_i, a_i, b_i \in \mathbb{Q}$. Clearly K_n is countable. Then let $K = \bigcup_n K_n$, which is countable. We claim K is dense in $\mathbb{R}^{[0,1]}$.

Let U be open in $\mathbb{R}^{[0,1]}$. There thus exists some basis set $\prod_{x\in[0,1]} U_x$ contained inside U. Let $x_1, \ldots, x_n$ be the points for which $U_x \neq \mathbb{R}$, and let them be ordered. Within each, there exists a rational q_{x_i}. For each x_i, pick distinct $a_i, b_i \in \mathbb{Q}$ such that $a_i \leq x_i \leq b_i$ (we can modify the procedure accordingly when $x_i = 0, 1$) and such that $b_i < a_{i+1}$. Then the function $q_{x_1}\chi_{[a_1,b_1]} + \cdots + q_{x_n}\chi_{[a_n,b_n]}$ lies in the basis set, and thus U, as desired.